20/01/26

Unit - 2.

Applications of OP-AMP :~ ...

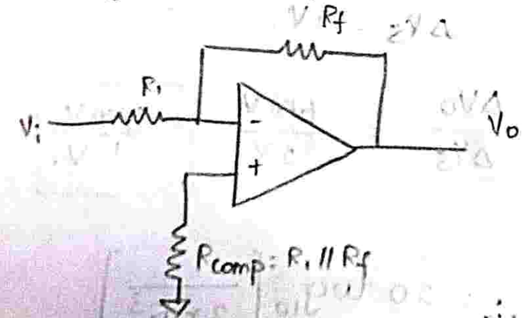
2 types:

1. Linear application. [Linear relationship b/w i/p & o/p]
2. Non-linear application.

Linear →
- current to voltage convertor.
- Voltage to current convertor.
- Adder.
- subtractor.
- summing amplifier.
- Instrumentation amplifier.

Non-Linear →
- Differentiator.
- Integrator
- Rectifier.
- Low pass filter.
- high pass filter.
- peak detector.
- Logarithmic amplifier.

① Inverting amplifier.



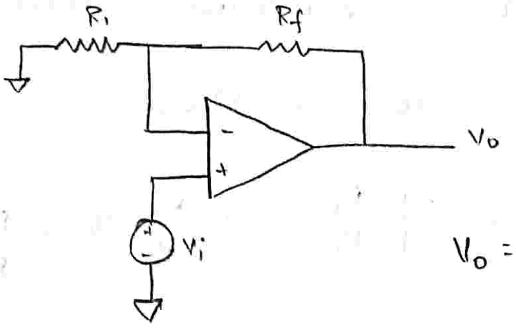
$$\boxed{V_o = -\left(\frac{R_f}{R_i}\right) V_i}$$

$\therefore R_f = R_i = R$.

then $\boxed{V_o = -V_i}$

$V_o = -V_i \longrightarrow$ act as scale changer / invertor.

②. Non-inverting amplifier.

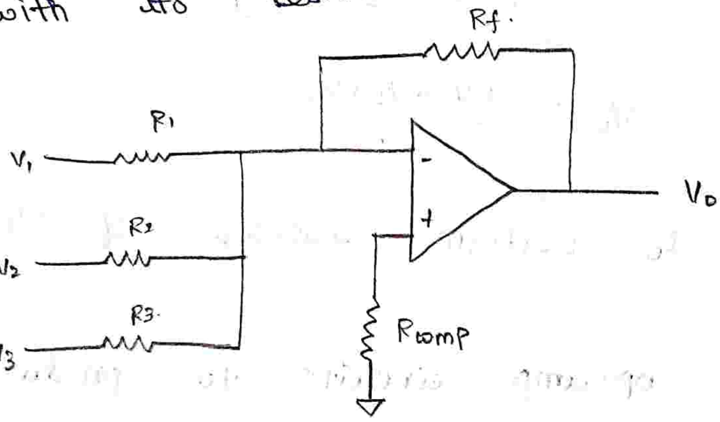


$$V_o = \left[1 + \frac{R_f}{R_1}\right] V_i.$$

Inverting summing amplifier.

=> to get this, we modify the inverting amplifier.

*. By applying multiple inputs along with its resistance.



*. It performs addition and amplification.

$V_{o1} \Rightarrow V_o$ due to 'V_1' alone $= -\frac{R_f}{R_1} V_1$

$V_{o2} \Rightarrow$ o/p (V_o) due to 'V_2' alone $= \frac{-R_f}{R_2} V_2$.

$V_{o3} \Rightarrow V_o$ due to 'V_3' alone $= -\frac{R_f}{R_3} V_3$.

by superposition theorem,

Total output is equal to output of each individual source.

$$V_o = V_{o1} + V_{o2} + V_{o3}.$$

$$V_o = -\left[\frac{R_f}{R_1} V_1 + \frac{R_f}{R_2} V_2 + \frac{R_f}{R_3} V_3 \right]$$

if $R_1 = R_2 = R_3 = R_f$

It performs addition of i/p's.

then, $\boxed{V_o = -(V_1 + V_2 + V_3)}$

[It has 180° phase shift btw i/p & o/p.]

if $R_1 = R_2 = R_3 = 3R_f$.

$$V_o = -\left[\frac{V_1}{3} + \frac{V_2}{3} + \frac{V_3}{3} \right]$$

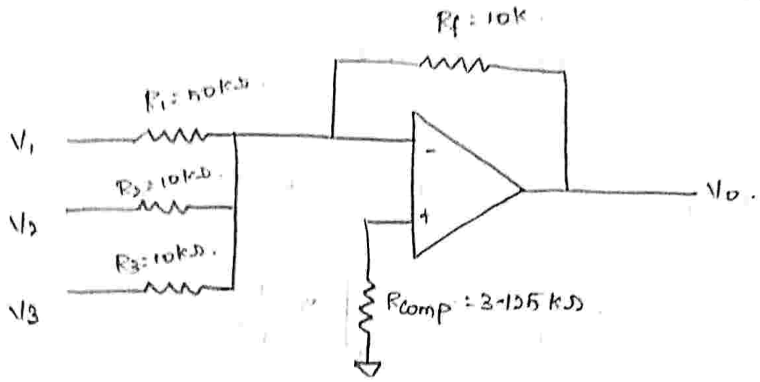
$$\boxed{V_o = -\frac{(V_1 + V_2 + V_3)}{3}}$$

It can able to perform average of i/p's.

①. design as op-amp circuits to produce the output expression of $V_o = -(0.2V_1 + V_2 + V_3)$

Sol: Inverting summing amplifier.

$$V_o = -\left[\frac{R_f}{R_1} V_1 + \frac{R_f}{R_2} V_2 + \frac{R_f}{R_3} V_3 \right]$$



compare equation,

$$\frac{R_f}{R_1} V_1 = 0.2 V_1$$

$$\frac{R_f}{R_1} = 0.2$$

$$R_1 = \frac{R_f}{0.2}$$

Let choose $R_f = 10k$.

$$R_1 = \frac{10k}{0.2} \boxed{= 50 k\Omega}$$

IIIly.
$$\frac{R_f}{R_2} V_2 = V_2$$

$$R_f = R_2$$

$$\boxed{\therefore R_2 = 10 k\Omega}$$

$$\frac{R_f}{R_3} V_3 = V_3$$

$$R_3 = R_f$$

$$\boxed{\therefore R_3 = 10 k\Omega}$$

$$R_{comp} = R_1 \parallel R_2 \parallel R_3 \parallel R_f .$$

$$\frac{1}{R_{comp}} = \frac{1}{50k} + \frac{1}{10k} + \frac{1}{10k} + \frac{1}{10k} .$$

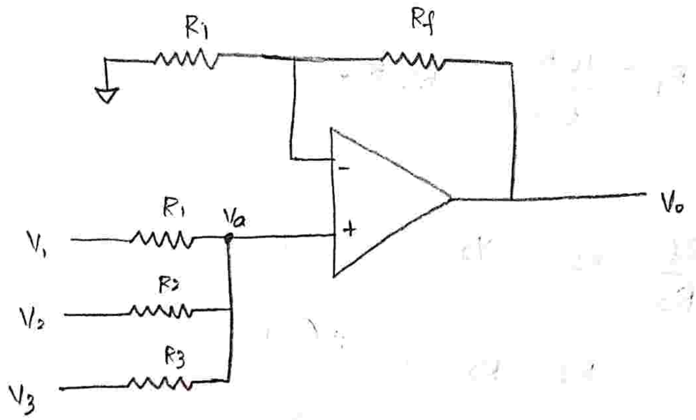
$$= \left[\frac{1}{50} + \frac{3 \times 5}{10 \times 5}\right] \times \frac{1}{10^3} .$$

$$\frac{1}{R_{comp}} = \frac{16}{50} \times \frac{1}{10^3} .$$

$$R_{comp} = \frac{50}{16} \times 10^3 .$$

$$\boxed{R_{comp} = 3.125 \times 10^3 \Omega}$$

Non Inverting - summing Amplifier.



$$V_o = \left[1 + \frac{R_f}{R_1}\right] V_a \rightarrow (1).$$

Apply nodal analysis at V_a.

$$\frac{V_a - V_1}{R_1} + \frac{V_a - V_2}{R_2} + \frac{V_a - V_3}{R_3} = 0 .$$

$$V_a\left[\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}\right] = \frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3} .$$

$$\therefore V_a = \frac{\frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3}}{\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}} \rightarrow (2).$$

Substitute (2) in equ (1).

$$V_o = \left[1 + \frac{R_f}{R_i}\right] \frac{\left[\frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3}\right]}{\left[\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}\right]}$$

Let us assume that

$$R_1 = R_2 = R_3 = R$$

$$V_o = \left[1 + \frac{R_f}{R_i}\right] \frac{\frac{V_1 + V_2 + V_3}{\cancel{R}}}{\frac{3}{\cancel{R}}}$$

$$V_o = \left(1 + \frac{R_f}{R_i}\right)\left(\frac{V_1 + V_2 + V_3}{3}\right)$$

choose, $R_i = \frac{R_f}{2}$.

$$V_o = \left[1 + \frac{2\cancel{R_f}}{\cancel{R_f}}\right]\left[\frac{V_1 + V_2 + V_3}{3}\right]$$

$$V_o = (\cancel{3}) \frac{V_1 + V_2 + V_3}{\cancel{3}}.$$

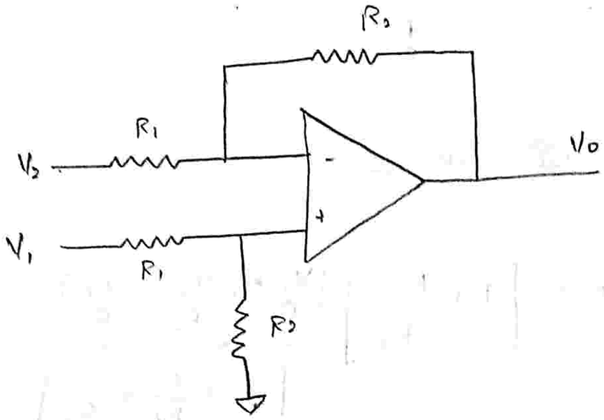
$$\boxed{\therefore V_o = V_1 + V_2 + V_3.}$$

$\therefore$ input and output of circuit is same phase.

$\frac{V_3}{R_3}$.

Subtractor:

* Difference Amplifier is used here.



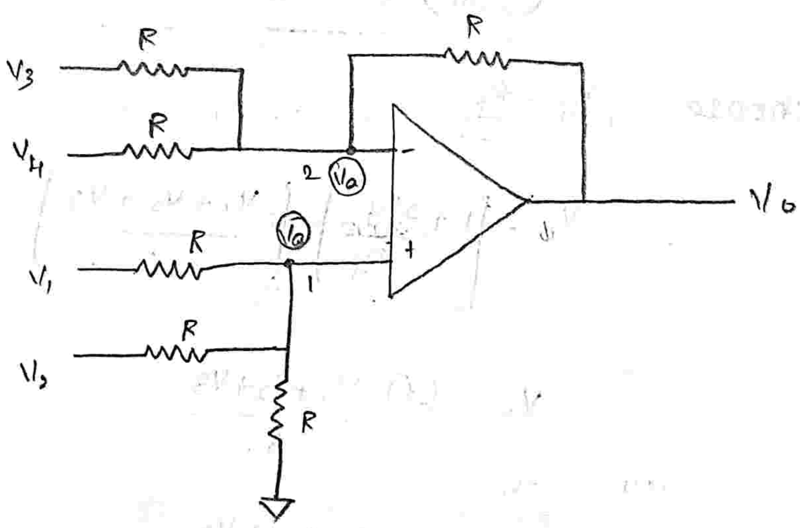
$$V_o = \frac{R_2}{R_1}(V_1 - V_2)$$

if $R_2 = R_1$,

then $V_o = (V_1 - V_2)$

$\Rightarrow$ it performs subtractor.

Adder - subtractor Circuit:



Apply nodal analysis at node 1

$$\frac{V_a - V_1}{R} + \frac{V_a - V_2}{R} + \frac{V_a - 0}{R} = 0.$$

$$\therefore V_a\left[\frac{1}{R} + \frac{1}{R} + \frac{1}{R}\right] = \frac{V_1}{R} + \frac{V_2}{R}$$

nero.

$$V_a = \frac{\frac{V_1+V_2}{R}}{\frac{3}{R}}.$$

$$V_a\left[\frac{3}{R}\right] = \frac{V_1+V_2}{R} \longrightarrow (1).$$

Apply nodal analysis at node 2.

$$\frac{V_a-V_3}{R} + \frac{V_a-V_4}{R} + \frac{V_a-V_0}{R} = 0.$$

$$V_a\left[\frac{1}{R}+\frac{1}{R}+\frac{1}{R}\right] = \frac{V_3}{R}+\frac{V_4}{R}+\frac{V_0}{R}.$$

$$V_a\left[\frac{3}{R}\right] = \frac{V_3+V_4+V_0}{R} \longrightarrow (2).$$

equate (1) & (2).

$$\frac{V_1+V_2}{\cancel{R}} = \frac{V_3+V_4+V_0}{\cancel{R}}$$

$$\boxed{\therefore V_0 = (V_1+V_2) - (V_3+V_4)}$$

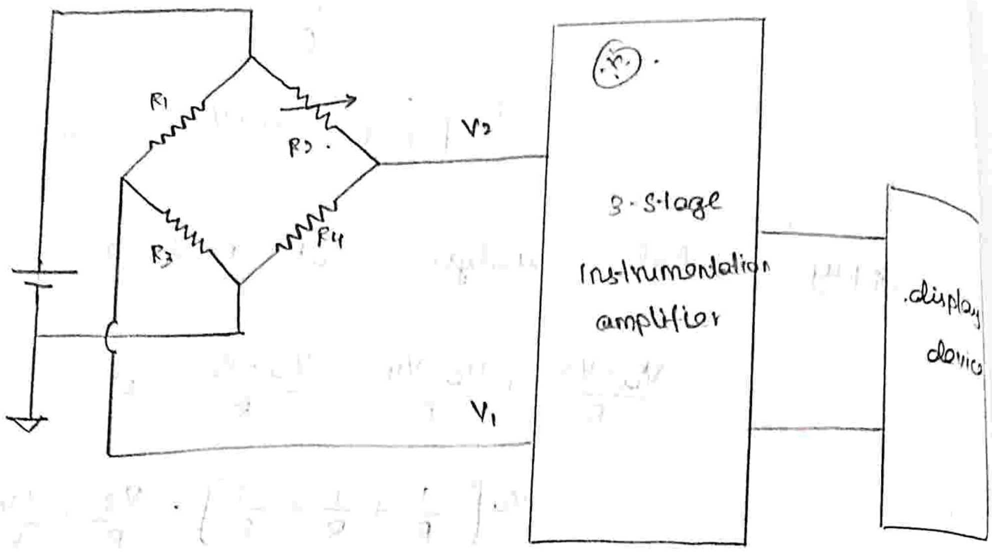
Instrumentation Amplifier: [transducer bridge Based Instrumentation Amplifier]

transducer - convert non electrical signal to electrical signal.

- It can able to measure any physical quantity.

eg: temp, humidity, water flow, pressure

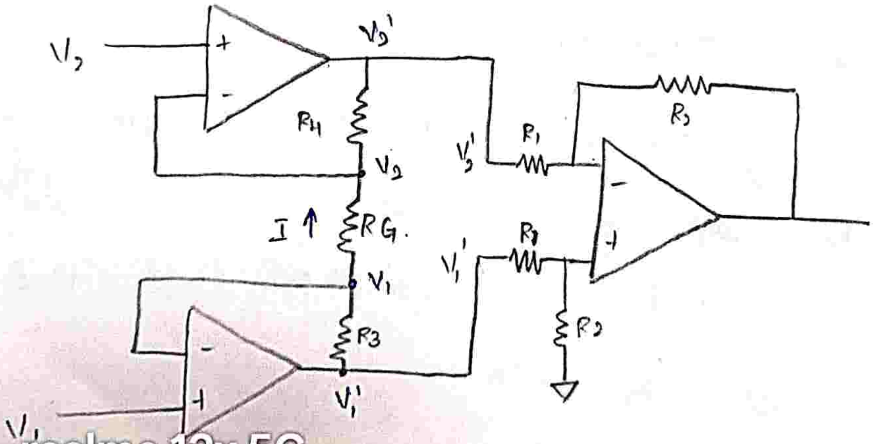
The o/p of transducer is very weak it is connected to amplifier in that we can see the o/p.



V_2 & V_1 is weak so it connected to instrumentation amplifier.

* In difference amplifier loading effect is created bcoz the input impedance are low to amplify.

* For instrumentation amplitude, the input sources given through the voltage (or) buffer follower circuit to the difference amplifier, by this loading effect is overcome.



$\underbrace{\text{2 Voltage follower circuits}}$ + $\underbrace{\text{1 Difference amplifier}}$ = 3 stage instrumentation amplifier

Advantage:

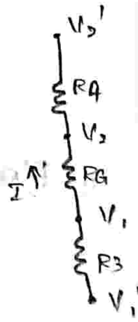
1. It offers high gain accuracy.
2. It offers high CMRR.
3. offers high stablity.
4. low offset voltages.
5. low output impedance.

$$V_o = \frac{R_2}{R_1}(V_1' - V_2') \rightarrow (1).$$

here $I = \dfrac{V_1 - V_2}{R_G} \Rightarrow V_1 - V_2 = IR_G \rightarrow (2)$.

no current flows throu input terminal.

so, R_3, R_4, R_G are connected in series



$$I = \frac{V_1' - V_2'}{R_3 + R_4 + R_G}$$

Let, $R_3 = R_4$

$$I = \frac{V_1' - V_2'}{2R_3 + R_G} \longrightarrow (3).$$

equate (2) & (3).

$$\frac{V_1 - V_2}{R_G} = \frac{V_1' - V_2'}{2R_3 + R_G}.$$

$$\therefore V_1' - V_2' = \frac{2R_3 + R_G}{R_G}(V_1 - V_2)$$

$$V_1' - V_2' = \left[1 + \frac{2R_3}{R_G}\right](V_1 - V_2) \longrightarrow (4).$$

Substitute (4) in eqn (1).

$$\therefore \boxed{V_o = \left(\frac{R_2}{R_1}\right)\left(1+\frac{2R_3}{R_G}\right)(V_1 - V_2)}$$

$\frac{R_2}{R_1} \rightarrow$ gain of difference amplifier.

$1+\frac{2R_3}{R_G} \rightarrow$ gain of 2 voltage follower amplifiers.

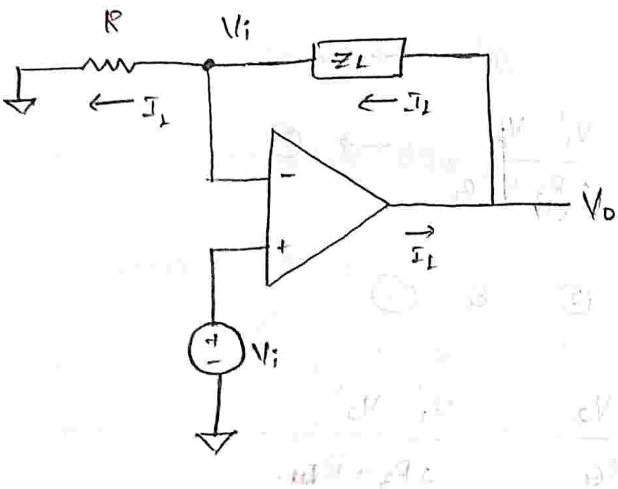
Voltage to Current Converter:

[Transconductance amplifier].

1. Floating load type V-I converter
2. Grounded load type V-I converter.

$\Rightarrow$ gain $= \frac{I}{V} = \frac{1}{R} = g_m$.

1. Floating load type V-I convertor.



Load impedance is not grounded, it is floating, we connected to o/p.

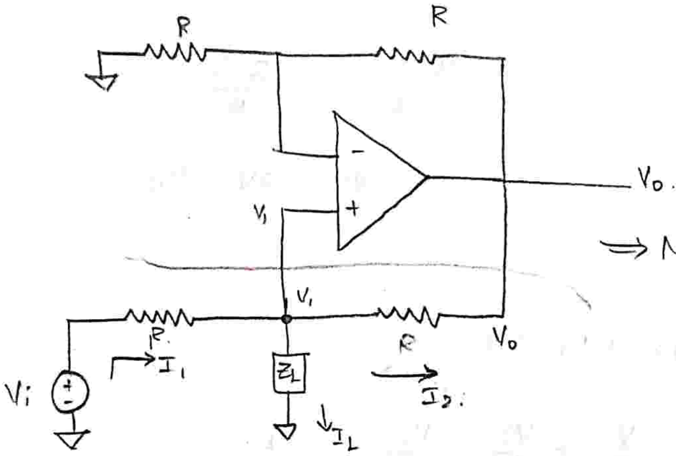
The applied voltage should decide

$$I_L = \frac{V_i}{R} \Rightarrow V_i = I_L R.$$

Precision resistor:

At any cost, value of impedance won't change.

2. Grounded load type V-I convertor.



⟹ Non-inverting amplifier.

Here the load impedance is connected to ground.

Two steps is followed.

1. Apply KCL w.r.t V_1 node.
2. Non-inverting amplifier.

∴ o/p of Non-inverting amplifier when V_1 is known.

$$V_0 = \left[1 + \frac{R}{R}\right] V_1$$

$$V_0 = 2V_1 \longrightarrow ①$$

Apply KCL eqn w.r.to Node 'V_1'

Let us assume $Z_L = R$.

$$\frac{V_0}{R} - \frac{V_1}{R} + \frac{V_i}{R} - \frac{V_1}{R} = \frac{V_1}{R}.$$

$$\frac{V_0 + V_i}{R} = \frac{3V_1}{R}$$

$$V_i = 3V_1 - V_0.$$

$$I_2 + I_1 = I_L.$$

$$\frac{V_0}{R} - \frac{V_1}{R} + \frac{V_i}{R} - \frac{V_1}{R} = I_L.$$

$$\frac{V_0 + V_i}{R} - \frac{2V_1}{R} = I_L.$$

$$V_0 = I_L R + 2V_1 - V_i. \rightarrow (2)$$

Sub (1) in (2).

$$\cancel{V_0} = I_L R + \cancel{2}\frac{\cancel{V_0}}{\cancel{2}} - V_i$$

$$\boxed{V_i = I_L R.} \quad (*)$$

Precision resistor

$$I_L = \frac{V_i}{R}.$$

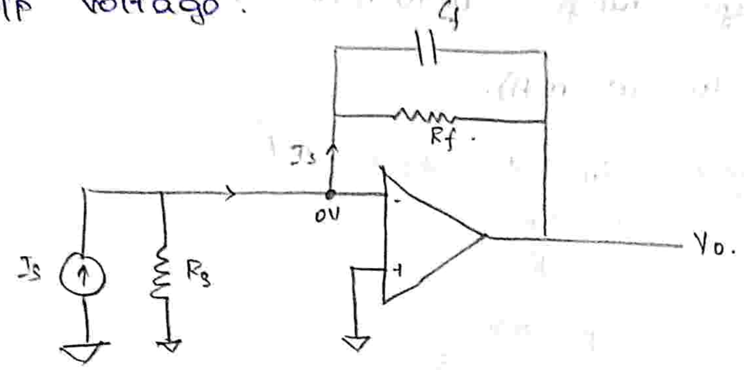
<u>V- I :</u>

* current produced w.r.t to voltage.

circuit, which produce constant current through load impedance.

Current to Voltage Converter:

Applied input current we get o/p voltage.



Assume, there is no feedback capacitance

$\therefore C_f \rightarrow$ eliminate high frequency noise signals.

Apply KCL at 0V node.

$$I_s = \frac{0 - V_o}{R_f}$$

$$\boxed{V_o = -I_s R_f}$$

$$\text{gain} \Rightarrow \frac{V_o}{I_s} = -R_f = R_f.$$

Gain of circuit expressed in resistance.

$\therefore$ It is also known as trans resistance Amplifier.

C_f is not needed when we using low frequency signals.

②. Design a V-I converter with grounded load using an ideal op-amp such that i/p voltage range (0 to 5)V, output current range (0 to 10 mA).

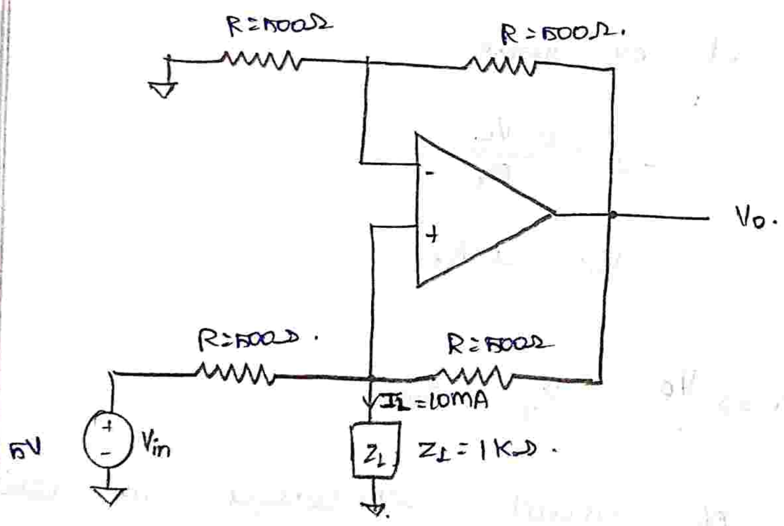
sol: Given, $V_{in} = 5V$, $I_L = 10mA$.

$$I_L = \frac{V_{in}}{R}$$

$$R = \frac{5V}{10 \times 10^{-3}}$$

$$R = 0.5 \, K\Omega$$

$$\boxed{R = 500 \, \Omega}$$



choose $Z_L = 1K\Omega$.

③. Design a grounded load V-I converter that produces o/p current is 2 mA/V of input signal, and maximum o/p votage is 4 V.

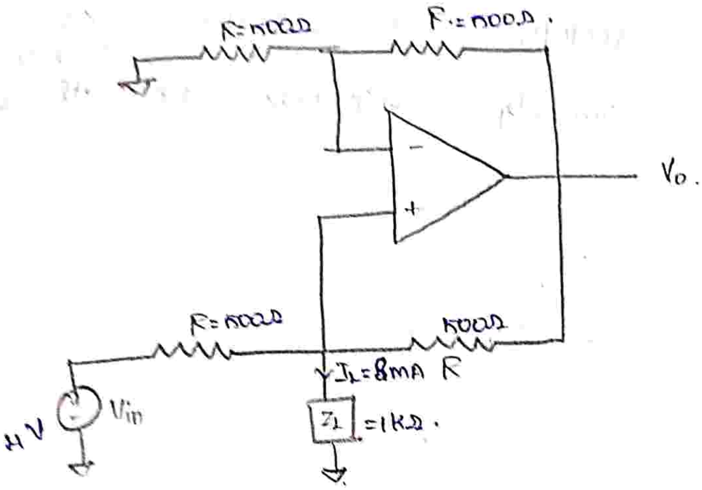
sol: $I_L = \frac{V_{in}}{R}$

$$\frac{2mA}{1V} = \frac{I_L}{V_{in}} = \frac{1}{R}$$

$$R = \frac{1}{2\times10^{-3}}$$

$$R = 0.5\ k\Omega.$$

$$\boxed{R = 500\Omega}.$$



when $V_{in} = 1V \rightarrow I_L = 2mA.$

$V_{in} = 4V \rightarrow I_L = 4\times2\times10^{-3}$

$I_L = 8\ mA.$

Resistance remains same. (that's why it is precision resistance).

⇒ when we use the input less than cut-off voltage of diode ($<0.7V$), the rectifier can't rectify. then we can go for op-amp based rectifier circuit which performs rectification.

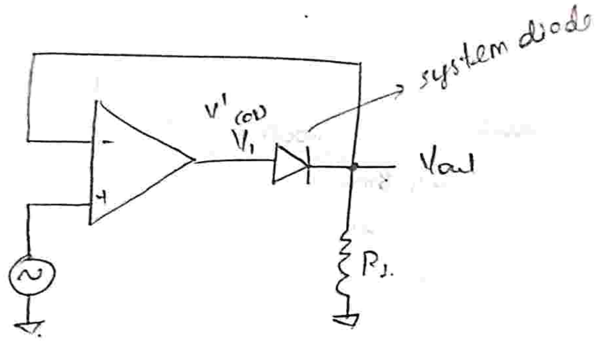
Super-diode:

whenever the combination of two diodes and op-amp.

Precision Rectifier:

Due to it's open loop gain, the diode rectify the signal ($< 0.7\,V$) ($< \mu V$)

V_{out} strongly depends on the diode



$V_1 = A(V_{in} - V_{out})$ (or) $V_1 = A(V_+ - V_-)$

$V_1 = 0.7 + V_{out}$.

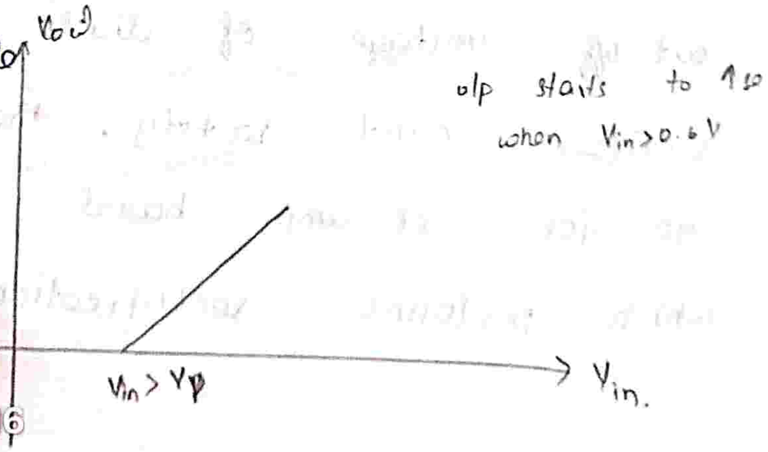
$0.7 + V_{out} = A(V_{in} - V_{out})$

$\frac{0.7 + V_{out}}{A} = V_{in} - V_{out}$.

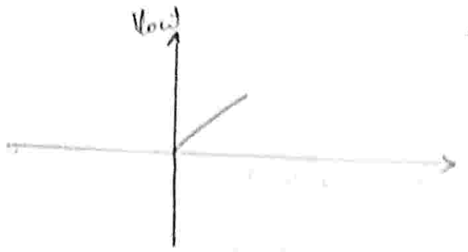
$\boxed{V_{in} = V_{out}}$, if $A_{OL} = \infty$.

For ideal diode → No knee voltage & built-in voltage.

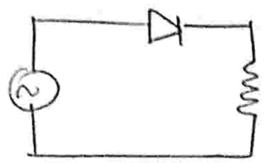
1) Normal diode

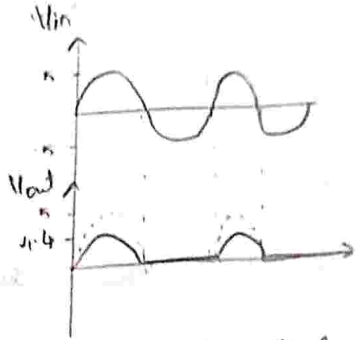


2). Super diode:



o/p comes at V=0 itself.





for non-ideal diode

*. Usually, diode has internal built-in voltage of 0.6 V - Silicon, 0.3 V - Germanium. This diode is non-ideal diode.

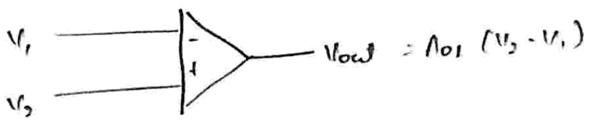
*. It give the o/p by reducing 0.6 V. It doesn't rectify when i/p is less than 0.6 V.

*. In graph, it reduces the amplitude, whereas in ideal there is no change in amplitude.

*. If we want o/p for (i/p < 0.6 V) then we use op-amp with diode.

$$\text{so}, \; V_{out} = \frac{0.6}{A_{OL}} = \frac{0.6}{10^4} = 60\mu V.$$

*. Now at 60×10^{-6} V the diode can rectify



$V_2 > V_1$: $+V_{out}$

$V_2 < V_1$ = $-V_{out}$.

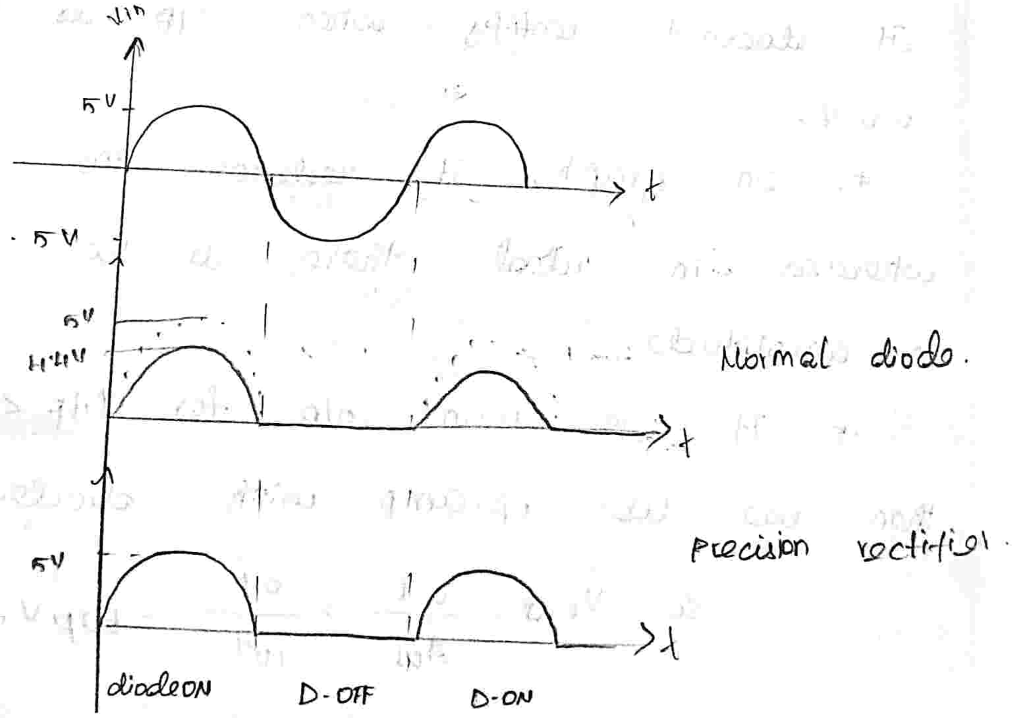
when, $V_i < 0$.

$V_1 = -V_{\cancel{out}\ sat}$.

The diode is off & reverse bias

$V_0 = 0$.

~~in~~ precision rectifier, it perform rectification with no drop voltage, in output.



<u>Limitation</u>:

*. It is only suitable to perform for low frequency signal. [due to it's slow Rate it may not get o/p]

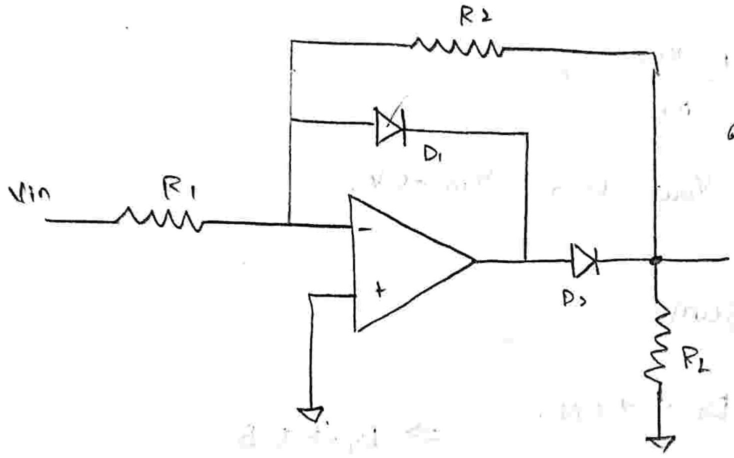
$V_{in} \rightarrow$ during +ve half cycle.

$V_o' = +V_{sat}$.

$V_{in} \rightarrow$ during -ve half cycle.

$V_o' = -V_{sat}$.

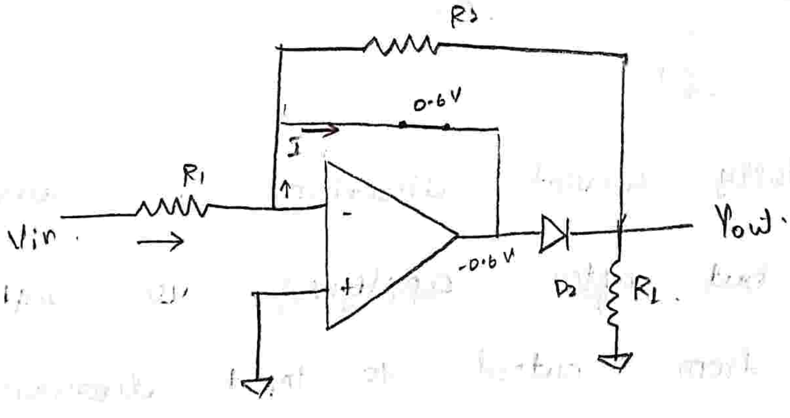
Modified precision half wave rectifier.



=) to overcome the drawback of precision rectifier we use this

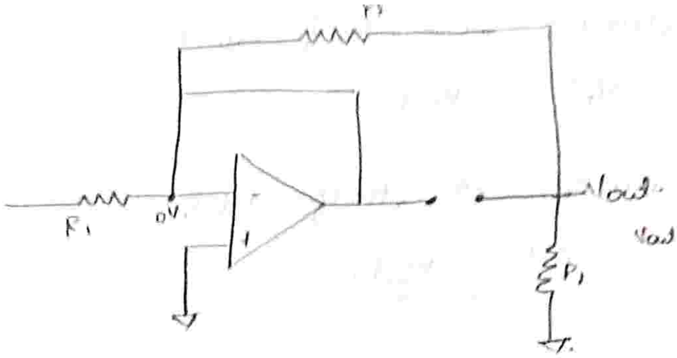
Let us assume, $D_1 \rightarrow$ ON, when $V_{in} > 0V$.

D_1 & D_2 is not directly connected to o/p of op-amp.



$D_2 \rightarrow$ OFF [bcoz -0.6V to +ve terminal connected so Reverse bias occurs].

Due to virtual ground concept -ve terminal has 0V. The circuit, $V_{out} = 0$.



V_{out} is corroded to Ground so 0V

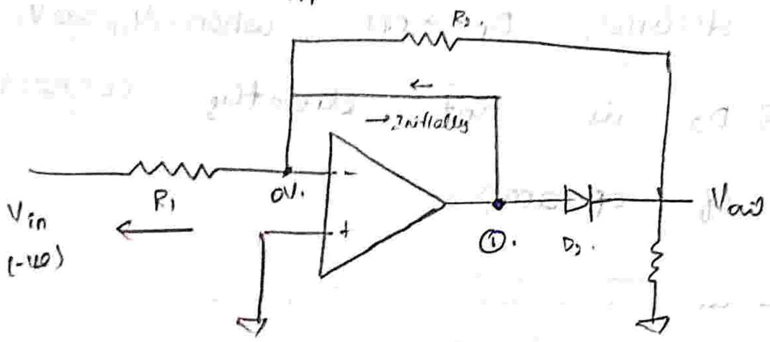
$$\frac{V_{out} - V_{in}}{R_2} = 0.$$

$\boxed{V_{out} = 0,}$ $V_{in} > 0V,$

Let us Assume,

$D_1 \longrightarrow ON,$ $\Rightarrow D_1 \rightarrow R.B$

$V_{in} = -ve.$



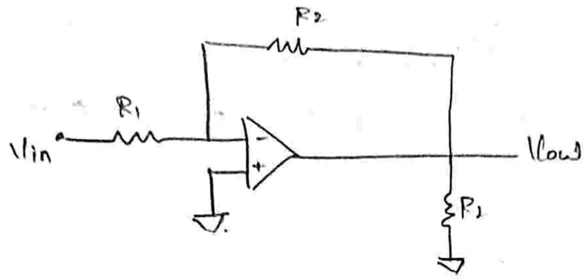
Initially current direction is towards V_{out} D-F.B but after applying -ve voltage I flows from output to input direction, direction changes, so D_1 is R.B, finally $D_1 \rightarrow OFF$.

∴ Voltage at 1 is greater than 0V (V_{in}

so +ve is connected to +ve of D_2

so, $D_2 \rightarrow ON.$

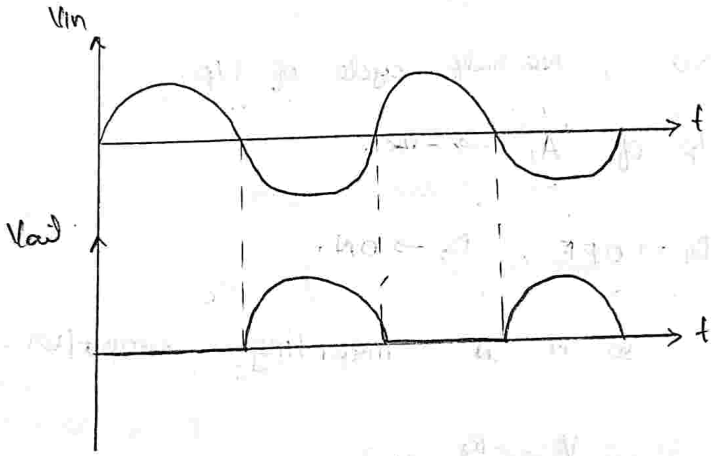
If ~~to~~ $V_{in} = -ve$, it is ~~b~~ Inverting amplifier.



$$V_{out} = \left(-\frac{R_2}{R_1}\right)(-V_{in})$$

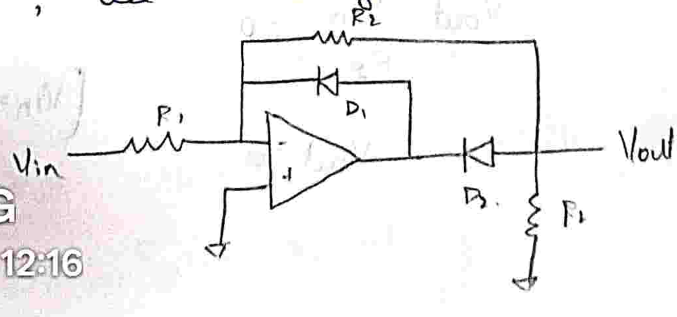
$$V_{out} = \frac{R_2}{R_1} V_{in}.$$

~~Ex~~ o/p of modified precision Rectifier circuit.



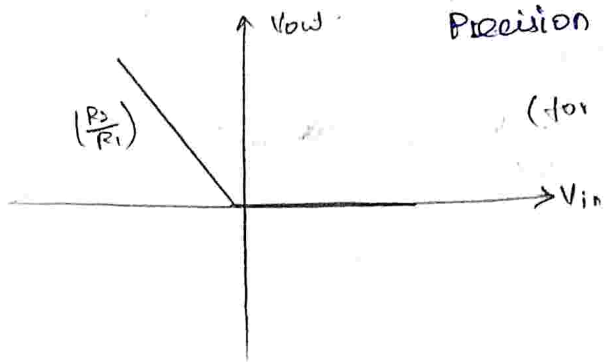
*. only +ve hc of circuit is rectified.

*. If we want to rectify -ve half cycle, we change the bias of D_1, D_2



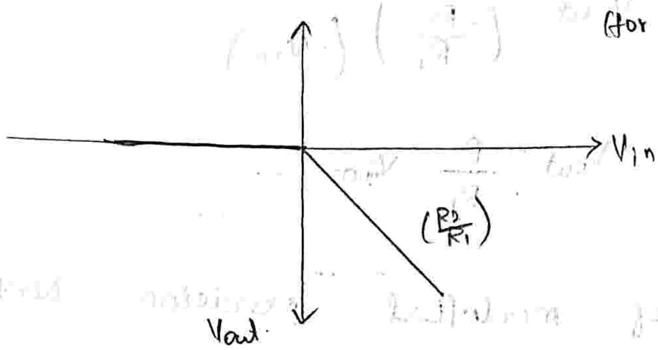
(it rectify -ve half cycle) o/p

Input & output characteristics of Modified Precision rectifier.



(for +ve hc)

(for -ve hc)



$V_{in} > 0$, +ve half cycle of i/p.

o/p of 'A₁' → -ve.

$D_1 \to$ OFF , $D_2 \to$ ON.

So it is inverting amplifier.

$V_0 = -\frac{R_2}{R_1} V_{in}$. It is rectified with amplitude $\frac{R_2}{R_1}$

$V_{in} < 0$, -ve half cycle of i/p.

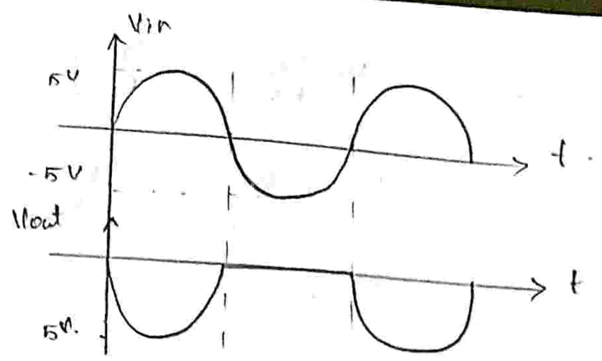
o/p of 'A₁' → +ve.

$D_1 \to$ ON , $D_2 \to$ OFF.

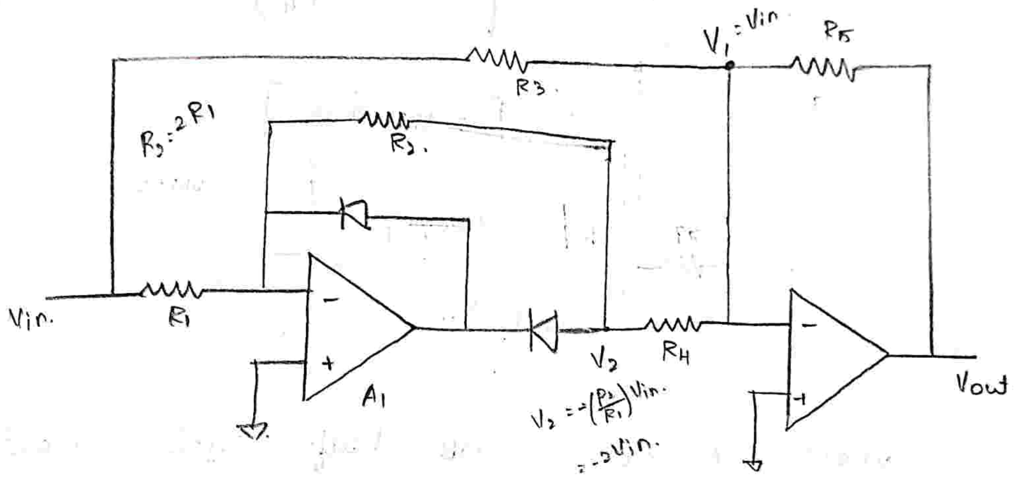
$$\frac{V_{out} - V_{in}}{R_2} = 0.$$

$[V_{in} = 0]$

$V_{out} = 0.$



Full-wave Precision Rectifier.



It is the combination modified half-wave rectifier followed by inverting summing amplifier.

when $V_{in} > 0$, [+ve half cycle i/p]

O/P of 'A1' → -ve.

$D_1 \rightarrow$ OFF (R.B)

$D_2 \rightarrow$ ON (F.B).

$$V_o = V_2 = -\frac{R_2}{R_1} V_{in} \quad (\text{-ve}).$$

$$V_1 = V_{in} = +ve.$$

when, $V_1 = V_2$ with opp polarity.

$R_3 = R_5 = R_4 = R$.

$$V_{out} = -\left(V_1 \frac{R_5}{R_4} + V_2 \frac{R_5}{R_3} \right)$$

$$= -\left(-V_{in} \frac{R_5}{R_4} + V_{in} \frac{R_5}{R_3} \right)$$

$$= -(-V_{in} + V_{in})$$

$$= 0.$$

$(R_5 = R_4 = R_3 = R)$

$$V_{out} = -\left[-2V_{in}\left(\frac{R_5}{R_4}\right) + V_{in}\left(\frac{R_5}{R_3}\right) \right]$$

$$= -[-2V_{in} + V_{in}]$$

$$= -(-V_{in})$$

$$\boxed{V_{out} = V_{in}.}$$

when $V_{in} < 0$, -ve half cycle input.

o/p of 'A_1' $\rightarrow$ +ve.

$D_1 \rightarrow$ ON, $D_2 \rightarrow$ OFF

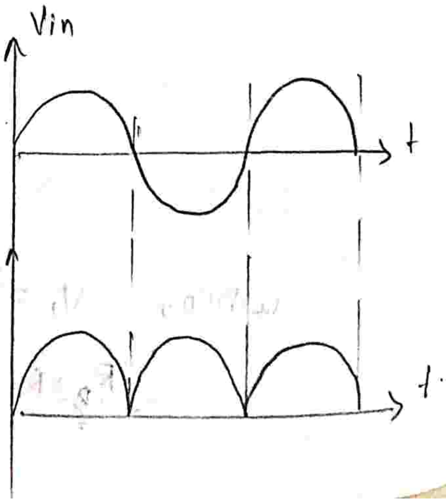
$V_{out} = 0.$

$\therefore V_2 = 0, \quad V_1 = -V_{in}.$

$$V_o = -\left[V_2\left(\frac{R_5}{R_4}\right) + V_1\left(\frac{R_5}{R_3}\right) \right]$$

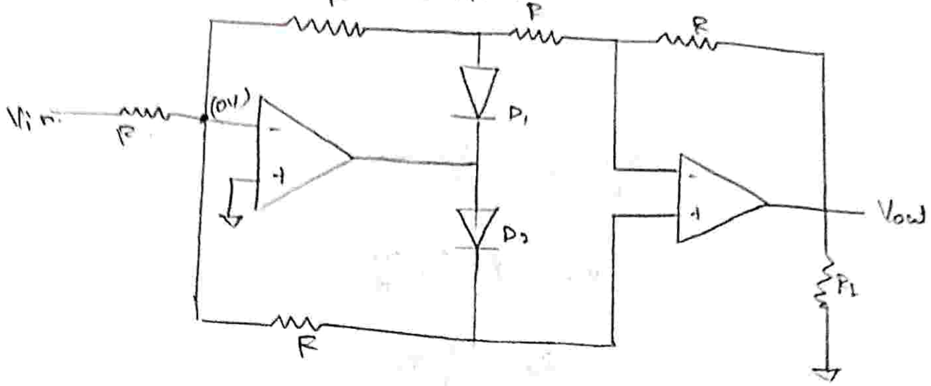
$$= -[0 - V_{in}]$$

$$V_o = +V_{in}$$



full wave rectifier

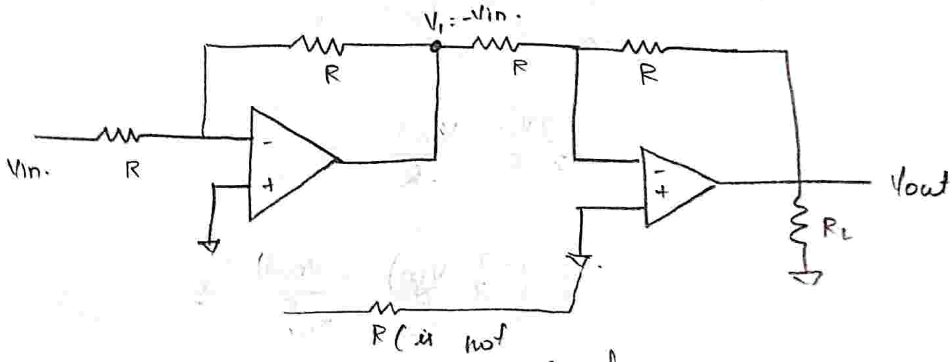
Full-wave Precision Rectifier.



$V_{in} > 0$, +ve half cycle of i/p.

$D_1 \rightarrow$ ON (short circuit)

$D_2 \rightarrow$ OFF (open circuit).



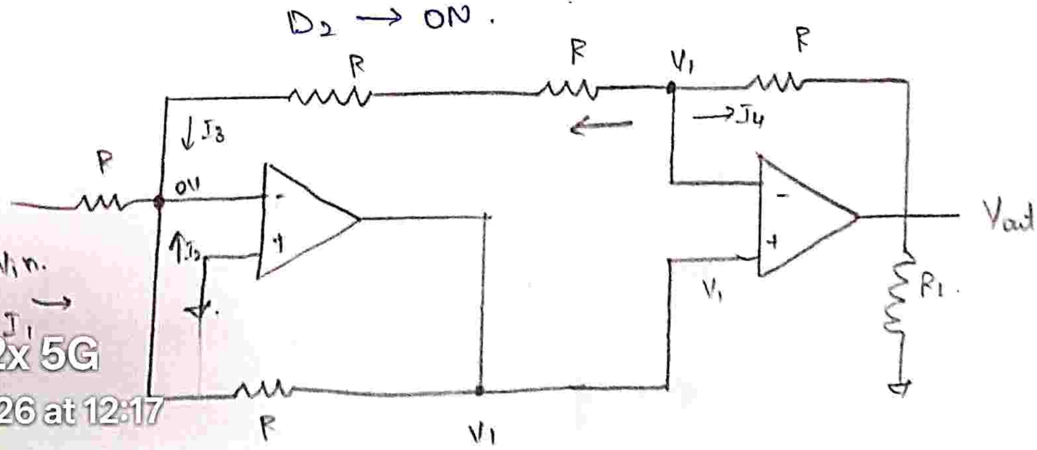
$V_1 = V_{o1} = -V_{in}$.

$V_{out} = -(-V_{in}) = V_{in}$.

$V_{in} < 0$, NHC of i/p.

$D_1 \longrightarrow$ OFF

$D_2 \longrightarrow$ ON.



$$\frac{V_{in}-0}{R} + \frac{V_1-0}{R} + \frac{V_1}{2R} = 0.$$

$$\frac{V_{in}}{R} + \frac{3}{2}\frac{V_1}{R} = 0.$$

$$V_{in} = -\cancel{R}\left(\frac{3}{2}\right)\frac{V_1}{\cancel{R}}.$$

$$\boxed{V_{in} = -\frac{3}{2}V_1}$$

$$V_1 = -\frac{2}{3}V_{in}.$$

$$I_3 + I_4 = 0.$$

$$\frac{V_1-0}{2R} + \frac{V_1 - V_{out}}{R} = 0.$$

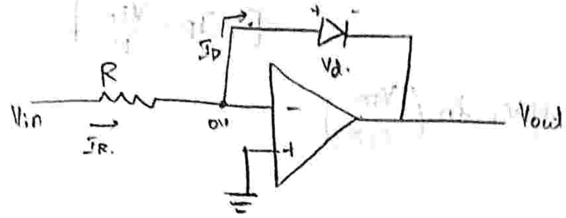
$$\frac{3}{2}\frac{V_1}{R} - \frac{V_{out}}{R} = 0.$$

$$\frac{\cancel{3}}{\cancel{2}}\left(-\frac{\cancel{2}}{\cancel{3}}\frac{V_{in}}{R}\right) - \frac{V_{out}}{R} = 0.$$

$$\frac{-V_{in}}{\cancel{R}} = \frac{V_{out}}{\cancel{R}}$$

$$\boxed{V_{out} = +V_{in}.}$$

Logarithmic - Amplifier.



$I_S \rightarrow$ reverse saturation current

$I_D \rightarrow$ diode current.

$V_D \rightarrow$ diode voltage.

$V_T \rightarrow$ Volt equivalent temperature

diode current equation $I_D = I_S\left[e^{V_D/\eta V_T} - 1\right]$ $\rightarrow$ ①.

where,

$$V_T = \frac{kT}{q}$$

$k \rightarrow$ boltzman's constant $= 1.38 \times 10^{-23}$ J/K

$q \rightarrow 1.602 \times 10^{-19}$ C [coloumb]

$T \rightarrow$ temperature.

$$V_T = \frac{kT}{q} \simeq 26\,mV \text{ at room temp } (25°C)$$

from the given circuit.

$$0 + V_d + V_{out} = 0.$$

$$V_{out} = -V_D. \longrightarrow ②.$$

From equ ①,

$$\frac{I_D}{I_S} = \left[e^{V_D/\eta V_T} - 1\right]$$

$$\frac{I_D}{I_S} + 1 = e^{V_D/\eta V_T}.$$

$\therefore \frac{I_D}{I_S} >> 1$, neglect 1.

$$\frac{I_D}{I_S} = e^{V_D/\eta V_T}$$

apply ln on b/s.

$$\frac{V_D}{\eta V_T} = \ln\left(\frac{I_D}{I_S}\right)$$

$$V_D = \eta V_T \ln\left(\frac{I_R}{I_S}\right)$$

$\left[\because I_R = \frac{V_{in}}{R}\right]$

$$V_D = \eta V_T \ln\left(\frac{V_{in}}{I_S R}\right)$$

from equ ①,

$$V_{out} = -V_D$$

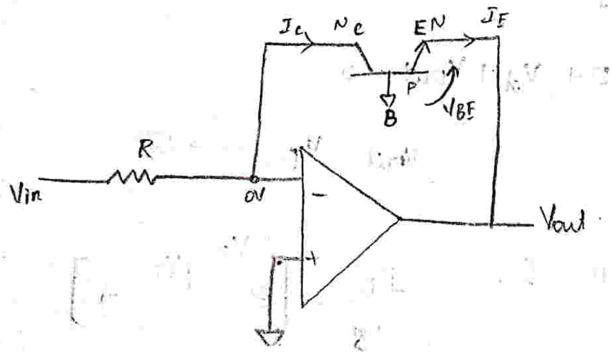
$$\boxed{V_{out} = -\eta V_T \ln\left(\frac{V_{in}}{I_S R}\right)}$$

$\therefore$ output $\propto$ logarithm of V_{in}

V_{out} depends on η, we don't need this so we construct transistor base logarithm op.amp.

Drawback : presence of η, dependence of temperature parameter

Transistor Base Log - Amplifier:



$\because$ $I_E = I_C + I_B$ [$\because$ Base is grounded $I_B = 0$]

$$\boxed{I_E \approx I_C.}$$

collector current, $I_C = I_S\left[e^{V_E/V_T} - 1\right] \rightarrow ①$

from the circuit.

$0 + V_E + V_{out} = 0.$

$$V_{out} = -V_E \rightarrow (2).$$

from equ (1).

$$\frac{I_c}{I_s} = \left[e^{V_E/V_T} - 1 \right]$$

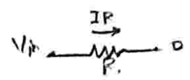
$$\frac{I_c}{I_s} + 1 = e^{V_E/V_T}.$$

$\because \frac{I_c}{I_s} >> 1$, 1 is neglected.

$$\frac{I_c}{I_s} = e^{V_E/V_T}$$

Apply natural log on b/s.

$$\frac{V_E}{V_T} = \ln\left(\frac{I_c}{I_s}\right)$$



$$\frac{V_E}{V_T} = \ln\left(\frac{I_R}{I_s}\right) \qquad \left[\because I_R = \frac{V_{in}}{R}\right]$$

$$V_E = V_T \ln\left(\frac{V_{in}}{I_s \cdot R}\right) \rightarrow (3).$$

sub (3) in (1).

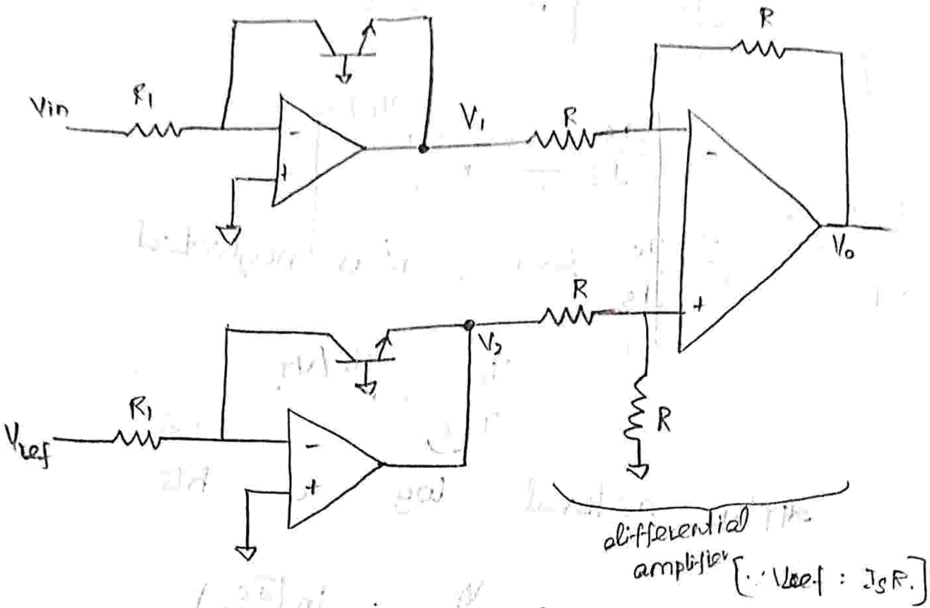
$$V_{out} = -V_T \ln\left(\frac{V_{in}}{I_s \cdot R}\right)$$

$$\boxed{V_{out} = -\frac{KT}{q} \ln\left(\frac{V_{in}}{I_s R}\right)}$$

Drawback:

Here, still V_{out} is dependent of temperature.

Temperature compensated transistor Based Logarithmic Amplifier.



$$V_1 = \frac{-KT}{q} \ln\left[\frac{V_{in}}{I_S R_1}\right]$$

$$V_2 = \frac{-KT}{q} \ln\left[\frac{V_{ref}}{I_S R_1}\right]$$

By differential amplifier we know,

$$V_o = +\frac{R_2}{R_1} (V_2 - V_1)$$

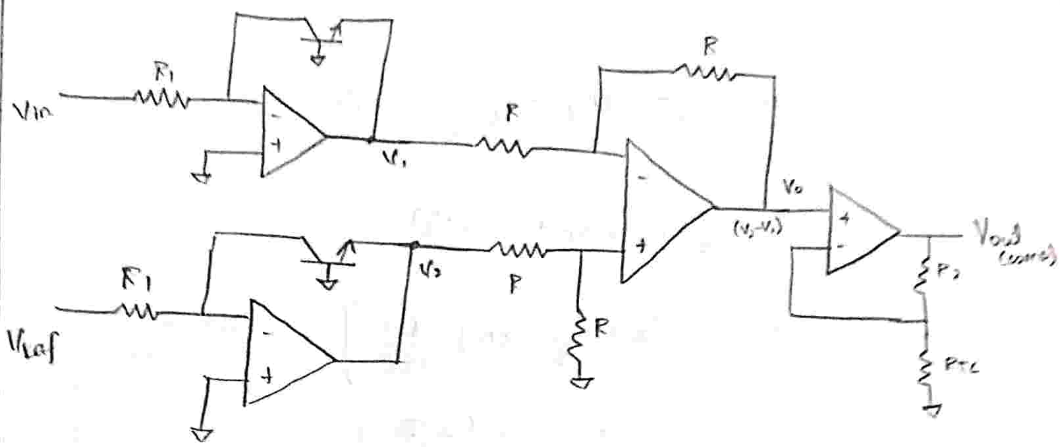
$$\therefore V_{out} = \frac{\cancel{R}}{\cancel{R}} (V_2 - V_1)$$

$$V_{out} = \frac{-KT}{q} \ln\left(\frac{V_{ref}}{I_S R_1}\right) + \frac{KT}{q} \ln\left(\frac{V_{in}}{I_S R_1}\right)$$

$$V_{out} = \frac{KT}{q} \ln\left[\frac{V_{in}}{V_{ref}}\right]$$

Inorder to compensate the T

we append one more circuit, is non-inverting amplifier.

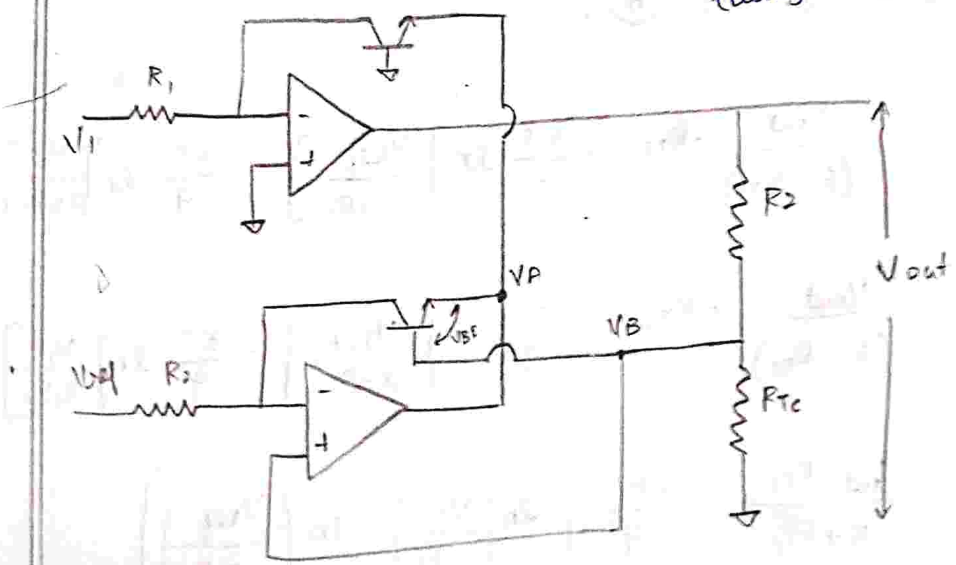


R_{Tc} → temperature dependent resistor

$$V_{o(comp)} = \left[1+\frac{R_2}{R_{Tc}}\right]\frac{KT}{q}\ln\left[\frac{V_{in}}{V_{ref}}\right]$$

when T ↑se R_{TC} also ↑se both cancel each other, so temperature effect is cancelled.

Temperature compensated log-amplifier: (using 2 log-amp).



V_A - Collector terminal of Transistor & collector terminal of Transistor.

$$V_A = -\frac{KT}{q} \ln\left[\frac{V_1}{I_S R_1}\right] \rightarrow (1).$$

$$\therefore V_B - V_E = V_A. \rightarrow (3)$$

here, $V_E = \frac{KT}{q} \ln\left[\frac{V_{ref}}{I_S R_1}\right] \Rightarrow$ derived from transistor base log amp. $\hookrightarrow (2)$.

sub (2) in (3).

$$V_B - \frac{KT}{q} \ln\left[\frac{V_{ref}}{I_S R_1}\right] = -\frac{KT}{q} \ln\left[\frac{V_1}{I_S R_1}\right] \rightarrow (5).$$

here V_B = voltage across 'R_{TC}'

By voltage - divider theorem, $V_B = \frac{V_{out}}{(R_2 + R_{TC})} R_{TC}$ $\hookrightarrow (4)$.

sub (4) in (5),

$$\frac{V_{out}}{(R_2+R_{TC})} \cdot R_{TC} - \frac{KT}{q} \ln\left[\frac{V_{ref}}{I_S R_1}\right] = -\frac{KT}{q} \ln\left[\frac{V_1}{I_S R_1}\right]$$

$$\frac{V_{out}}{(R_2+R_{TC})} \cdot R_{TC} = \frac{KT}{q} \ln\left[\frac{V_{ref}}{I_S R_1}\right] - \frac{KT}{q} \ln\left[\frac{V_1}{I_S R_1}\right]$$

$$\frac{V_{out} \cdot R_{TC}}{R_2 + R_{TC}} = -\frac{KT}{q} \left[\ln\left(\frac{V_1}{I_S R_1}\right) - \ln\left(\frac{V_{ref}}{I_S R_1}\right)\right]$$

V_B depends upon V_{out}.

$$V_{out} = \left(\frac{R_2 + R_{TC}}{R_{TC}}\right)\left(\frac{-KT}{q}\right) \ln\left[\frac{\frac{V_1}{\cancel{I_S R_1}}}{\frac{V_{ref}}{\cancel{I_S R_1}}}\right]$$

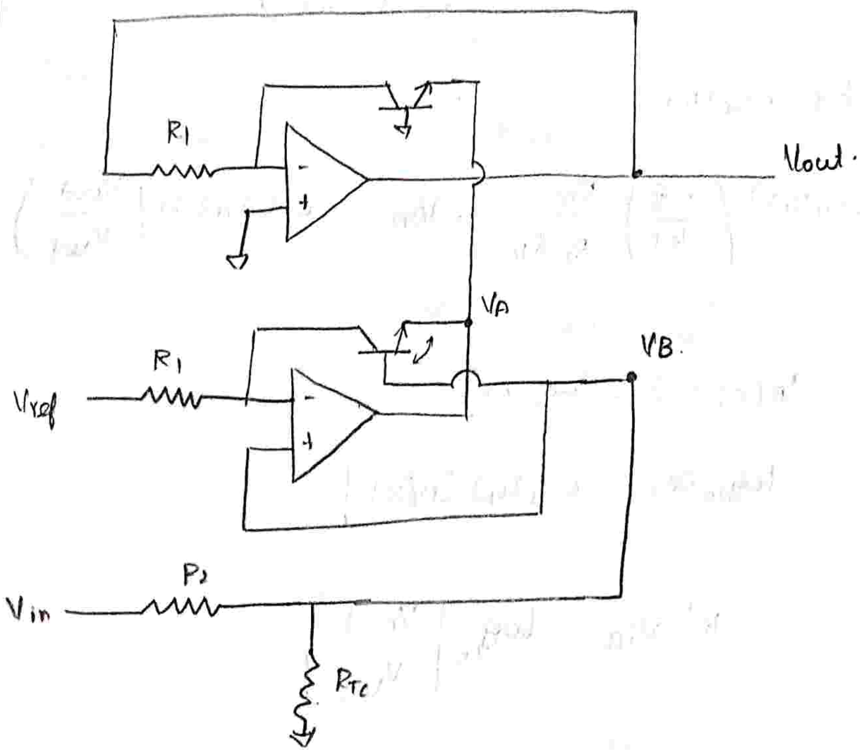
$$\boxed{\therefore V_{out} = \left[1 + \frac{R_2}{R_{TC}}\right]\left(\frac{-KT}{q}\right) \ln\left(\frac{V_1}{V_{ref}}\right)}$$

This is how we can design a temperature compensated log-amplifier using only two logarithmic op-Amp.

eg: $\log(1000) = 3$.

$\text{antilog}(10^3) = 1000$

Anti-log amplifier:



$$V_A = \frac{-KT}{q} \ln\left[\frac{V_{out}}{I_S R_1}\right] \longrightarrow (1).$$

$$V_B - V_E = V_A.$$

$$V_B - \frac{KT}{q} \ln\left[\frac{V_{ref}}{I_S R_1}\right] = -\frac{KT}{q} \ln\left[\frac{V_{out}}{I_S R_1}\right]$$

$$\frac{V_{in} \cdot R_{TC}}{R_2 + R_{TC}} - \frac{KT}{q} \ln\left[\frac{V_{ref}}{I_S R_1}\right] = -\frac{KT}{q} \ln\left[\frac{V_{out}}{I_S R_1}\right]$$

$$\frac{V_{in} \cdot R_{TC}}{R_2 + R_{TC}} = \frac{KT}{q} \ln\left[\frac{V_{ref}}{I_S R_1}\right] - \frac{KT}{q} \ln\left[\frac{V_{out}}{I_S R_1}\right]$$

$$\frac{V_{in} R_{TC}}{R_2 + R_{TC}} = -\frac{KT}{q}\left[\ln\left(\frac{V_{out}}{I_S R_1}\right) - \ln\left(\frac{V_{ref}}{I_S R_1}\right)\right]$$

$$\left(\frac{-q}{KT}\right)\left(\frac{R_{TC}}{R_2 + R_{TC}}\right) V_{in} = \ln\left(\frac{V_{out}}{V_{ref}}\right)$$

Mly by 0.4343,

$$\underbrace{(0.4343)\left(\frac{-q}{KT}\right)\frac{R_{TC}}{R_2 \cdot R_{TC}}}_{\text{constants Replace by } k'} \cdot V_{in} = 0.4343 \ln\left(\frac{V_{out}}{V_{ref}}\right).$$

$$\left[\because \ln(x) = 2.3 \log_{10}(x)\right.$$

$$\left. \log_{10}(x) = 0.4343 \ln(x)\right]$$

$$-k' V_{in} = \log_{10}\left[\frac{V_{out}}{V_{ref}}\right]$$

$$\therefore \frac{V_{out}}{V_{ref}} = 10^{-k' V_{in}}.$$

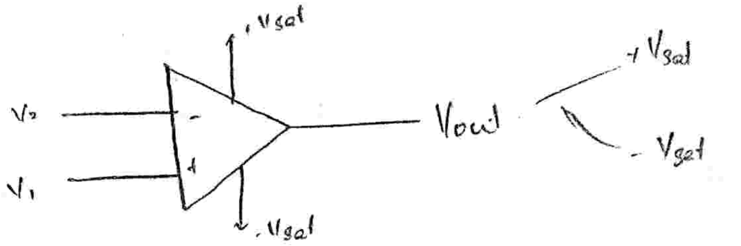
$$\boxed{\therefore V_{out} = V_{ref}\, 10^{-k' V_{in}}}$$

eg: If $R' = V_{in} = 1$, then.

$$V_{out} = \frac{V_{ref}}{10}.$$

V_{ref} ↓se by 10 times.

COMPARATOR :



$V_{out} = A_{OL}(V_1 - V_2)$ [∵ If $(V_1 - V_2)$ is small

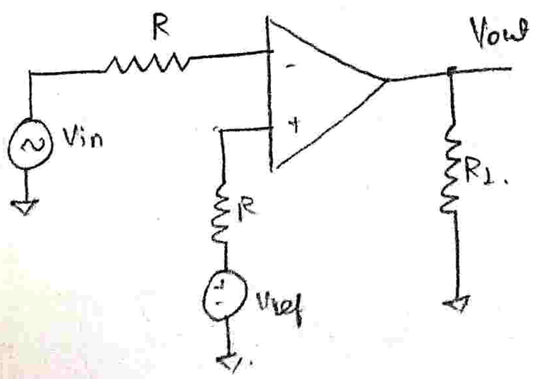
$= +V_{sat}$ (or) $-V_{sat}$. due A_{OL} (10^6) we get this]

It compare two terminal voltage, higher voltage terminal, the o/p go to corresponding V_{sat}.

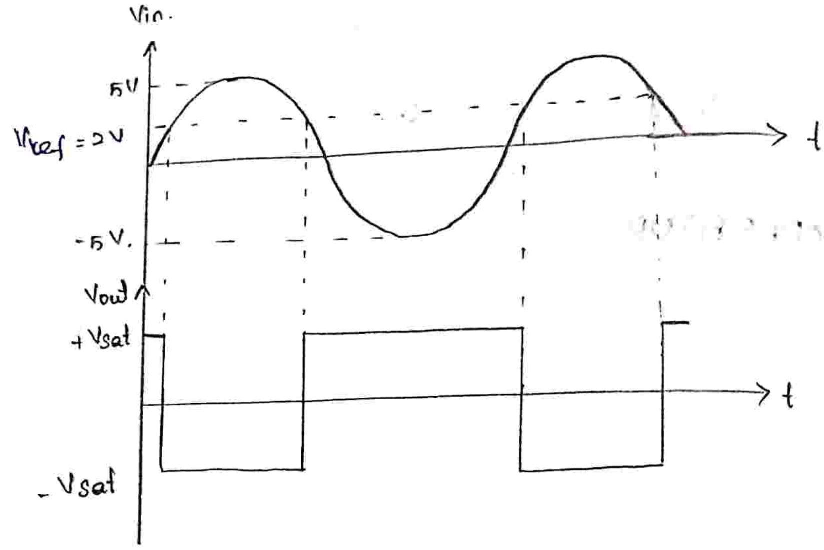
when, $V_1 > V_2$, o/p : $+V_{sat}$

$V_2 > V_1$, o/p : $-V_{sat}$.

Inverting mode comparitor :



Assume, $V_{ref} = 2V$.



when, $V_{in} < 2V$, $V_{out} = +V_{sat}$.

$(V_- < V_+)$

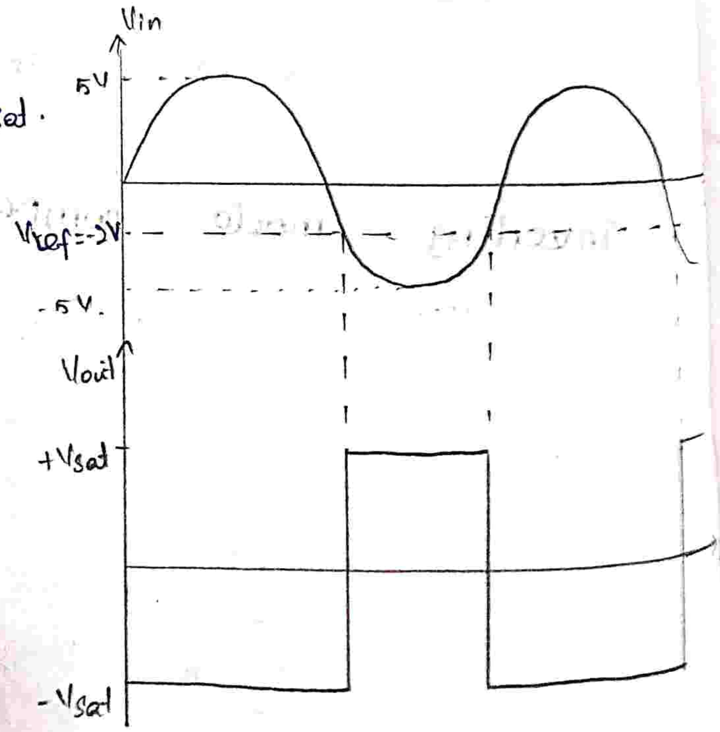
when, $V_{in} > 2V$, $V_{out} = -V_{sat}$.

$(V_- > V_+)$

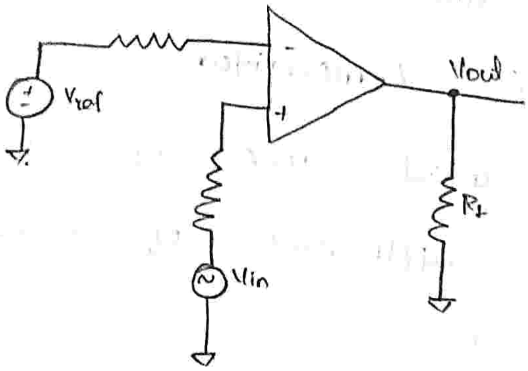
Assume, $V_{ref} = -2V$.

when, $V_{in} > 2V$, $V_o = -V_{sat}$.

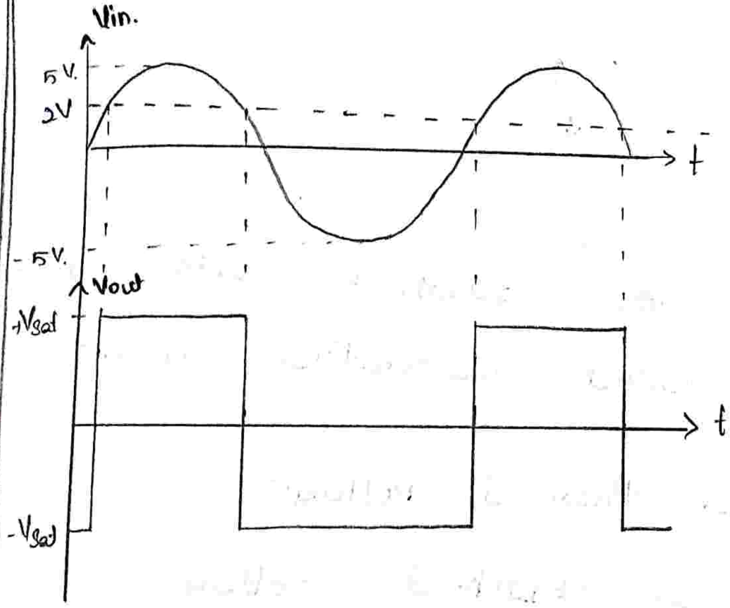
$V_{in} < 2V$, $V_o = +V_{sat}$



Non-Inverting Mode comparitor.



Assume, $V_{ref} = 2V$.



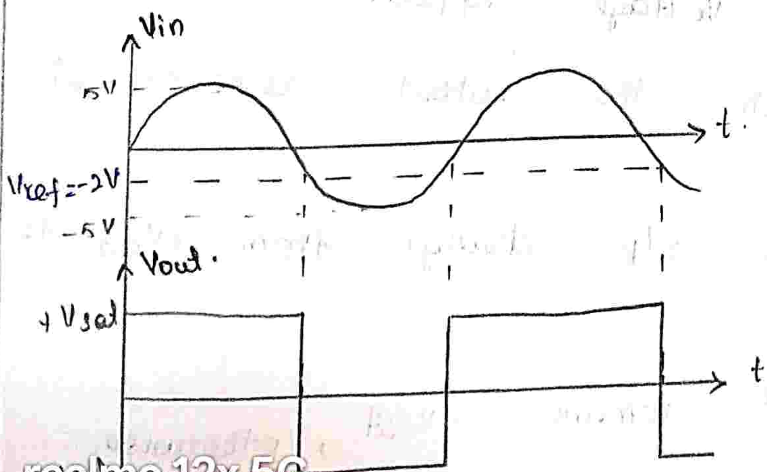
$V_{in} < 2V \Rightarrow V_{out} = -V_{sat}$

$V_+ < V_-$

$V_{in} > 2V \Rightarrow V_{out} = +V_{sat}$.

$V_+ > V_-$

Assume, $V_{ref} = -2V$.



$V_{in} > -2V \Rightarrow V_o = +V_{sat}$.

$V_{in} < 2V \Rightarrow V_o = -V_{sat}$.

$V_+ \quad V_-$

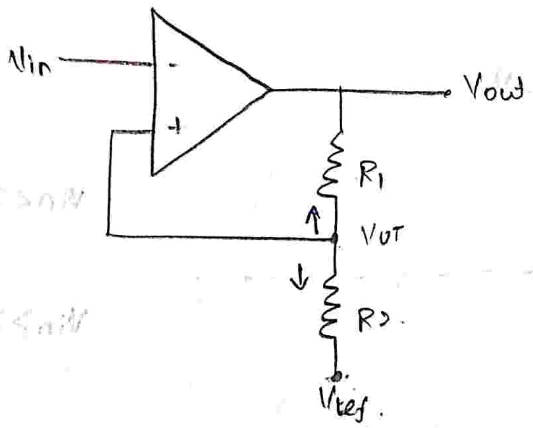
If we kept, $V_{ref} = 0V$ then it is zero crossing integrator circuit.

~~Reo~~ Regenerative Comparitor:

⇒ Using closed loop op-amp.

⇒ It is application of comparitor.

[also called as schmitt trigger]



whenever the feedback exist in the circuit is called regenerative circuit.

V_{UT} → upper threshold voltage.

V_{LT} → lower threshold voltage.

V_{UT} ⇒ Max voltage appear at Non-inverting terminal [upto which the output remains at +Vsat]

(or)

[when the o/p change from +Vsat to -Vsat]

V_{LT} ⇒ output remains -Vsat, $V_{in} < V_{LT}$. otherwise change

Minimum voltage, It remains $-V_{sat}$, otherwise the o/p transition from $-V_{sat}$ to $-V_{sat}$.

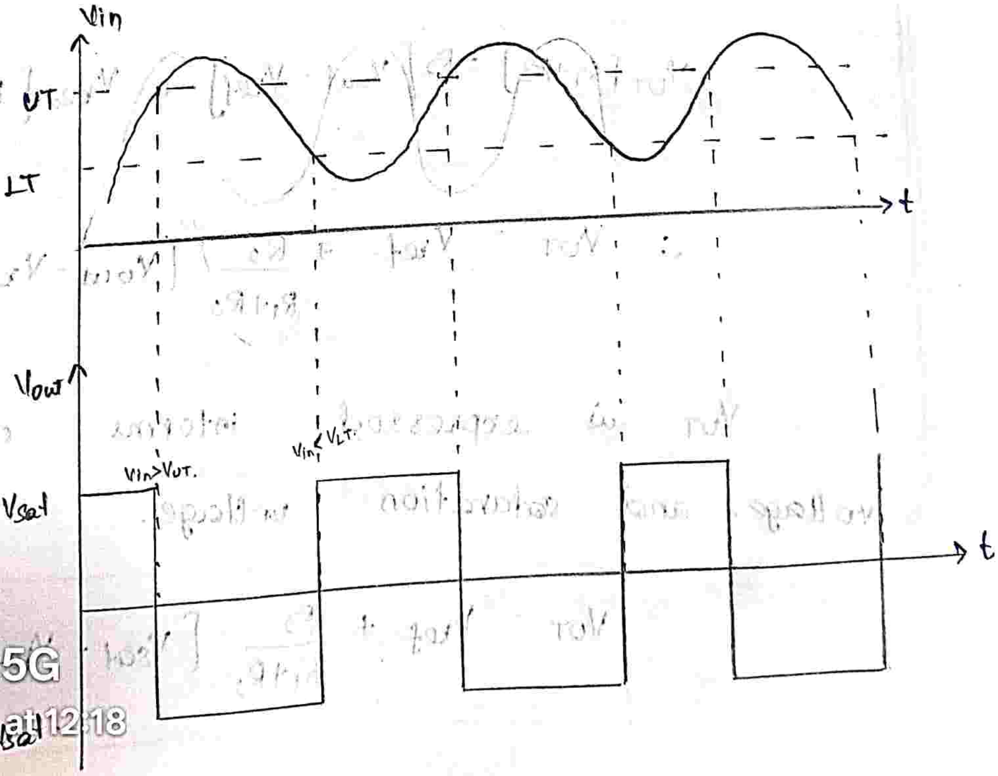
[Draw back of using open loop op-amp:
⇒ It has one threshold voltage]

⇒ o/p depends upon multiple threshold voltage [here we use 2].

when the i/p lies btw V_{UT} & V_{LT} the o/p simply ~~maintains~~ go its previous value. → hysteresis value.

V_{UT} → is a point max voltage at which the o/p change from : $+V_{sat} \xrightarrow{} \text{to} -V_{sat}$.

V_{LT} → Max voltage at which o/p changes from $-V_{sat}$ to $+V_{sat}$.



when, $V_{out} = +V_{sat}$, $V_+ = V_{UT}$

Apply nodal analysis, w.r.t "V_{UT}"

$$\frac{V_{UT} - V_{out}}{R_1} + \frac{V_{UT} - V_{ref}}{R_2} = 0.$$

$$V_{UT}\left[\frac{1}{R_1} + \frac{1}{R_2}\right] - \frac{V_{out}}{R_1} - \frac{V_{ref}}{R_2} = 0.$$

$$V_{UT}\left[\frac{R_1 + R_2}{R_1 R_2}\right] = \frac{V_{out}}{R_1} + \frac{V_{ref}}{R_2}$$

$$V_{UT}\left[R_1 + R_2\right] = \cancel{R_1} R_2\left[\frac{V_{out}}{\cancel{R_1}}\right] + R_1 \cancel{R_2}\left[\frac{V_{ref}}{\cancel{R_2}}\right]$$

$$V_{UT}[R_1 + R_2] = R_2 V_{out} + R_1 V_{ref}.$$

Add & sub by $R_2 V_{ref}$:

$$V_{UT}[R_1 + R_2] = \underline{R_2 V_{out}} + R_1 V_{ref} + R_2 V_{ref} - \underline{R_2 V_{ref}}.$$

$$V_{UT}[R_1 + R_2] = \frac{R_2\left[V_{out} - V_{ref}\right]}{R_1 + R_2} + \frac{V_{ref}\left[\cancel{R_1 + R_2}\right]}{\cancel{R_1 + R_2}}$$

$$\therefore \quad V_{UT} = V_{ref} + \frac{R_2}{R_1 + R_2}\left[V_{out} - V_{ref}\right]$$

V_{UT} is expressed interms of reference voltage and saturation voltage.

$$\boxed{V_{UT} = V_{ref} + \frac{R_2}{R_1 + R_2}\left[V_{sat} - V_{ref}\right]}$$

when, $V_{out} = -V_{sat}$, $V_+ = V_{LT}$

$$\cancel{\frac{V_{out} - V_{LT}}{R_1} + \frac{V_{ref} - V_{LT}}{R_2} = 0}$$

apply nodal analysis w.r.to V_{LT}.

$$\frac{V_{LT} - V_{out}}{R_1} + \frac{V_{LT} - V_{ref}}{R_2} = 0.$$

$$\frac{V_{LT} + V_{sat}}{R_1} + \frac{V_{LT} - V_{ref}}{R_2} = 0.$$

$$V_{LT}\left[\frac{R_1+R_2}{R_1R_2}\right] = \frac{-V_{sat}}{R_1} + \frac{V_{ref}}{R_2}.$$

xly by $R_1 R_2$:

$$V_{LT}\left[R_1+R_2\right] = -R_2 V_{sat} + R_1 V_{ref} + R_2 V_{ref}.$$

$$-R_2 V_{ref}.$$

$$V_{LT}\left[R_1+R_2\right] = -R_2\left[V_{sat} + V_{ref}\right] + V_{ref}\left[R_1+R_2\right]$$

$$V_{LT} = \frac{-R_2}{R_1+R_2}\left[V_{sat} + V_{ref}\right] + \frac{V_{ref}\left[\cancel{R_1+R_2}\right]}{\cancel{R_1+R_2}}$$

$$\boxed{\therefore V_{LT} = V_{ref} - \frac{R_2}{R_1+R_2}\left[V_{sat} + V_{ref}\right]}$$

Hysteresis Voltage:

$$V_H = V_{UT} - V_{LT}.$$

$$= \cancel{V_{ref}} + \frac{R_2}{R_1+R_2}\left[V_{sat} - V_{ref}\right] - \cancel{V_{ref}} + \frac{R_2}{R_1+R}(V_{sat} + V_{ref})$$

$$= \frac{R_2}{R_1+R_2}\left[V_{sat} - \cancel{V_{ref}} + V_{sat} + \cancel{V_{ref}}\right]$$

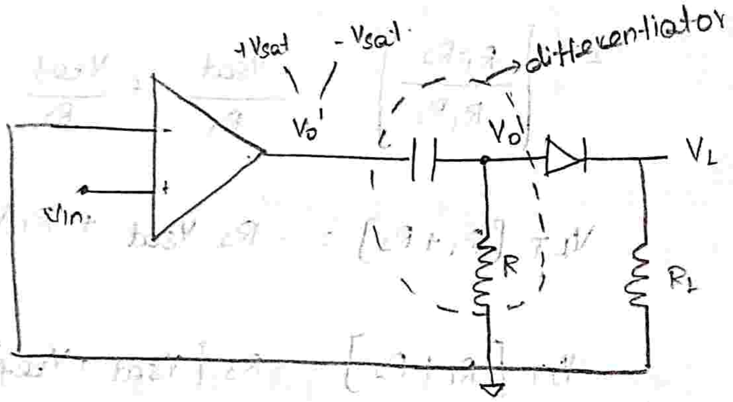
$$\boxed{V_H = \frac{2R_2}{R_1+R_2}(V_{sat})}$$

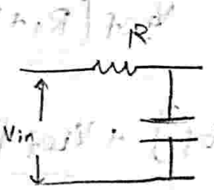
Advantage:

1. Noise unity offered by regenerative comparator is high.
2. switch speed is more.
3. stability is more

Application:

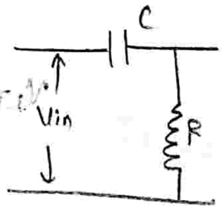
1. Time marker generative circuit.





passive low pass filter

[integrator]



[differentiator]

V_o : +Vsat   D → ON

V_o : -Vsat   D → OFF

MULT- VIBRATORS:

Types of multivibrators.

1. Astable → No stable states / both are quasi stable states.

2. Monostable → 1 stable state & 1 Quasi stable state.

3. Bistable → 2 stable states.

Astable => no need to apply external i/p pulse to change it's o/p states.

eg: oscillator, Relaxation oscillator.

- also called Free - Running multivibrator.

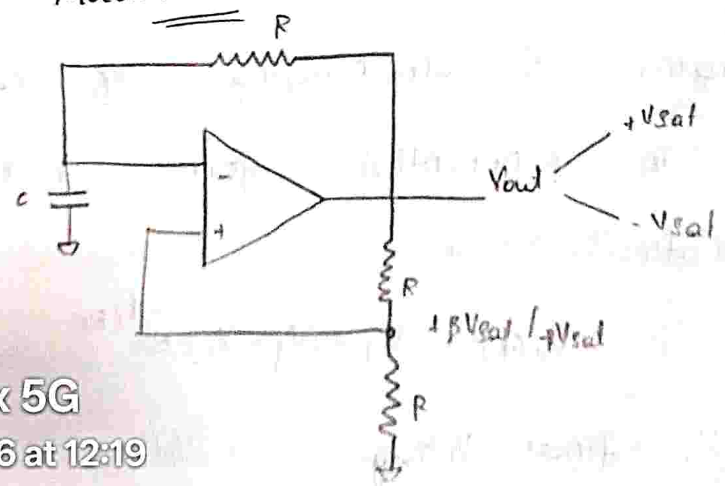
Monostable => it requires -ve trigger pulse to change o/p from stable to quasi-stable stable.

- also called one -shot multivibrator.

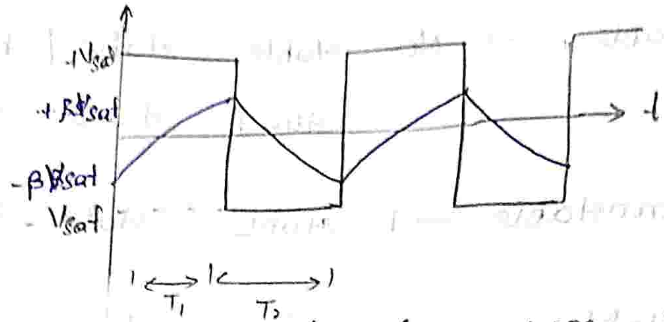
Bistable => It requires two trigger pulse

- also called binary circuit bcoz it has 2 stable state.

Astable Multivibrator:



→ Based on o/p the capacitor charged & discharged



→ o/p voltage depends upon threshold voltage & charging / discharging of capacitor.

→ Initially Assume, $V_o = +V_{sat}$, voltage at the terminal $+\beta V_{sat}$.

→ capacitor → charged beyond β voltage charge $V_- > V_+$ then output is $-V_{sat}$

Assume, $V_o = -V_{sat}$.

$V_+ = -\beta_{sat}$.

C → discharge upto $-\beta V_{sat}$ when it goes lower than $-\beta V_{sat}$

$V_- < V_+$ then the o/p is $+V_{sat}$.

Charging & discharging of capacitor occur in exponential form, which is mathematically as.

$$V_c(t) = V_f - (V_f - V_i)\, e^{-t/RC}.$$

V_f → final voltage. (Max voltage "c" can be charged is V_{sat})

V_i → initial voltage.

T_1 — time period when capacitor charged exponentially

T_2 — time period when capacitor discharged exponentially.

By Mathematical;

$$V_c(t) = V_{sat} - (V_{sat} - (-\beta V_{sat}))\, e^{-t/RC}$$

$$V_c(t) = V_{sat} - (V_{sat} + \beta V_{sat})\, e^{-t/RC}$$

at $t = T_1$, $V_c(t) = +\beta V_{sat}$.

$$+\beta V_{sat} = V_{sat} - (V_{sat} + \beta V_{sat})\, e^{-T_1/RC}.$$

$$\beta V_{sat} - V_{sat} = -(V_{sat} + \beta V_{sat})\, e^{-T_1/RC}.$$

$$\cancel{-V_{sat}}\,(1-\beta) = \cancel{-V_{sat}}\,(1+\beta)\, e^{-T_1/RC}.$$

$$\frac{1-\beta}{1+\beta} = e^{-T_1/RC}.$$

Apply ln on B/S, $\quad -T_1/RC = \ln\left(\frac{1-\beta}{1+\beta}\right)$

$$\frac{T_1}{RC} = -\ln\left(\frac{1-\beta}{1+\beta}\right)$$

$$\boxed{\therefore\ T_1 = RC \ln\left(\frac{1+\beta}{1-\beta}\right)}$$

capacitor required same time for charging & discharging.

If o/p is symmetrical,

$$T = T_1 + T_2 = 2T_1$$

$$T = 2RC \ln\left(\frac{1+\beta}{1-\beta}\right).$$

$\therefore$ Time period depends on R, C & β.

If we don't want V_{sat} in o/p we can use clipping circuit by zener diode.

Connected (1 by 1 zener diode) in V_{out}, it limits the amplitude.

$$\therefore \beta = \frac{R_2}{R_1+R_2}$$

If $R_1 = R_2$.

$$\beta = \frac{\cancel{R_1}}{2\cancel{R_1}} = \frac{1}{2} = 0.5.$$

$$T = 2RC \ln\left[\frac{1+0.5}{1-0.5}\right] \Rightarrow T = 2RC \ln[3]$$

If $R_1 = 1.16 R_2$.

$$\beta = \frac{\cancel{R_2}}{2.16\cancel{R_2}} = \frac{1}{2.16} = 0.463.$$

$$T = 2RC \ln\left[\frac{1+0.463}{1-0.463}\right] = 2RC \ln[2.724]$$

$$T = 2RC(\underbrace{1.002}_{\approx 1})$$

$T_t = 2RC.$

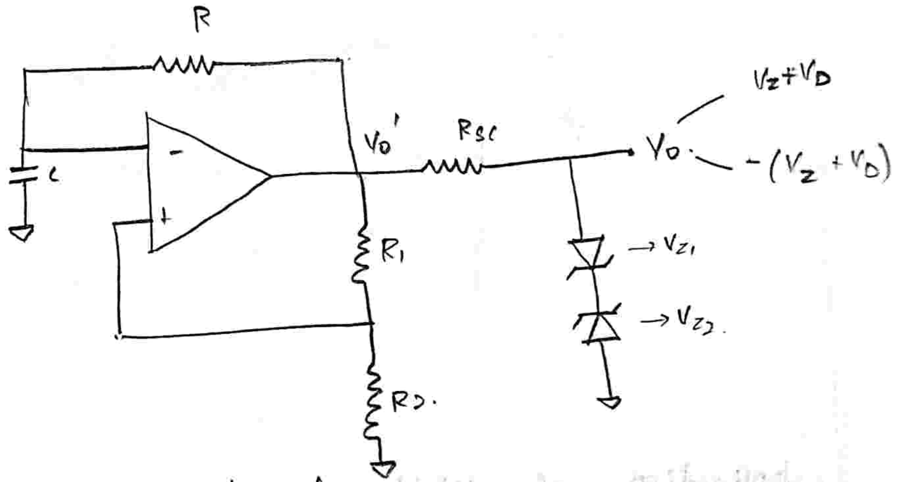
In this case time period is completly depends on external R, C.

$$\boxed{\therefore \; f_0 = \frac{1}{T} = \frac{1}{2RC}.}$$

It is a symmetrical square wave generator circuit.

symmetric - on & off time is same.

<u>For asymmetry square wave:</u>



I am going to fix V_o value by taking zener diode back to back.

standard voltage of zener diode $\approx 5.6V$.

$R_{sc} \rightarrow$ current limiting resistor (inorder to prevent the breakdown of zener by allowing to excessive current in R.B).

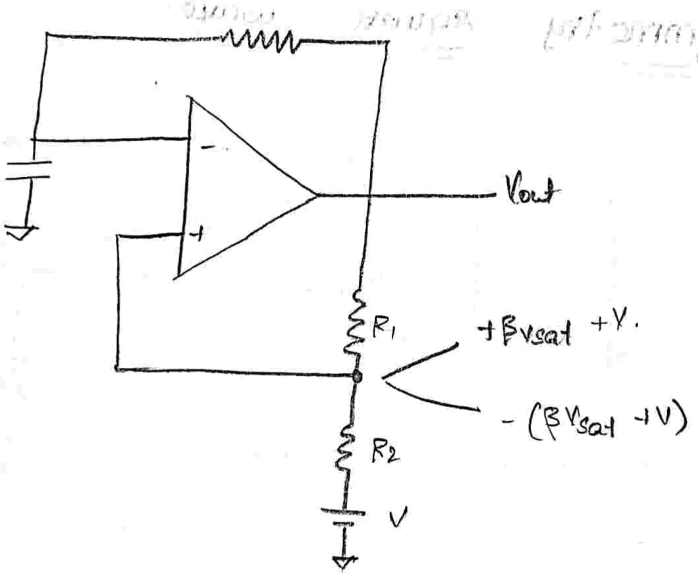
By connecting zenerdiode we can able to fix voltage (V_z+V_D) / $-(V_z+V_D)$ rather than $+V_{sat}$, $-V_{sat}$.

$$I = \frac{V_{out} - V_z}{R_{sc}}$$ $V_z \rightarrow$ breakdown voltage

$V_{z1} = V_{z2} \Rightarrow$ generate symmetric square wave.

$V_{z1} \neq V_{z2} \Rightarrow$ generate asymmetric " "

asymmetric square wave generator.



[Application of Astable :

*. Voltage to frequency converter.]

$\Rightarrow$ We can control the frequency by changing the dc voltage.

T_p of astable depends on external dc supply.

4) Design a square wave oscillator for $f_0 = 1kHz$ using an IC741 op-amp and dc supply voltage of $\pm 12V$

Sol: Given, $f_0 = 1kHz$.

Design a symmetric square wave.

$$f_0 = \frac{1}{2RC}$$

$R_1 = 1.16 R_2$.

Let $R_2 = 10k\Omega$.

$R_1 = 11.6k\Omega$.

$$f_0 = \frac{1}{2RC} = 1kHz.$$

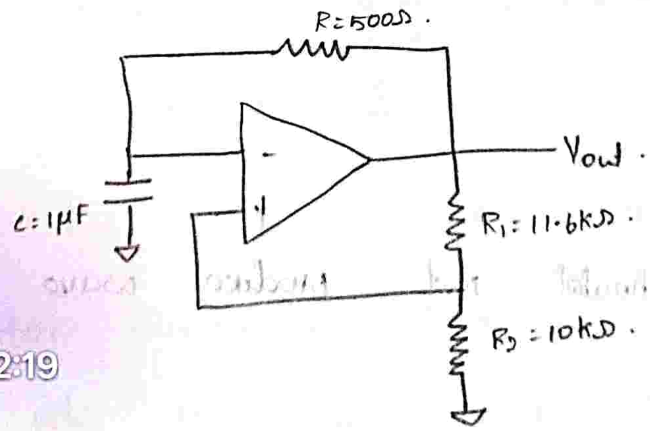
let us choose, $C = 1\mu F$.

$$\frac{1}{2 \times 1 \times 10^{-6} \times 10^3} = R.$$

$$\frac{1}{2 \times 10^{-3}} = R.$$

$R = 0.5 k\Omega$

$\boxed{R = 500\Omega.}$



MONOSTABLE MULTIVIBRATOR:

1). It has 1 stable stable & 1 quasi stable state.

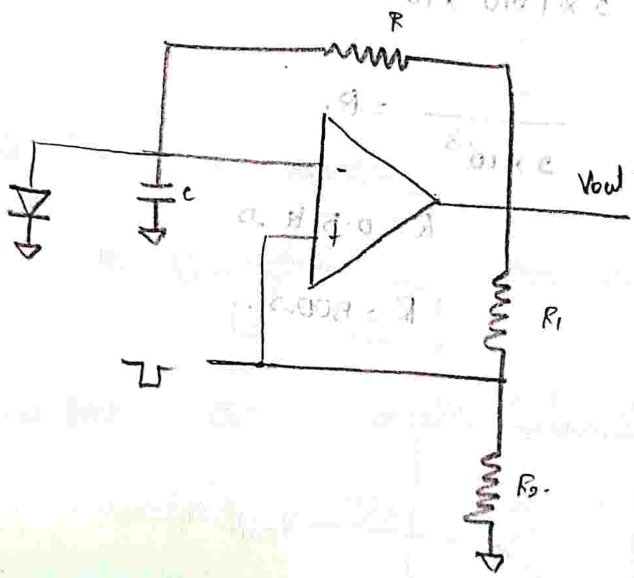
⇒ quasi stable state → Non-permanent state

2). It requires 1 triggering pulse extremely. (to change state of o/p)

3). It is called 1-shot multi-vibrator.

4). Gating circuits - It can generate a pulse (or) square wave in specified time period (fixed time period).

5). Time delay circuit.



It should not produce wave form continuously.

By placing the diode $\parallel^{el}$ to capacitor, when 0.7 V the capacitor stop chaging (so that it doesn't produce continuous).

→ $V_{out} = +V_{sat}$. (Before applying -ve trigger)

The voltage at capacitor fixed at 'V_D'
↓
0.7V.

Now, I need to apply with suitable amplitude. (giving -ve max voltage.)

eg: $-4.00V < 0.00$.
↓ -ve trigger pulse.
↓ inverting terminal voltage.

[trigger → short duration pulse]

∵ Inverting terminal > Non-Inverting voltage, the [olp → $-V_{sat}$.]

⇒ Now capacitor start discharge upto $-V_{sat}$.

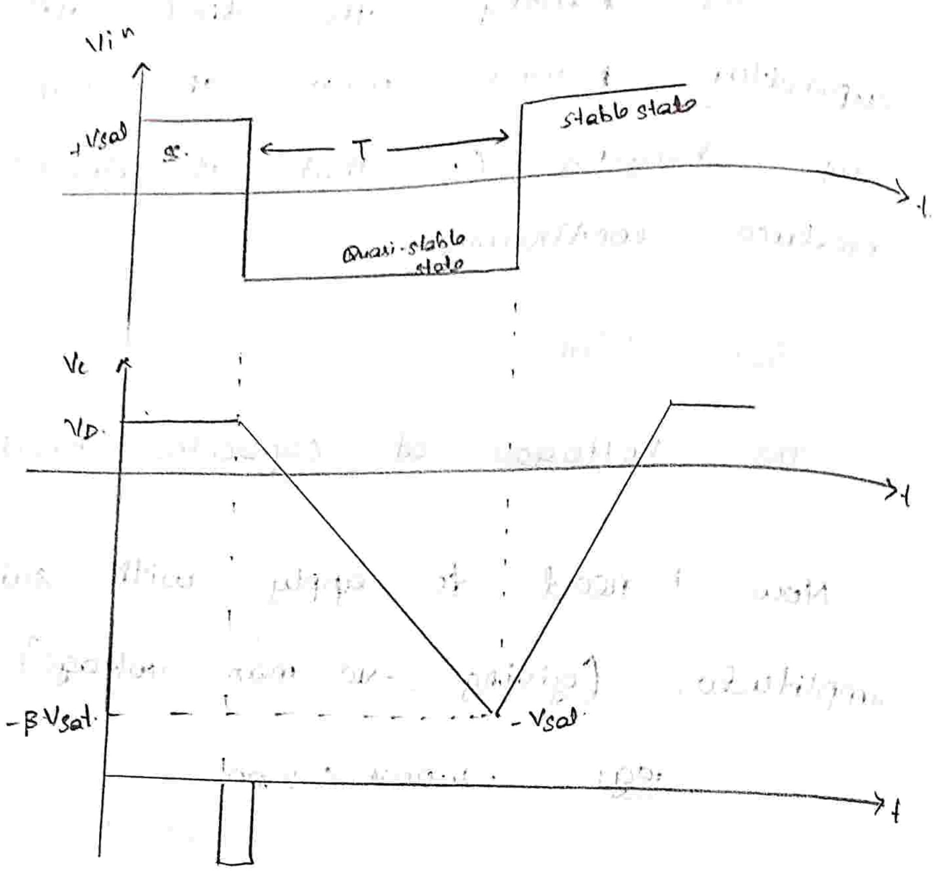
⇒ when it goes below $-V_{sat}$,

$V_- < V_+$ (-ve trigger) like $(-5.00 < -4.00)$,
V_- , V_+ (-ve trigger)

olp → $+V_{sat}$

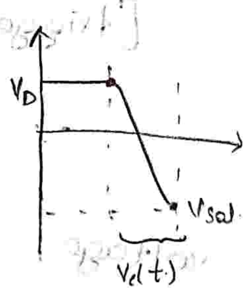
⇒ due to $\parallel^{el}$ connection of diode, capacitor discharge linearly.

After it charge upto $+V_{sat}$, but due to diode charge upto 0.7V after that remains V_D.



$$V_c(t) = V_f - (V_f - V_i)\, e^{-t/RC}$$

$$V_c(t) = -V_{sat} - (-V_{sat} - V_D)\, e^{-t/RC}$$



$$\Rightarrow \left[\because V_c(t) = -\beta V_{sat},\ \text{at } t = T\right]$$

$$-\beta V_{sat} = -V_{sat} - [-V_{sat} - V_D]\, e^{-T/RC}$$

$$-\beta V_{sat} + V_{sat} = [V_{sat} + V_D]\, e^{-T/RC}$$

$$\frac{V_{sat}[1-\beta]}{V_{sat} + V_D} = e^{-T/RC}$$

$$\frac{V_{sat}(1-\beta)}{V_{sat}\left(1+\frac{V_D}{V_{sat}}\right)}$$

$$e^{-T/RC} = \left[\frac{1-\beta}{1+\frac{V_D}{V_{sat}}}\right]$$

$$e^{T/RC} = \frac{1+\frac{V_D}{V_{sat}}}{1-\beta}$$

take ln on b/s,

$$\frac{T}{RC} = \ln\left(\frac{1+\frac{V_D}{V_{sat}}}{1-\beta}\right)$$

$$T = RC \ln\left(\frac{1+\frac{V_D}{V_{sat}}}{1-\beta}\right)$$

∵ $V_D \rightarrow$ diode drop. (less $= 0.7V$).

$V_{sat} \rightarrow$ very high

$V_D < V_{sat}$, $\frac{V_D}{V_{sat}} << 1$, so it is neglected.

$$T = RC \ln\left[\frac{1}{1-\beta}\right]$$, here $\beta = \frac{R_2}{R_1+R_2}$.

if $R_1 = R_2$.

$$\beta = \frac{\cancel{R_2}}{2\cancel{R_2}} = 0.5$$

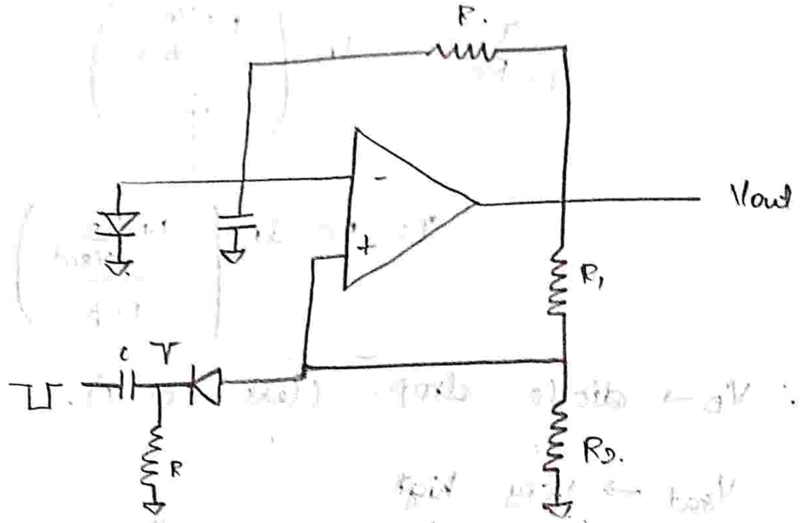
$$T = RC \ln\left[\frac{1}{1-0.5}\right] = RC \ln\left[\frac{1}{1/2}\right]$$

$$T = RC \ln(2)$$

$$\boxed{T = 0.693\, RC.}$$

By choosing RC we can ~~determine~~ generate a pulse of T.

diode is connected to ensure the -ve trigger is applied, when +ve trigger apply diode become OFF.



(5). Design a monostable multivibrator with trigger pulse shaping which will drive a LED ON for 0.5 sec, each time it is pulsed.

Sol:- Given, $T = 0.5$ sec.

Famula, $T = 0.69\ RC$.

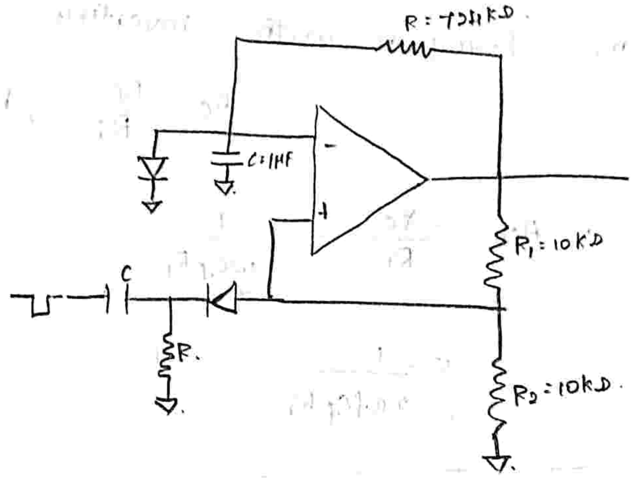
Let use choose, $C = 1\,\mu F$

$$R = \frac{0.5}{0.69 \times 1 \times 10^{-6}}$$

$$R = 0.7246 \times 10^{6}$$

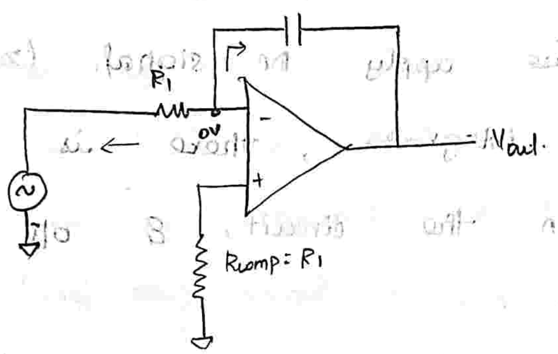
$$\boxed{R = 724.63\ K\Omega}$$

assume, $R_1 = R_2 = 10K\Omega$.



Integrator:

$C_f \Rightarrow X_C = \frac{1}{\omega C} = \frac{1}{2\pi f C}$



$Q = CV$

$I = \frac{dQ}{dt} = C\frac{dv}{dt}$

Apply nodal analysis at inverting terminal node.

$$\frac{0 - V_{in}}{R_1} + \frac{0 - C\, dV_{out}}{dt} = 0$$

$$\frac{-C\, dV_{out}}{dt} = \frac{V_{in}}{R_1}$$

$$\frac{dV_{out}}{dt} = \frac{-V_{in}}{R_1 C_f}$$

$$\boxed{\therefore V_{out} = \frac{-1}{R_1 C_f}\int V_{in} \cdot dt + V_c(0)}$$

$V_c(0) \rightarrow$ initial voltage at capacitor.

Gain, [compare with inverting amplifier, $A_c = -\frac{R_f}{R_i}$, here $R_f = X_{C_f}$]

$$\therefore A_{CL} = \frac{-X_c}{R_1} = -\frac{1}{\omega C_f R_1}$$

$$= -\frac{1}{2\pi f C_f R_1}$$

$$\boxed{|A_{CL}| = \frac{1}{2\pi f R_1 C_f}.}$$

If we apply DC signal (zero frequency) to the integrator, there is no feedback exist in the circuit, & o/p become saturated. (bcoz it become open loop)

Due to the high gain it can't provide good stability.

open loop gain practical $\approx 10^6$

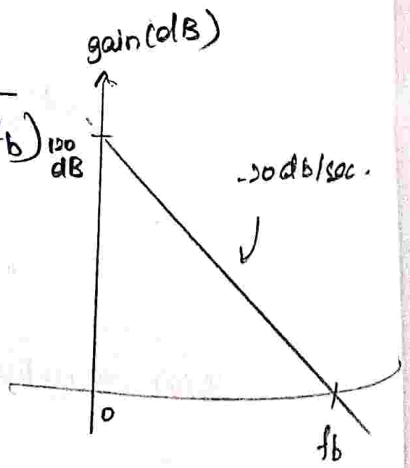
gain in dB $= 20 \log_{10}(10^6)$

$= 20 \times 6$

$= 120\,dB.$

$$|A_{CL}| = \frac{1}{2\pi f R_1 C_f} = \frac{1}{(f/f_b)}$$

$$\because f_b = \frac{1}{2\pi R_1 C_f}.$$



cut off frequency at which gain of circuit touch '0' dB.

Practical I
To overco
one mode
Vin
No
'C' Impeda
circuit,
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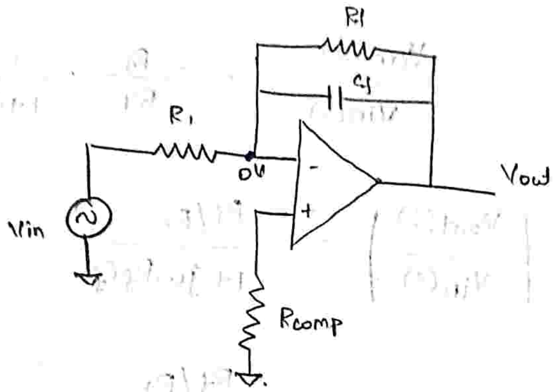
Practical Integrator : (Lossy integrator).
To overcome the drawback we add one more resistor in feedback.



Now, if we give '0' Hz as frequency, 'C' impedance is ∞ it become open circuit, Now it is similar as inverting amplifier. so o/p doesn't saturated. Now the gain is $\frac{-R_f}{R_1}$

Apply Nodal Analysis.

$$\frac{0-V_{in}}{R_1} + \frac{0-V_{out}}{R_f} - C\frac{dV_{out}}{dt} = 0.$$

$$-C\frac{dV_{out}}{dt} = \frac{V_{in}}{R_1} + \frac{V_{out}}{R_f}$$

take laplace transform.

$$-sC_f V_{out}(s) = \frac{V_{in}(s)}{R_1} + \frac{V_{out}(s)}{R_f}.$$

$$\frac{V_{out}(s)}{R_f} + sC_f V_{out}(s) = -\frac{V_{in}(s)}{R_1}$$

$$V_{out}(s)\left[\frac{1}{R_f} + sC_f\right] = -\frac{1}{R_1} V_{in}(s)$$

$$V_{out}(s)\left[\frac{1+sR_fC_f}{R_f}\right] = -\frac{1}{R_1}V_{in}(s)$$

$$\frac{V_{out}(s)}{V_{in}(s)} = -\frac{R_f}{R_1}\cdot\frac{1}{1+sR_fC_f}$$

$$\left|\frac{V_{out}(s)}{V_{in}(s)}\right| = \frac{R_f/R_1}{1+j\omega R_fC_f}$$

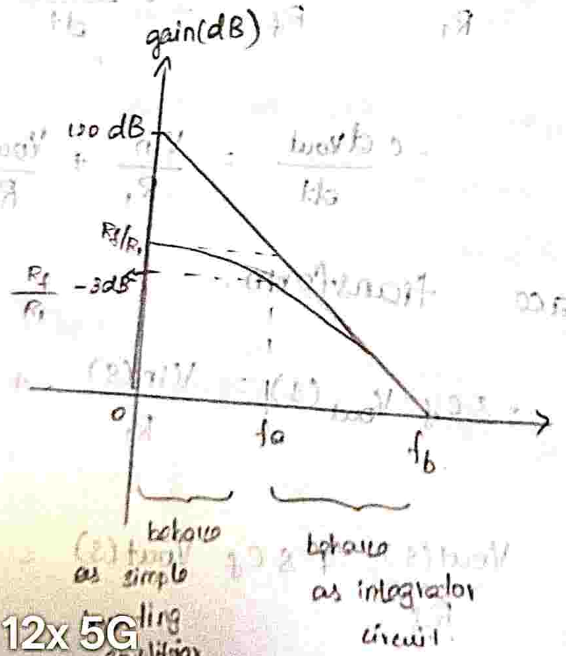
$$= \frac{R_f/R_1}{1+j2\pi f R_fC_f}$$

$$\left|\frac{V_{out}(s)}{V_{in}(s)}\right| = \frac{R_f/R_1}{1+j\left[f/f_a\right]}$$

$$\therefore f_a = \frac{1}{2\pi R_fC_f}$$

If $f=0$, gain is $\frac{R_f}{R_1}$

when, $f=f_a$, the gain is reduce to -3 dB.
input



when we design practical integrator we need to follow three rules:

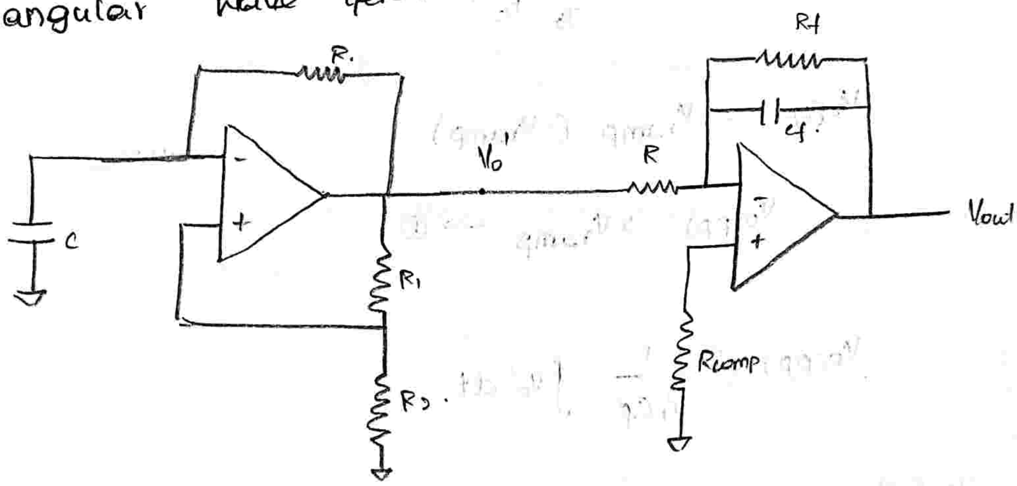
1. Relationship blw R_f & R_1 $\Rightarrow R_f = 10R_1$
2. $f_{in} = 10 f_a$.
3. $f_a < f \leq f_1$

$\Rightarrow$ without R_f, C_f charge completly full)

$\Rightarrow$ with R_f, amount of charging by capacitor is reduce, no possible for full (or) complete charge.

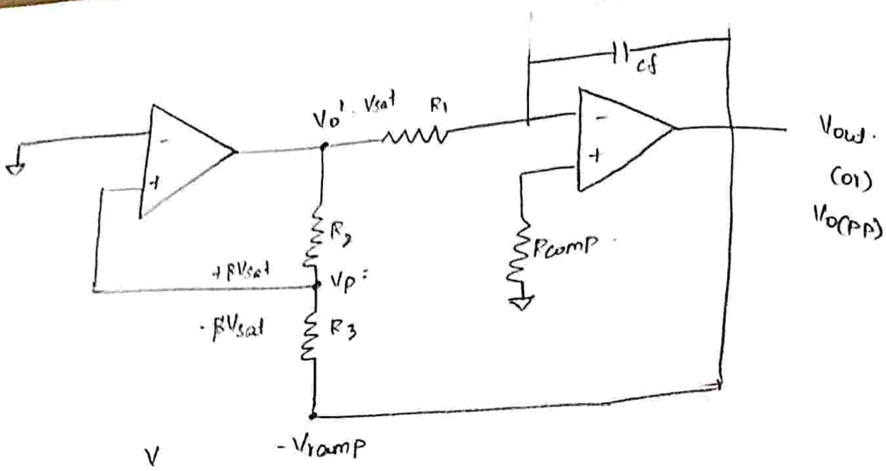
$\therefore$ current divide & pass through R_f & C_f so leakage of current occurs. so it is called lossy .(loss of power due to R_f) .

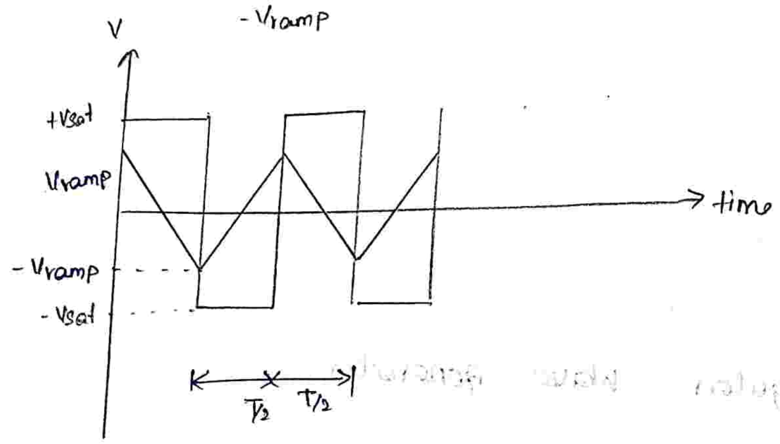
Triangular Wave Generator:



Inorder to dertermene the frequency of square wave generator. remove R & C.

Inorder to change voltage of square of wave generator. connect output with it.





$$V_{o(PP)} = V_{ramp} - (-V_{ramp})$$

$$V_{o(PP)} = 2V_{ramp} \rightarrow ①.$$

$$V_{o(PP)} = \frac{-1}{R_1 C_f} \int V_o' \, dt.$$

Initially we assume, $V_o' = +V_{sat}$

$V_{out} = -V_{ramp}$

Apply nodal analysis with respect to V_p

$$\frac{V_p - V_{sat}}{R_2} + \frac{V_p - (-V_{ramp})}{R_3} = 0.$$

$$V_p \left[\frac{1}{R_2} + \frac{1}{R_3} \right] = \frac{V_{sat}}{R_2} - \frac{V_{ramp}}{R_3}$$

Vout. (or) $V_{o(PP)}$

time

$\rightarrow V_p$

$\frac{V_{ramp}}{R_3}$

$$V_p\left(\frac{R_2+R_3}{R_2R_3}\right) = \frac{V_{sat}}{R_2} - \frac{V_{ramp}}{R_3}.$$

$$V_p(R_2+R_3) = R_3V_{sat} - R_2V_{ramp}.$$

$$V_p(R_2+R_3) = R_3V_{sat} - R_2V_{ramp} + R_3V_{ramp} - R_3V_{ramp}$$

$$V_p(R_2+R_3) = R_3(V_{sat}+V_{ramp}) - V_{ramp}(R_2+R_3)$$

$$V_p = \frac{R_3}{R_2+R_3}(V_{sat}+V_{ramp}) - V_{ramp}.$$

$V_p \rightarrow$ is point at which the transition from $+V_{sat}$ to $-V_{sat}$.

make, $\therefore V_p = 0$.

$$\frac{R_3}{R_2+R_3}[V_{sat}] + \frac{R_3}{R_2+R_3}V_{ramp} - V_{ramp} = 0.$$

$$\frac{R_3}{R_2+R_3}\cdot V_{sat} + V_{ramp}\left[\frac{R_3}{R_2+R_3} - 1\right] = 0.$$

$$V_{ramp}\left[\frac{\cancel{R_3} - R_2 - \cancel{R_3}}{R_2+R_3}\right] + \left(\frac{R_3}{R_2+R_3}\right)V_{sat} = 0.$$

$$R_3V_{sat} = R_2V_{ramp}.$$

$$\boxed{\therefore V_{ramp} = \frac{R_3}{R_2}V_{sat}.}$$

$$\boxed{\therefore V_{o(PP)} = 2V_{ramp} = \frac{2R_3}{R_2}V_{sat}} \quad 1$$

$$V_{out(PP)} = \frac{-1}{R_1C_f}\int_0^{T/2} V_o'\,dt.$$

$$= \frac{-1}{R_1 C_f} \int_0^{T/2} (-V_{sat}) \, dt$$

$$= \frac{1}{R_1 C_f} V_{sat} \cdot [t]_0^{T/2}$$

$$V_{out(PP)} = \frac{T \cdot V_{sat}}{2 R_1 C_f} \longrightarrow (3)$$

equate (2) & (3).

$$\frac{2R_3}{R_2} \cancel{V_{sat}} = \frac{T \cancel{V_{sat}}}{2 R_1 C_f}$$

$$\boxed{T = \frac{4 R_1 R_3 C_f}{R_2}}$$

This is time period of triangular wave generator.

$$\boxed{\therefore f = \frac{1}{T} = \frac{R_2}{4 R_1 R_3 C_f}}$$

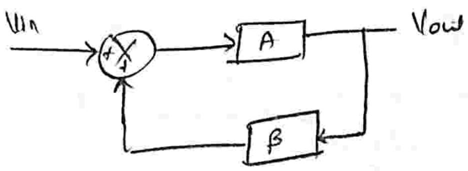
OSCILLATOR:

It is a electronic device, to produce a sinusoidal (or) Non-sinusoidal signal periodically (continuosly) with fixed amplitude without applying external input.

-ve F.B $\rightarrow \frac{V_o(s)}{V_i(s)} = \frac{G}{1+GH}$ (or) $\frac{A}{1+A\beta}$

+ve F.B $\rightarrow \frac{V_o(s)}{V_i(s)} = \frac{G}{1-GH}$ (or) $\frac{A}{1-A\beta}$

*. crystal oscillator provide high stability.



Barkhuaxan criterian: i) $|A\beta| = 1$

$\phi = 0° / 360°$

* if oscillator has -ve F.B, we won't get constant amplitude. (it can't generate continuosly)

*. To design a oscillator, the circuit should has +ve Feedback.

to get continuosly → should be inphase.

if $A\beta < 1$, $A\beta > 1$; we won't get continuous wave without apply external i/p.

RC oscillators:

The feedback network is designed only with the help (use) of resistor & capacitor.

Drawback:

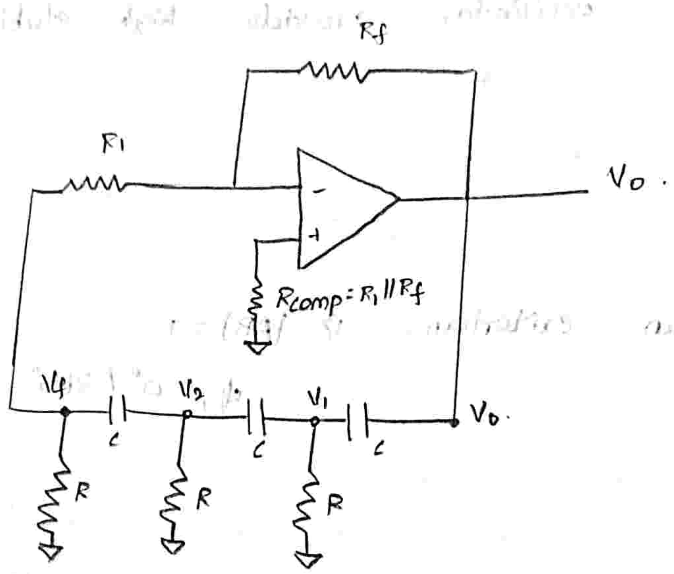
*. The o/p of signal the frequency is in Hz (or) 1kHz.

*. stability is low

①. R-C phase shift oscillator.

②. Wien-bridge oscillator.

①. RC phase shift oscillator.



op-amp provide -180° phase shift.

feedback network provide - 180° phase shift

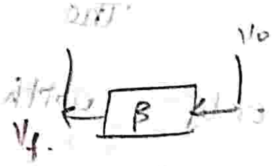
= 360°

In F.B network, each RC network provide 60° phase shift

to satisfy, $|A\beta| = 1$.

$\beta = \frac{V_f}{V_o}$ [feedBack gain]

$\therefore V_f = \beta V_o$.



Sustained oscillation:

It generates waveform with constant amplitude & frequency.

$|A\beta| = 1$.

If $|A\beta| < 1$, the amplitude decreases.
If $|A\beta| > 1$, the output becomes saturated.

write down KCL with respect to node 1 'V_1'.

$$\frac{V_1 - V_0}{X_C} + \frac{V_1 - 0}{R} + \frac{V_1 - V_2}{X_C} = 0.$$

$$X_C = \frac{1}{SC}$$

$$(V_1 - V_0)SC + \frac{V_1}{R} + (V_1 - V_2)SC = 0.$$

$$V_1 SC + \frac{V_1}{R} + V_1 SC - V_0(SC) - V_2(SC) = 0.$$

$$V_1\left[2SC + \frac{1}{R}\right] - V_0(SC) - V_2(SC) = 0.$$

$$V_1\left[\frac{1+2SCR}{R}\right] - V_0(SC) - V_2(SC) = 0.$$

$$V_1(1+2SCR) - V_0(RSC) - V_2(SCR) = 0.$$

$$V_1[1+2SRC] - V_2[SCR] = V_0[SRC]$$

$$V_0 = \frac{V_1(1+2SCR)}{SCR} - V_2 \longrightarrow (1).$$

write down KCL with respect to node 2 'V_2'

$$\frac{V_2 - V_1}{X_C} + \frac{V_2 - 0}{R} + \frac{V_2 - V_f}{X_C} = 0.$$

$$(V_2 - V_1)(SC) + \frac{V_2}{R} + (V_2 - V_f)(SC) = 0.$$

$$V_2 SC + \frac{V_2}{R} + V_2 SC - V_1(SC) - V_f(SC) = 0.$$

$$V_2\left[2SC + \frac{1}{R}\right] - V_1(SC) - V_f(SC) = 0.$$

$$V_2\left[\frac{1+2SCR}{R}\right] - V_1(SCR) - V_f(SCR) = 0.$$

$$V_f(SCR) = V_2(1+2SCR) - V_1(SCR)$$

$$V_f = \frac{V_2(1+2SCR)}{SCR} - V_1 \longrightarrow (2).$$

write down KCL with respect to node 3 'Vf'

$$\frac{V_f - V_2}{X_C} + \frac{V_f - 0}{R} = 0.$$

$$(V_f - V_2)SC + \frac{V_f}{R} = 0.$$

$$V_f\left[SC + \frac{1}{R}\right] - V_2(SC) = 0.$$

$$V_f\left[\frac{1+RSC}{R}\right] - V_2(SC) = 0.$$

$$V_f(1+RSC) - V_2(SCR) = 0.$$

$$V_2 = \frac{V_f(1+RSC)}{SCR} \longrightarrow (3).$$

sub (3) in (2)

$$V_f = \frac{V_f(1+RSC)}{SCR} \cdot \frac{(1+2SCR)}{SCR} - V_1$$

$$V_f = \frac{V_f(1+RSC)(1+2SCR)}{S^2C^2R^2} - V_1$$

$$V_f = \frac{V_f(1+2sCR + sRC + 2s^2R^2C^2)}{s^2R^2C^2} - V_1$$

$$-V_f + \frac{V_f(1+3sCR + 2(sRC)^2)}{(sRC)^2} = V_1$$

$$\frac{-V_f(sRC)^2 + V_f(1+3sCR+2(sRC)^2)}{(sRC)^2} = V_1$$

$$\frac{V_f[-(sCR)^2 + 1 + 3sCR + 2(sCR)^2]}{(sRC)^2} = V_1$$

$$V_1 = \frac{V_f[(sRC)^2 + 1 + 3sCR]}{(sRC)^2}$$

$$V_1 = \frac{V_f[1 + 3sCR + (sRC)^2]}{(sRC)^2}$$

sub V_1, V_2 in ①.

$$V_0 = \frac{V_f[1+3sCR+(sRC)^2]}{(sRC)^2} \cdot \frac{(1+2sCR)}{(sRC)} - \frac{V_f(1+RsC)}{sCR}.$$

$$= \frac{V_f(1+3sCR+(sRC)^2)(1+2sCR)}{(sRC)^3} - \frac{V_f(1+sRC)}{sCR}$$

$$= \frac{V_f[1+3sCR+(sRC)^2](1+2sCR) - V_f(1+sRC)(sRC)^2}{(sCR)^3}.$$

$$= \frac{V_f[1+2sCR + 3CsR + 6(sCR)^2 + (sRC)^2 + 2(sRC)^3] - V_f[(sRC)^2 + (sRC)^3]}{(sCR)^3}.$$

$$V_o = \frac{V_f\left[1+5SCR+7(SRC)^2+2(SRC)^3\right]-V_f\left[(SRC)^2+(SRC)^3\right]}{(SCR)^3}$$

$$V_o = \frac{V_f\left[1+5SCR+7(SRC)^2+2(SRC)^3-(SRC)^2-(SRC)^3\right]}{(SCR)^3}$$

$$V_o = \frac{V_f\left[1+5SCR+(SRC)^3+6(SRC)^2\right]}{(SRC)^3}$$

$$\frac{V_f}{V_o} = \frac{(SRC)^3}{\left[1+5SRC+6(SRC)^2+(SRC)^3\right]}$$

$$\beta = \frac{(SRC)^3}{(SRC)^3\left[\frac{1}{(SRC)^3}+5\frac{1}{(SRC)^2}+\frac{6}{SRC}+1\right]}$$

$$\beta = \frac{V_f}{V_o} = \left[\frac{1}{1+\frac{5}{(SRC)^2}+\frac{6}{SRC}+\frac{1}{(SRC)^3}}\right]$$

Replace s with jω and ω with 2πf.

$$\beta = \frac{1}{1+\frac{6}{j2\pi fRC}+\frac{5}{4\pi^2f^2R^2C^2}-\frac{1}{j(2\pi fRC)^3}}$$

$$\beta = \frac{1}{\left[1-5\left(\frac{f_1}{f}\right)^2\right]+\frac{1}{j}\left[6\left(\frac{f_1}{f}\right)-\left(\frac{f_1}{f}\right)^3\right]}$$

$$\therefore f_1 = \frac{1}{2\pi RC}.$$

$$= \frac{1}{\left[1 - 5\left(\frac{f_1}{f}\right)^2\right] - j\left[6\left(f_1/f\right) - \left(f_1/f\right)^3\right]}$$

The imaginary term is zero.

$$\beta = \frac{1}{\left(1 - 5\left(\frac{f_1}{f}\right)^2\right)} \rightarrow (3).$$

$$\left[6\left(\frac{f_1}{f}\right) - \left(\frac{f_1}{f}\right)^3\right] = 0.$$

$$6\left(\frac{\cancel{f_1}}{\cancel{f}}\right) = \left(\frac{f_1}{f}\right)^{\cancel{3}\,2}$$

$$\frac{f_1}{f} = \sqrt{6}. \rightarrow (11).$$

$$f_0 = \frac{1}{\sqrt{6}\,2\pi RC}.$$

sub (11) in (3),

$$\beta = \frac{1}{1 - 5(6)}$$

$$= \frac{1}{1-30} = \frac{1}{-29}$$

$$A_V \cdot \beta = 1$$

$$A_V \cdot \left(-\frac{1}{29}\right) = 1.$$

$$A_V \geq -29$$

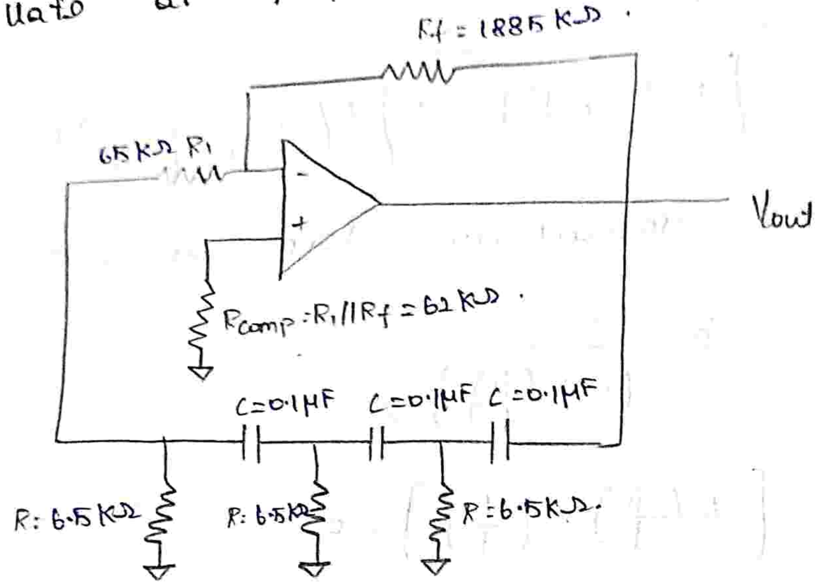
To find R_1 & R_f.

$$1 - \frac{R_f}{R_1} = -29.$$

$$R_f = 29 R_1$$

$$R_1 = 10 R.$$

6). Design an RC phase shift oscillator to oscillate at frequency $f_0 = 100\,Hz$.



$$f_0 = \frac{1}{\sqrt{6}\, 2\pi RC}.$$

Let $\boxed{C = 0.1\,\mu F.}$

$$100 = \frac{1}{\sqrt{6} \times 2 \times 3.14 \times R \times 0.1 \times 10^{-6}}$$

$$100 = \frac{1}{1.53822 \times 10^{-6} \times R}$$

$$\frac{1}{R} = 153.82 \times 10^{-6}$$

$$R = 6.501 \times 10^{6} \times 10^{-3}$$

$$\boxed{R = 6.5\,k\Omega.}$$

$$R_1 = 10(R)$$
$$= 10 \times 6.5 \times 10^{3}$$

$$\boxed{R_1 = 65\,k\Omega.}$$

$$R_f = 29\,(65\,k\Omega)$$

$$\boxed{R_f = 1885\,k\Omega.}$$

$$\frac{1}{R_{comp}} = \frac{1}{R_1} + \frac{1}{R_f}.$$

$$\frac{1}{R_{comp}} = \frac{1}{65000} + \frac{1}{1885000}$$

$$\frac{1}{R_{comp}} = \frac{1885000 + 65000}{(65000)(1885000)}$$

$$= \frac{1950000}{1.225 \times 10^{11}}$$

$$= \frac{1950 \times 10^{3}}{1.225 \times 10^{11}}$$

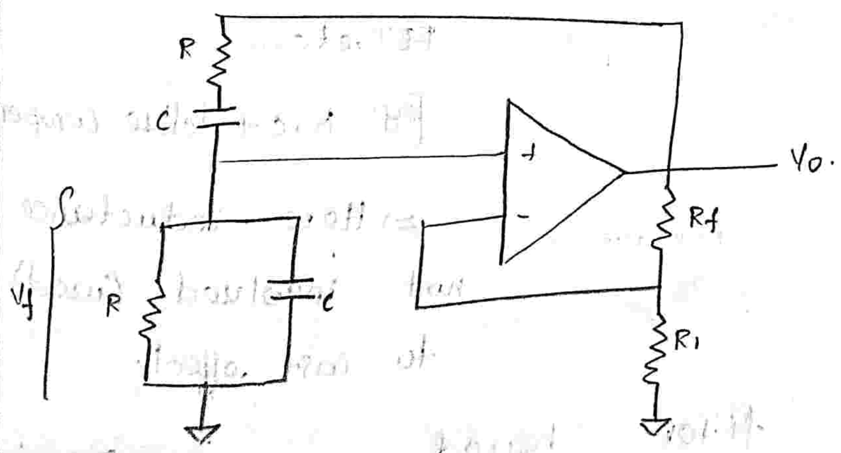
$$\frac{1}{R_{comp}} = 1591.83 \times 10^{-8}$$

$$R_{comp} = 6.282 \times 10^{-4} \times 10^{8}$$

$$= 6.282 \times 10^{4}$$

$$\boxed{R_{comp} = 62 K\Omega.}$$

WIEN - BRIDGE OSCILLATOR: 2m (or) 8m.



Active Filters:

Filter → Frequency selective Network.

*. Based on nature of signals as i/p.

Filter is classified in.

1. Analog filter.
2. Digital filter.

*. In Analog filter is classified based on the components used.

Analog filter

- Passive filter
  [eg: R, L, C]
- Active filter.
  ⇒ along with passive components active components added like op-amp, FET etc...
  [eg: R, C + active component]
  ⇒ Here inductance is not involved (used) due to cost effect.

*. Active filter based on suppression of i/p signal is classified as,

1. Low pass filter - LPF
2. High pass filter - HPF
3. Band pass filter - BPF.

4). BEF / BRF / BSF

*. Based on frequency response, the filter classification.

PB → Pass band.
SB → Stop band

reduction in gain occurs in montonically dec.

1. Butterworth filter - flat PB, flat SB.

[You may not find the ripple

2. chebystor filter

3. Bessel function.

⇒ We use Butterwort filters.

Parameters :

1). Pass Band.

2). Stop Band.

3). Cut-off frequency.

Let us consider Low pass filter.

f_h → maximum range of freq. that is transmitted by LPF filter.

stop band } - Range of frequency of i/p signal that can be supressed / eliminated / attenuated by filter circuit.

case (ii): $f >> f_L$

$$\left|\frac{V_o(s)}{V_i(s)}\right| > A_F$$

$$V_o(s) > A_F\, V_{in}(s)$$

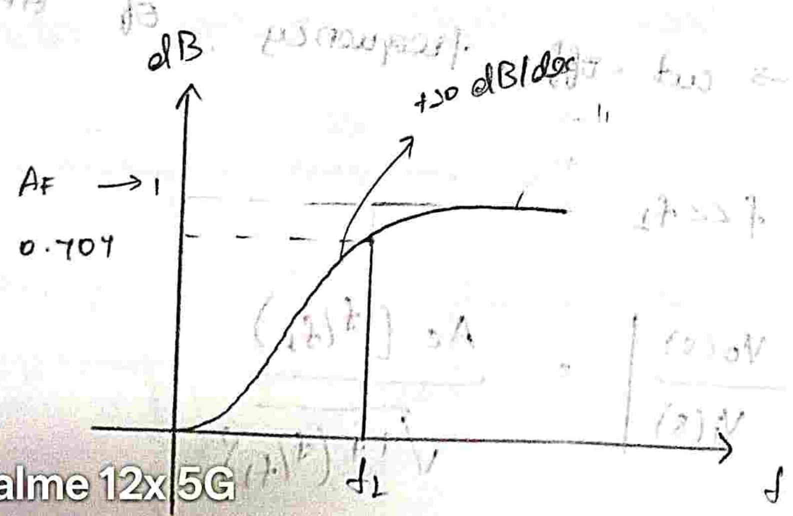
It allows all the signal frequency greater than cut-off frequency.

case (iii): $f = f_L$

$$\left|\frac{V_o(s)}{V_i(s)}\right| = \frac{A_F(f/f_L)}{\sqrt{1+(f/f_L)^2}}$$

$$= \frac{A_F \cdot (1)}{\sqrt{1+1}} = \frac{A_F}{\sqrt{2}}.$$

$$\left|\frac{V_o(s)}{V_i(s)}\right| = 0.707\ A_F$$



(9). Design a 1st HPF the cut off frequency of 2KHz.

$$f_L = 2KHz.$$

$$f_L = 2\times10^3 \text{ Hz}.$$

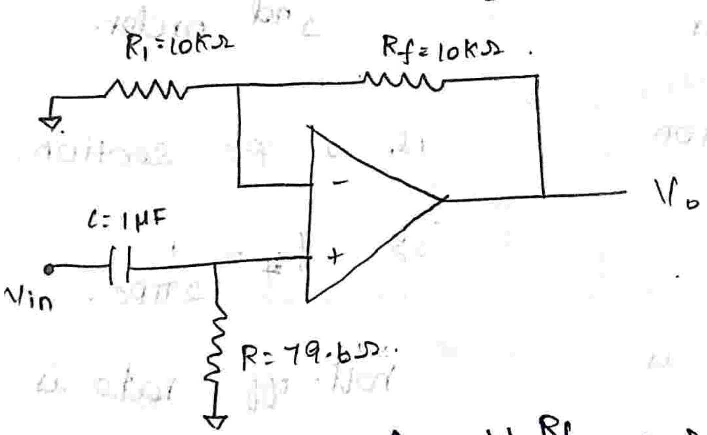
$$f_L = \frac{1}{2\pi RC} = 2\times10^3$$

Let $C = 1\mu F$, $R = \frac{1}{4\times\pi\times1\times10^{-6}\times10^{3}}$

$$= \frac{1}{12.5663}$$

$$R = 0.07957 K\Omega$$

$$\boxed{R = 79.57\Omega}$$



$$A_F = 1+\frac{R_f}{R_1} = 2.$$

$$\frac{R_f}{R_1} = 1$$

$$\therefore R_f = R_1$$

Let us assume $\boxed{R_1 = R_f = 10K\Omega.}$

Note:

*. For n^{th} order high pass filter, the generalized expression of gain is .

$$\left|\frac{V_o(s)}{V_i(s)}\right| = |H(s)| = \frac{A_F}{\sqrt{1+(f_L/f)^{2n}}}$$

*. For n^{th} order of LPF.

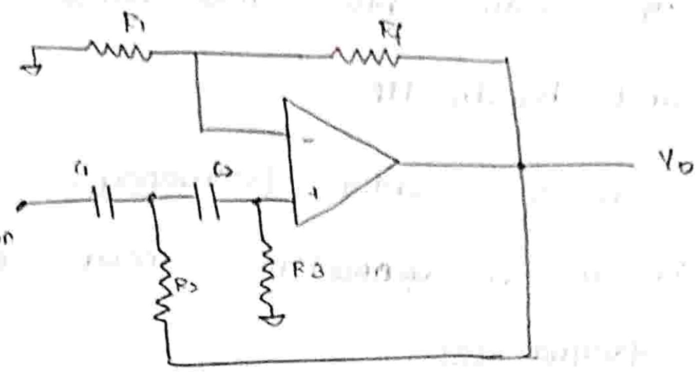
$$\left|\frac{V_o(s)}{V_i(s)}\right| = H(s) = \frac{A_F}{\sqrt{1+(f/f_H)^{2n}}}$$

Difference btw 1^{st} & 2^{nd} order HPF.

1^{st} order	2^{nd} order.
1). 1 RC section	1). 2 - RC section.
2). $f_L = \frac{1}{2\pi RC}$	2). $f_H = \frac{1}{2\pi RC}$.
roll-off rate is +20 db/dec.	roll-off rate is +40 db/dec.

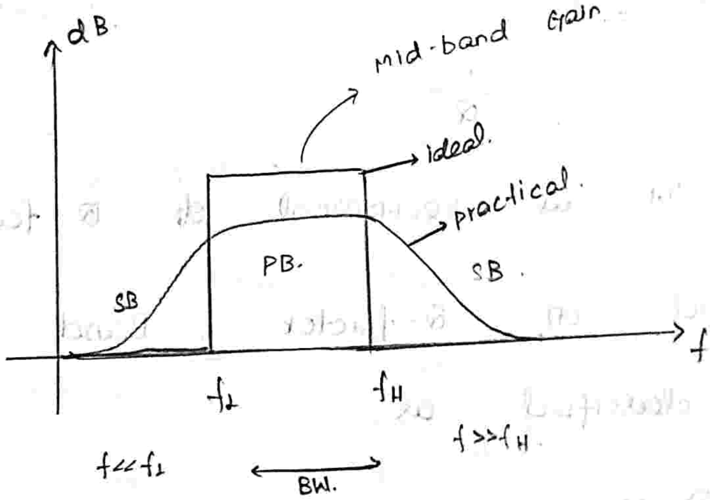
*. For n^{th} order HPF, is $+n\,20$ db/dec.

2nd order high-pass filter:



Band-pass filter (BPF).

It allows all the signal frequency which is lies between two cut-off frequency. (f_L & f_H)



=>. The signal is attenuated when $f<<f_L$, $f>>f_H$.

=> btw f_L & f_H is called "mid band frequencies".

Bandwidth:

*. The range of frequency for which (the gain is maximum) or (the circuit is providing maximum gain)

*. It is used to analyse the performance of the active filters

Quality factor:

* The ratio b/w mid-band frequency (f_0) and bandwidth.

$\therefore f_0 = f_c$ = center frequency.

⇒ It is a geometric mean of two cut-off frequencies.

$$Q = \frac{f_0}{BW} = \frac{f_c}{BW}$$

$$\boxed{f_0 = f_c = \sqrt{f_L \cdot f_H}}$$

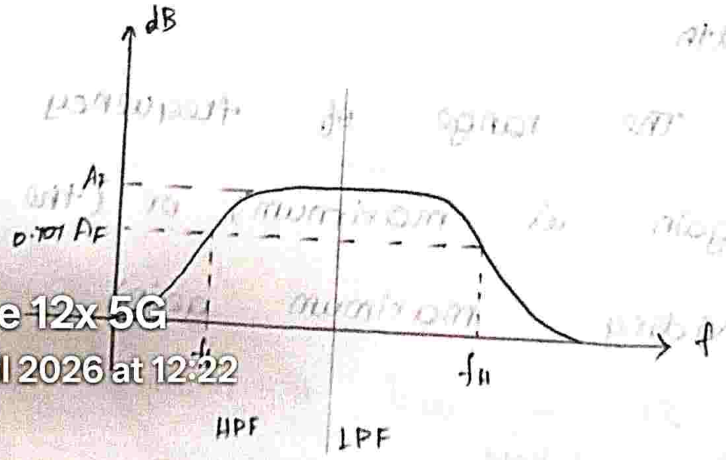
selectivity:

$$\boxed{= \frac{1}{Q}}$$

It is reciprocal of Q-factor.

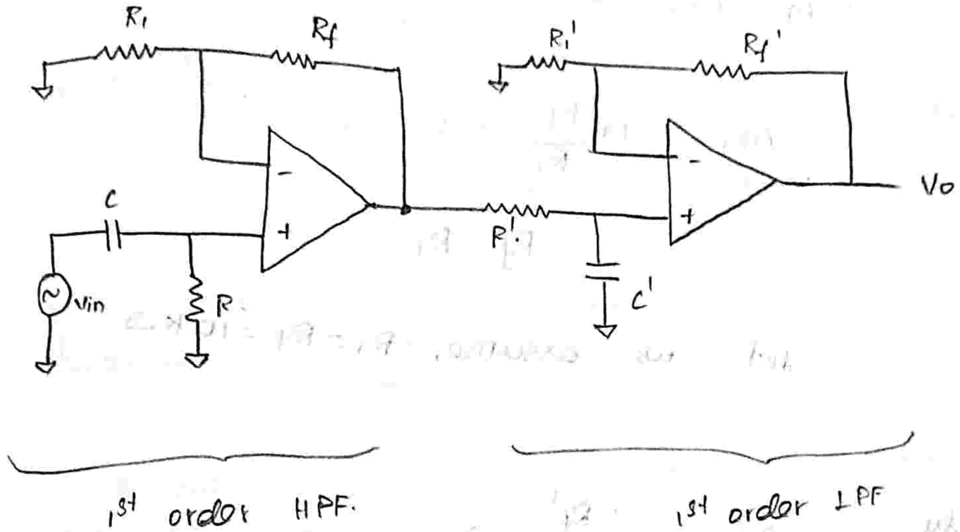
- Based on Q-factor, Band pass filter is classified as,

1. if $Q > 10 \rightarrow$ NBPF - Narrow band pass filter.
2. if $Q < 10 \rightarrow$ WBPF - wide band pass filter.



1^{st} order wide Band Pass filter (WBPF).

*. The HPF is cascaded with LPF.



$$\left|\frac{V_o(s)}{V_{in}(s)}\right| = |H(s)| = \frac{A_{F_1}\cdot(f/f_L)}{\sqrt{1+(f/f_L)^2}} \cdot \frac{A_{F_2}}{\sqrt{1+(f/f_H)^2}}.$$

[∵ gain of WBPF = Product of HPF & LPF]

(10). Design a wide BPF having $f_L = 400\ Hz$. and $f_H = 2\ KHz$ and pass band gain of 4. also find the quality factor of ~~fact filter~~ circuit.

Sol:- $f_L = 400\ Hz$.

$f_H = 2\ KHz$.

WPBF gain of '4'.

$Q = ?$

$$A = A_{F1} \cdot A_{F2}.$$

$$\therefore A_{F1} = A_{F2} = 2$$

here,

$$A_{F1} = 1 + \frac{R_f}{R_1} = 2.$$

$$R_f = R_1$$

let us assume, $\boxed{R_1 = R_f = 10\,k\Omega.}$

IIIly,

$$A_{F2} = 1 + \frac{R_f'}{R_1'} = 2.$$

$$\frac{R_f'}{R_1'} = 1$$

$$R_f' = R_1'$$

let us assume $\boxed{R_1' = R_f' = 10\ k\Omega.}$

for HPF, $f_L = 400$ Hz.

$$f_L = \frac{1}{2\pi R'C} = 400.$$

let $\boxed{C = 1\,\mu F}$

$$R = \frac{1}{400 \times 2 \times \pi \times 10^{-6}}$$

$$= \frac{1}{2513.27} \times 10^{6}$$

$$= 3.9788 \times 10^{-4} \times 10^{6}.$$

$\boxed{R = 397.88\,\Omega.}$

realme 12x 5G
20 April 2026 at 12:23

for LPF, $f_H = 2\,KHz$.

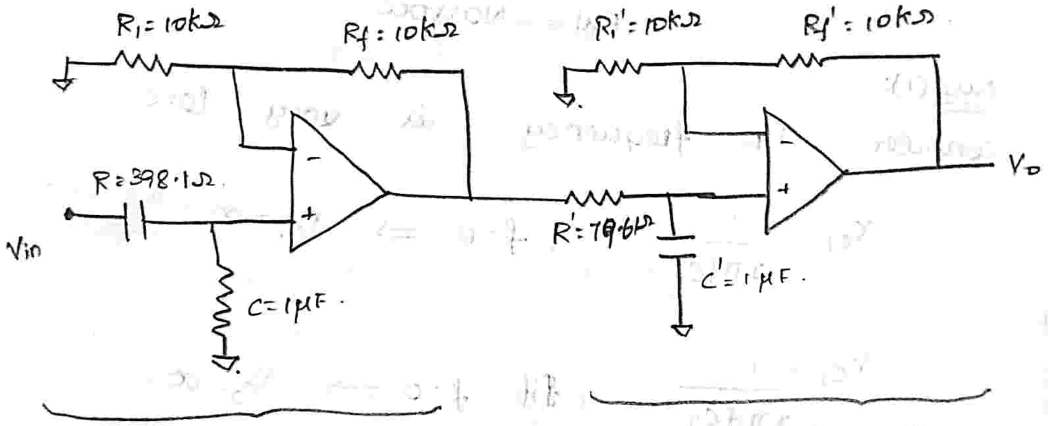
$$f_H = \frac{1}{2\pi R'C'} = 2\times10^3.$$

Let, $\boxed{C = 1\ \mu F}$

$$R' = \frac{1}{4\times\pi\times10^3\times1\times10^{-6}}$$

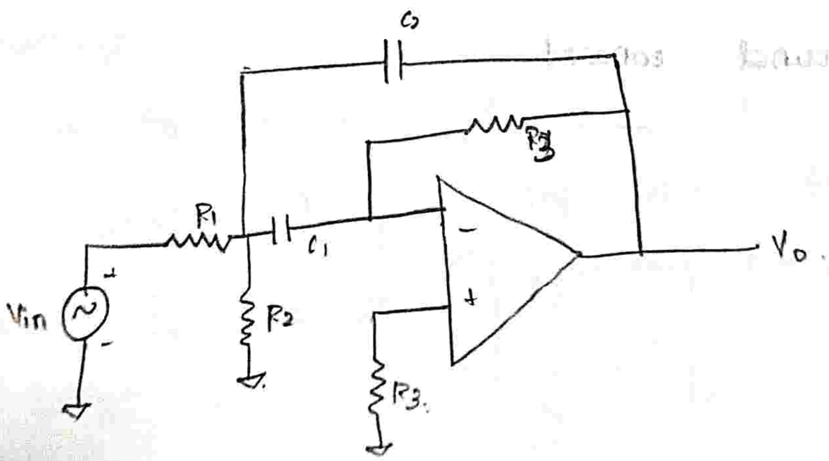
$$= 0.07961\times10^3$$

$$\boxed{R' = 79.61\ \Omega.}$$



Narrow Band pass filter: (NBPF)

=> The input is given in inverting terminal.



1. Op-amp is configured in inverting mode.

2. It has two feedback.

because of this feedback it is named as multiple feedback filter.

③. C_1 is responsible for make this circuit fn as LPF.

C_2 is responsible for make this circuit fn as HPF.

$C_1 \longrightarrow LPF$

$C_2 \longrightarrow HPF$.

④. $Q > 10 = \frac{f_0}{BW} \leftarrow$ Narrow

case (i):

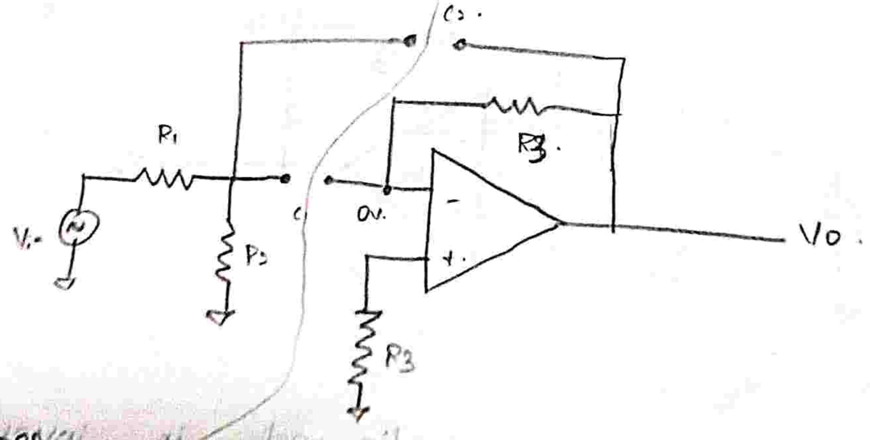
consider i/p frequency is very less.

$X_{C_1} = \frac{1}{2\pi f C_1}$, if $f=0 \Rightarrow X_{C_1} = \infty$.

$X_{C_2} = \frac{1}{2\pi f C_2}$, if $f=0 \Rightarrow X_{C_2} = \infty$.

$\therefore$ If $X_{C_1} = X_{C_2} = \infty$ it act as open circuit.

$\boxed{V_0 = 0}$ due to virtual ground concept.



$\Rightarrow$ It exhibits the behaviour of HPF.

$$\frac{V_o}{V_{in}} = \frac{-sR_fC_1}{(1+sR_fC_f)(1+sR_1C_1)}$$

$$\left|\frac{V_o}{V_{in}}\right| = \frac{sR_fC_1}{(1+sR_fC_f)(1+sR_1C_1)}$$

assume, $R_fC_f = R_1C_1$.

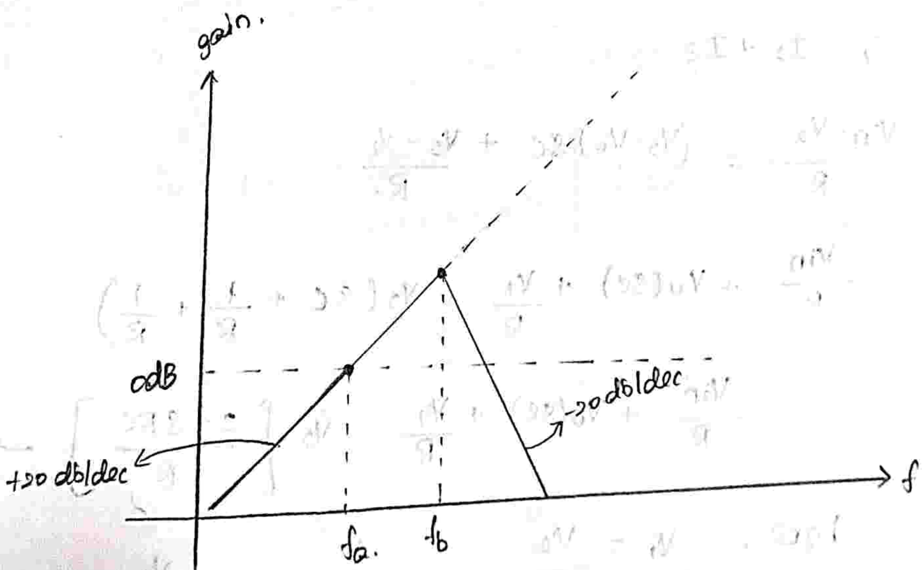
$$\left|\frac{V_o}{V_{in}}\right| = \frac{sR_fC_1}{[1+sR_1C_1]^2}$$

$s = j\omega$
$= j2\pi f$.

$$= \frac{sR_fC_1}{[1+j2\pi fR_1C_1]^2}$$

$$\left|\frac{V_o}{V_{in}}\right| = \frac{sR_fC_1}{[1+j(f/f_b)]^2}$$

$$\boxed{\text{here, } f_b = \frac{1}{2\pi R_1C_1} = \frac{1}{2\pi R_fC_f}.}$$



$f_a = f_b = 1$

$A = 0\ dB$.

conditions to design:

→ $f_b = 10 f_a$.

→ signal frequency $f_s = \frac{f_a}{20}$

→ $T \geq R_f C_1$

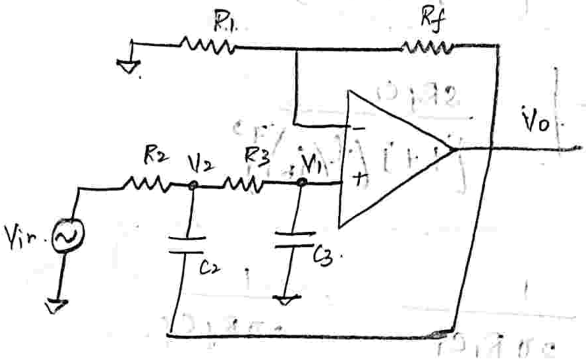
→ $R_1 C_1 = R_f C_f$.

⇒ use f_a find R_f

⇒ use $f_b = 10 f_a$ find f_b.

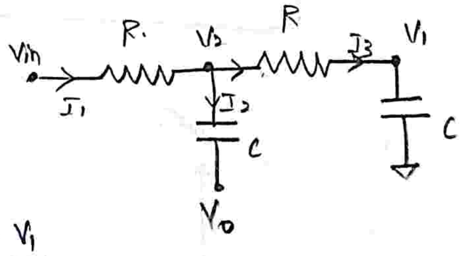
⇒ use f_b find R_1.

2nd order low pass filter:



$$V_o = \left[1 + \frac{R_f}{R_1}\right] V_1 \rightarrow (1).$$

let, $R_2 = R_3 = R$, $C_2 = C_3 = C$.



$$I_1 = I_2 + I_3$$

$$\frac{V_{in} - V_2}{R} = (V_2 - V_o) sC + \frac{V_2 - V_1}{R}.$$

$$\frac{V_{in}}{R} + V_o(sC) + \frac{V_1}{R} = V_2\left(sC + \frac{1}{R} + \frac{1}{R}\right)$$

$$\frac{V_{in}}{R} + V_o(sC) + \frac{V_1}{R} = V_2\left[\frac{2 + sRC}{R}\right] \rightarrow (2).$$

here, $V_1 = \frac{V_2}{R + 1/sC} \cdot \frac{1}{sC} = \frac{V_2}{1 + sRC}$.

$$\boxed{\therefore V_2 = V_1(1+SRC)}$$

apply this into equ (2).

$$\frac{V_{in}}{R} + V_o(SC) + \frac{V_1}{R} = \frac{V_1(1+SRC)(2+SRC)}{R}$$

$$\frac{V_{in}}{R} + V_o(SC) = \frac{V_1\left[2+SRC+2SRC+(SRC)^2\right]-V_1}{R}$$

$$\frac{V_{in}}{R} + V_o(SC) = \frac{V_1}{R}\left[2+3SRC+(SRC)^2-1\right]$$

$$\frac{V_{in}}{R} + V_o(SC) = \frac{V_1}{R}\left[1+3SRC+(SRC)^2\right]$$

$$V_{in} + V_o(SRC) = V_1\left[1+3SRC+(SRC)^2\right]$$

$$\boxed{\therefore V_1 = \frac{V_{in}+V_o(SRC)}{1+3SRC+(SRC)^2}}$$

apply this into equ (1).

$$V_o = A_f\left[\frac{V_{in}+V_o(SRC)}{1+3SRC+(SRC)^2}\right]$$

$$V_o = A_F\left[\frac{V_{in}}{1+3SRC+(SRC)^2}\right] + A_F\left[\frac{V_o(SRC)}{1+3SRC+(SRC)^2}\right]$$

$$V_o\left[1-\frac{A_F(SRC)}{1+3SRC+(SRC)^2}\right] = A_F\frac{V_{in}}{1+3SRC+(SRC)^2}$$

$$V_o\left[\frac{1+3SRC+(SRC)^2-A_F(SRC)}{1+3S\cancel{RC+(SRC)^2}}\right] = A_F\frac{V_{in}}{1+3S\cancel{RC+(SRC)^2}}$$

$$\therefore \frac{V_o}{V_{in}} = \frac{A_F}{1+SRC\left[3-A_F\right]+(SRC)^2}.$$

$$\frac{V_o}{V_{in}} = \frac{A_F}{1+sRC[3-A_F]+s^2R^2C^2}$$

$$= \frac{A_F}{R^2C^2\left[s^2+\frac{s}{RC}(3-A_F)+\frac{1}{R^2C^2}\right]}$$

$$= \frac{A_F/(RC)^2}{s^2+\left[\frac{3-A_F}{RC}\right]s+\frac{1}{R^2C^2}}$$

$$= \frac{A_F\,\omega_h^2}{s^2+\alpha\omega_h s+\omega_h^2} \qquad \left[\because \omega_h = \frac{1}{RC},\ \alpha = 3-A_F\right]$$

$$\frac{V_o}{V_{in}} = A_F \cdot \frac{1/R^2C^2}{s^2+\left[\frac{3-A_F}{RC}\right]s+\frac{1}{R^2C^2}} \longrightarrow (8)$$

we know that for 2nd order system.

$$\frac{V_o}{V_{in}} = \frac{\omega_n^2}{s^2+2\xi\omega_n s+\omega_n^2} = \frac{V_o}{s^2+\sqrt{2}\,s+1} \longrightarrow (11)$$

compare (8) & (11).

$$\frac{3-A_F}{RC} = \sqrt{2}$$

Let $\boxed{RC=1}$, $3-A_F=\sqrt{2}$.

$$A_F = 3-\sqrt{2}$$

$$\boxed{A_F = 1.586}$$

For nth order LPF, $\left|\frac{H(j\omega)}{A_o}\right| = \frac{1}{\sqrt{1+(\omega/\omega_h)^{2n}}}$

Also,

(12) design filter 1KHz.

sol:

Let

Vin.

Also,

$$\omega_h^2 = \omega_0^2 = \frac{1}{R^2C^2}.$$

$$\omega_0 = \frac{1}{RC}.$$

$$f_0 = \frac{1}{2\pi RC}$$

$$\text{If, } R_1 = R_2 = R, \quad C_1 = C_2 = C$$

$$f_0 = \frac{1}{2\pi\sqrt{R_1R_2C_1C_2}}.$$

(12) design a second order butterworth lowpass filter having upper cut-off frequency of 1KHZ.

sol:-

$$A_F = 1 + \frac{R_f}{R_1} = 1.586.$$

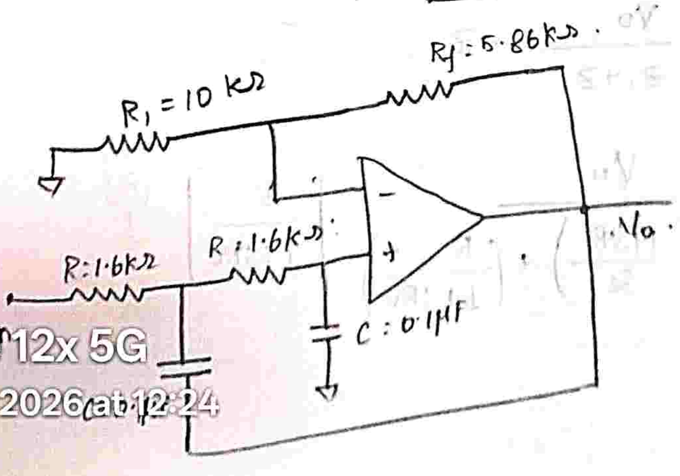
$$R_f = 0.586\,R_1.$$

Let $\boxed{R_1 = 10\,k\Omega, \quad R_f = 5.86\,k\Omega.}$

$$f_0 = \frac{1}{2\pi RC} = 1000$$

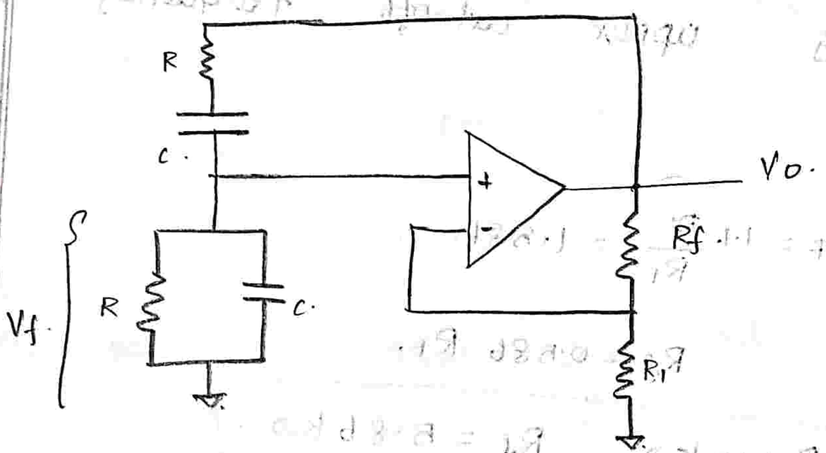
Let $C = 0.1\,\mu F$, $R = \dfrac{1}{2\pi \times 0.1 \times 10^{-6} \times 1000}$

$$\boxed{R = 1.6\,k\Omega.}$$



order	factors of polynomials.
1.	$s+1$
2.	$s^2+\sqrt{2}s+1$
3.	$(s+1)\ (s^2+s+1)$
4.	$(s^2+0.765s+1)\ (s^2+1.848s+1)$
5.	$(s+1)\ (s^2+0.618s+1)\ (s^2+1.618s+1)$

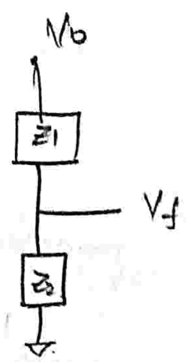
Wien- bridge oscillator.



let $Z_1 = R + \frac{1}{sC} = \frac{1+sRC}{sC}$

$$Z_2 = R \parallel \frac{1}{sC} = \frac{R \times 1/sC}{R + 1/sC} = \frac{R}{sRC+1}$$

here, V_f = Voltage across 'Z_2'



$$V_f = \frac{V_o}{Z_1+Z_2} \cdot Z_2 .$$

$$= \frac{V_o}{\left(\frac{1+sRC}{sC}\right) + \left(\frac{R}{1+sRC}\right)} \cdot \left[\frac{R}{1+sRC}\right]$$

$$= \frac{V_o(1+\cancel{s}RC)(sC)}{[C(1+sRC)^2 + sCR]} \cdot \frac{R}{(1+sRC)}$$

$$V_f = \frac{V_o\,(sRC)}{1+(sRC)^2+2sRC+sCR}$$

$$V_f = \frac{V_o\,(sRC)}{1+3sRC+(sRC)^2}$$

$$\therefore \beta = \frac{V_f}{V_o} = \frac{sRC}{1+3sRC+(sRC)^2}$$

$$= \frac{\cancel{sRC}}{\cancel{sRC}\left[3+sRC+\frac{1}{sRC}\right]}$$

$$= \frac{1}{3+j\omega RC+\frac{1}{j\omega RC}}$$

$$\beta = \frac{1}{3+j\underbrace{\left[\omega RC-\frac{1}{\omega RC}\right]}_{\to 0}}$$

to have sustained oscillations 'β' must be a Real.

$$\beta = \frac{V_f}{V_o} = \frac{1}{3}$$

equating imaginary part to '0'.

$$\omega RC - \frac{1}{\omega RC} = 0$$

$$\omega^2R^2C^2 = 1$$

$$\omega^2 = \frac{1}{R^2C^2} \Rightarrow \omega = \frac{1}{RC}$$

$$\boxed{f_0 = \frac{1}{2\pi RC}}$$

To satisfy the magnitude condition,

$A_v \beta = 1.$

$A_v\left(\frac{1}{3}\right) = 1.$

$\boxed{A_v = 3.}$

(13). Design a wien bridge oscillator for $f_0 = 1\,kHz$.

Sol:- Given, $f_0 = 1\,kHz = 1000\,Hz$.

W.K.T $f_0 = \frac{1}{2\pi RC} = 1000$

Let $C = 0.1\,\mu F$.

$$R = \frac{1}{2\pi \times 1000 \times 0.1 \times 10^{-6}}$$

$R = 1.59\,k\Omega.$

$\boxed{R \simeq 1.6\,k\Omega.}$

for wien-Bridge

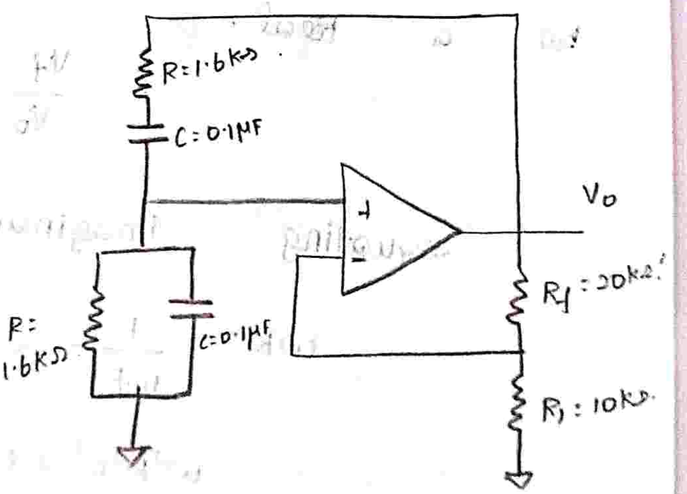
$A_v = 3.$

$\left[1 + \frac{R_f}{R_1}\right] = 3.$

$R_f = 2R_1.$

Let $\boxed{R_1 = 10\,k\Omega.}$

$\boxed{\therefore R_f = 20\,k\Omega.}$



Narrow

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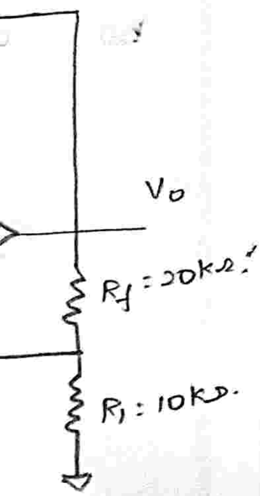
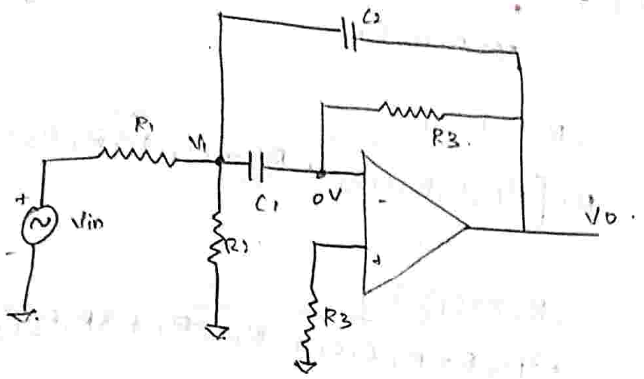


Narrow Band Pass Filter (NBPF).



Let, $Z_1 = 1/sC_1$, $Z_2 = 1/sC_2$

write Nodal equation w.r.t Node 'V1'

$$\frac{V_1 - V_{in}}{R_1} + \frac{V_1}{R_2} + (V_1 - V_0)\,sC_2 + V_1(sC_1) = 0$$

$$V_1\left[\frac{1}{R_1} + \frac{1}{R_2} + sC_2 + sC_1\right] - \frac{V_{in}}{R_1} - V_0(sC_2) = 0 \longrightarrow (1)$$

write Nodal eqn w.r.t Node '0'

$$(0 - V_1)\,sC_1 + \frac{(0 - V_0)}{R_3} = 0$$

$$\frac{V_0}{R_3} = -V_1 sC_1$$

$$V_1 = \frac{-V_0}{sR_3C_1} \longrightarrow (2)$$

apply equation (2) into equation (1).

$$\left[-\frac{V_0}{sR_3C_1}\right]\left[\frac{1}{R_1} + \frac{1}{R_2} + sC_2 + sC_1\right] - \frac{V_{in}}{R_1} - V_0(sC_2) = 0$$

$$V_0(sC_2) + \frac{V_0}{sR_3C_1}\left[\frac{1}{R_1} + \frac{1}{R_2} + sC_2 + sC_1\right] = -\frac{V_{in}}{R_1}$$

$$V_0\left[sC_2 + \frac{1}{sR_3C_1}\left(\frac{R_2 + R_1 + sR_1R_2C_2 + sR_1R_2C_1}{R_1R_2}\right)\right] = -\frac{V_{in}}{R_1}$$

$$V_o\left[\frac{s^2R_1R_2R_3C_1C_2 + R_2 + R_1 + sR_2R_1C_2 + sR_1R_2C_1}{sR_1R_2R_3C_1}\right] = -\frac{V_{in}}{R_1}$$

$$\frac{V_o}{V_{in}} = \frac{-sR_1R_2R_3C_1}{R_1\left[s^2R_1R_2R_3C_1C_2 + R_2 + R_1 + sR_1R_2C_2 + sR_1R_2C_1\right]}$$

$$\frac{V_o}{V_{in}} = \frac{-(R_2R_3C_1)s}{s^2R_2R_3R_1C_1C_2 + R_2 + R_1 + sR_1R_2C_2 + sR_1R_2C_1}$$

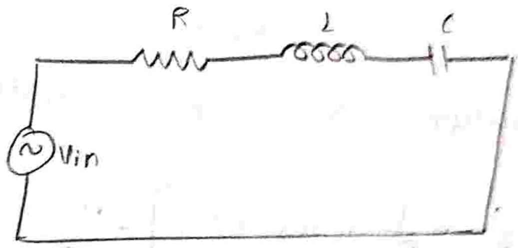
$$\frac{V_o}{V_{in}} = \frac{-(R_2R_3C_1)s}{R_1R_2R_3C_1C_2\left[s^2 + \frac{1}{R_1R_3C_1C_2} + \frac{1}{R_2R_3C_1C_2} + \frac{s}{R_3C_1} + \frac{s}{R_3C_2}\right]}$$

Let, $C_1 = C_2 = C$

$$\frac{V_o}{V_{in}} = \frac{-(s/R_1C)}{s^2 + \frac{1}{R_1R_3C^2} + \frac{1}{R_2R_3C^2} + \frac{2s}{R_3C}}$$

$$\boxed{\therefore \frac{V_o}{V_{in}} = \frac{(-s/R_1C)}{s^2 + \frac{2s}{R_3C} + \frac{1}{R_3C^2}\left[\frac{1}{R_2} + \frac{1}{R_1}\right]}} \longrightarrow (3)$$

Transfer $\underline{fn}$ of RLC series resonance circuit



overall impedance.

$$Z = R + sL + \frac{1}{sC} = \frac{sRC + s^2LC + 1}{sC}.$$

$$V_o = \frac{V_{in} \cdot A_F}{\left[\frac{sRC + s^2LC + 1}{sC}\right]}$$

A_F = Max Gain of resistance

$$= \frac{V_{in}(SR\ell)\cdot A_F}{LC\left[s^2+\frac{R}{L}s+\frac{1}{LC}\right]}$$

$$V_o : \frac{V_{in}\,(R/L)\,s\cdot A_F}{\left[s^2+(R/L)s+1/LC\right]}$$

Note:
=> at resonance Max o/p occurs across the resistor 'R'.
$\therefore V_o = V_R$.

$$\therefore \frac{V_o}{V_{in}} = \frac{-s\,(R/L)\cdot A_F}{s^2+\left(\frac{R}{L}\right)s+\frac{1}{LC}} \rightarrow (4)$$

here,

Quality factor, $Q = \frac{\omega_0 L}{R}$.

$$\therefore \frac{R}{L} = \frac{\omega_0}{Q} \rightarrow (5).$$

Also $\omega_0^2 = \frac{1}{LC}$.

$$\omega_0 = \frac{1}{\sqrt{LC}} \rightarrow (6).$$

apply equ (5), (6) into equ (4).

$$\therefore \frac{V_o}{V_{in}} = \frac{-s(\omega_0/Q)\cdot A_F}{s^2+\left(\frac{\omega_0}{Q}\right)s+\omega_0^2} \rightarrow (7)$$

The above equation (7) represents the transfer fn of RLC series resonant circuit.

Compare equ (7) with equ (3).

$$A_F\cdot\frac{\omega_0}{Q} = \frac{1}{R_1 C}.$$

$$\therefore R_1 = \frac{Q}{\omega_0 C\cdot A_F} = \frac{Q}{2\pi f_0 C\cdot A_F} \rightarrow (8).$$

IIIly $\frac{\omega_0}{Q} = \frac{2}{R_3 C}$.

$$R_3 = \frac{2Q}{\omega_0 C} = \frac{2Q}{2\pi f_0 C} = \frac{Q}{\pi f_0 C}.$$

$$R_3 = \frac{Q}{\pi f_0 C} \rightarrow (9).$$

Also,
$$\omega_0^2 = \frac{1}{R_3 C^2}\left[\frac{1}{R_2} + \frac{1}{R_1}\right]$$

$$\omega_0^2 [R_3 C^2] = \frac{1}{R_2} + \frac{1}{R_1}$$

$$\frac{1}{R_2} = \omega_0^2 [R_3 C^2] - \frac{1}{R_1}$$

$$\frac{1}{R_2} = \omega_0^2 [R_3 C^2] - \frac{1}{(Q/2\pi f_0 C A_F)}$$

$$= \omega_0^2 R_3 C^2 - \frac{2\pi f_0 C A_F}{Q}.$$

$$\frac{1}{R_2} = \frac{Q\omega_0^2 [R_3 C^2] - 2\pi f_0 C A_F}{Q}$$

$$\therefore R_2 = \frac{Q}{Q\omega_0^2 (R_3 C^2) - 2\pi f_0 C A_F}$$

$$= \frac{Q}{Q\omega_0^2 \left[\frac{Q}{\pi f_0 C}\right] C^2 - 2\pi f_0 C A_F}$$

$$= \frac{Q}{Q\omega_0^2 \left[\frac{2QC^{\cancel{2}}}{\cancel{2\pi f_0 C}}\right] - 2\pi f_0 C A_F}.$$

$$= \frac{Q}{2Q^2 \omega_0 C - 2\pi f_0 C A_F} \quad \because \omega_0 = 2\pi f_0$$

from

IIIly

equ

to

$$= \frac{Q}{\omega_0 C\left[2Q^2 - A_F\right]}.$$

$$\therefore R_2 = \frac{Q}{2\pi f_0 C\left[2Q^2 - A_F\right]} \rightarrow (10).$$

from equ (9).

$$f_0 = \frac{Q}{\pi R_3 C} \rightarrow (11)$$

Illly from equ (8).

$$f_0 = \frac{Q}{2\pi R_1 \cdot C \cdot A_F} \rightarrow (12).$$

equating (11) & (12).

$$\frac{\cancel{Q}}{\cancel{\pi} R_3 \cancel{C}} = \frac{\cancel{Q}}{2\cancel{\pi} R_1 \cancel{C} A_F}.$$

$$\therefore A_F = \frac{R_3}{2R_1} \rightarrow (13).$$

to change centre frequency from $f_0 \rightarrow f_0'$

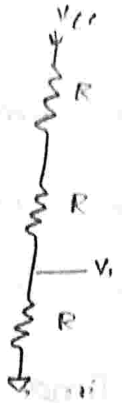
$$(X). \quad R_2' = R_2\left[f_0 / f_0'\right]^2.$$

All resistors used in this IC is 5 kΩ that's why is called 555 timer

$$V = \frac{V_{cc}}{R+2R}(2R)$$

$$= \frac{V_{cc}}{3R}(2R)$$

$$V_i = \frac{2}{3} V_{cc}$$



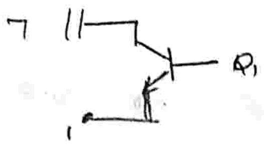
$$V_1 = \frac{V_{cc}}{3R}(R)$$

$$= \frac{V_{cc}}{3}.$$

⇒ If you want to alter the voltage at inverting terminal of UC we connect pin 5.

⇒ Capacitor can discharge via Q_1, to create low resistance path.

connect capacitor across the collector terminal of transistor



⇒ Reset - negative edge triggered pulse.

⇒ If Q_2 - ON make $Q' = 1$ o/p is 0. allow reset

⇒ If I want to disable Reset signal.

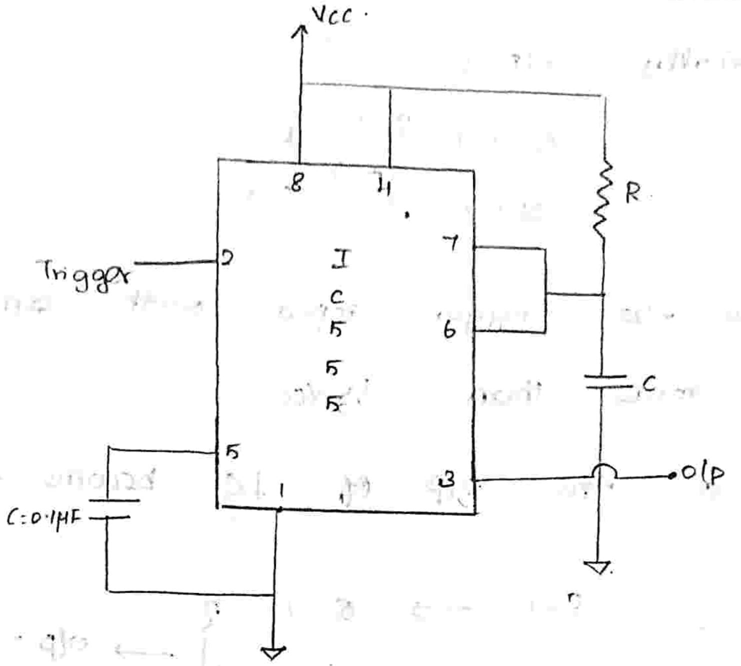
Simply Q_2 make OFF

If Q_2 - OFF not allowing Reset

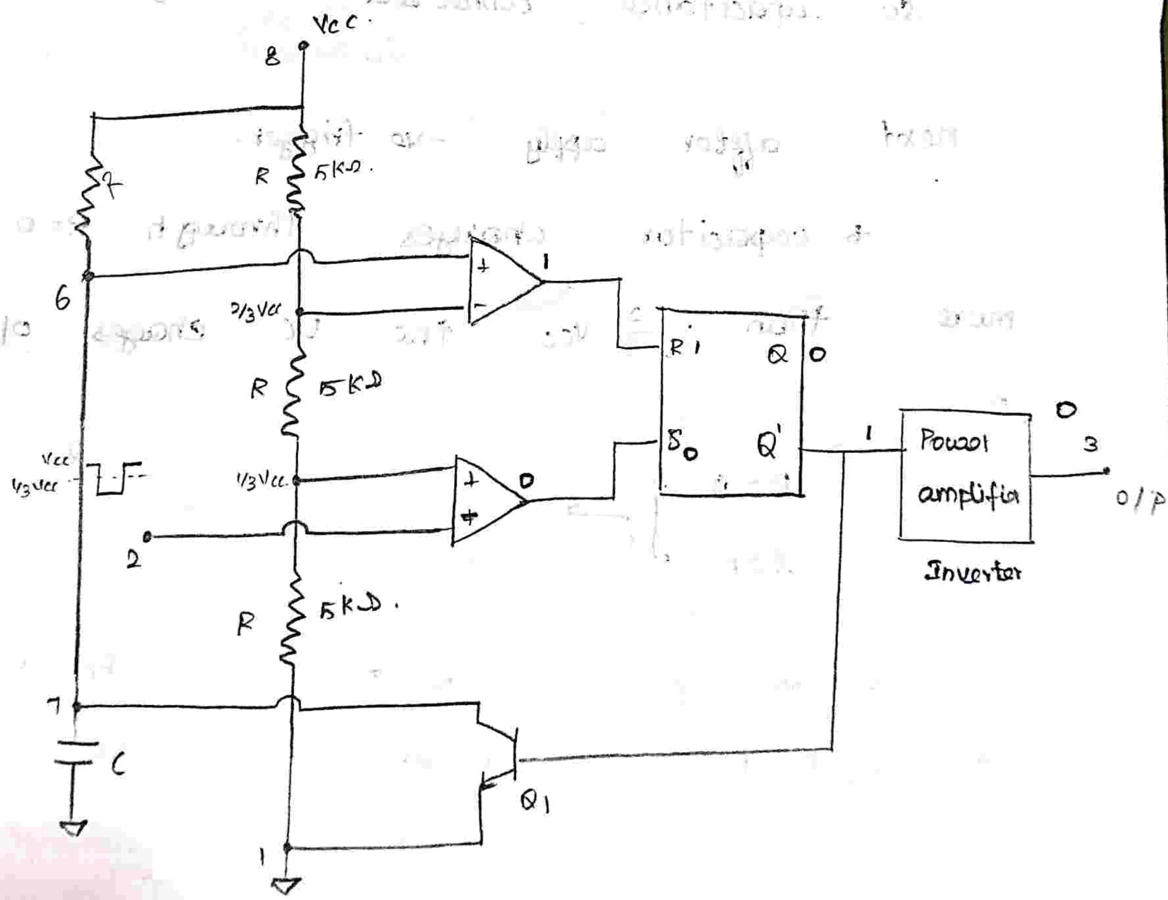
If we connect Vcc to Reset, Q_2 become of Reset signal is disabled.

MONOSTABLE MULTIVIBRATOR USING IC 555

TIMER:



This performs monostable operation (or) one-shot multivibrator.



Modified circuit to perform monostable operation.

Initially, o/p = 0.

$Q' = 1$
$Q = 0$ } → $R = 1$, $S = 0$.

[Apply -ve trigger input with amplitude more than 1/3 Vcc.

so the o/p of LC become = 1.

after short duration S become 0. so ← $S = 1$, $R = 1$ → $Q = 1$, $Q' = 0$ } → o/p = 1.]

initially, Q_1 → ON creates low Resistance path so capacitance connected to ground. (discharged).

next after apply -ve trigger.

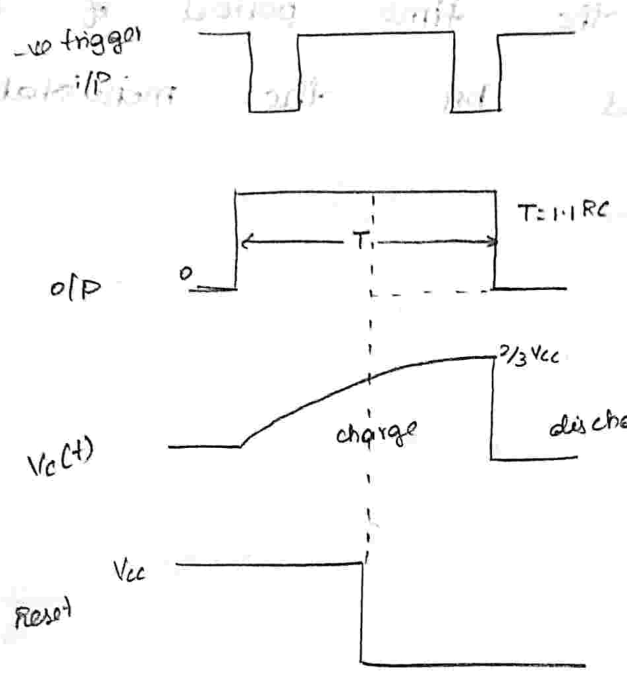
* capacitor charges through R = 0 more than $\frac{2}{3}$ Vcc the UC chages o/p 0.

$R = \phi$
$S = 1$ } → $Q = 0$, $Q' = \phi$ → o/p = 0

=> discharge sudden decrease bcoz Q_1 of C directly connect to ground without R.

both R = S = 1, so we need to apply short duration Pulse with amplitude more the $\frac{1}{3}$ Vcc, so S = 1 w... of long time it changes to S = 0.

... $S = 0$ } → $Q = 0$, $Q' = 1$ → o/p: 0.



Voltage across capacitance is expressed as.

$$V_c(t) = V_f - (V_f - V_i)\, e^{-t/RC}$$

$$= V_{cc} - (V_{cc} - 0)\, e^{-t/RC}. \quad \left[\because \text{during charging. } V_f = V_{cc},\ V_i = 0.\right]$$

$$= V_{cc} - V_{cc}\, e^{-t/RC}.$$

$$\boxed{V_c(t) = V_{cc}\left(1 - e^{-t/RC}\right)}$$

at $t = T$, $V_c(t) = \frac{2}{3} V_{cc}$.

$$\frac{2}{3} \cancel{V_{cc}} = \cancel{V_{cc}}\left[1 - e^{-T/RC}\right]$$

$$e^{-T/RC} = 1 - \frac{2}{3} = \frac{1}{3}.$$

take ln on b/s, $$\frac{-T}{RC} = \ln\left(\frac{1}{3}\right)$$

$$+\frac{T}{RC} = +1.098$$

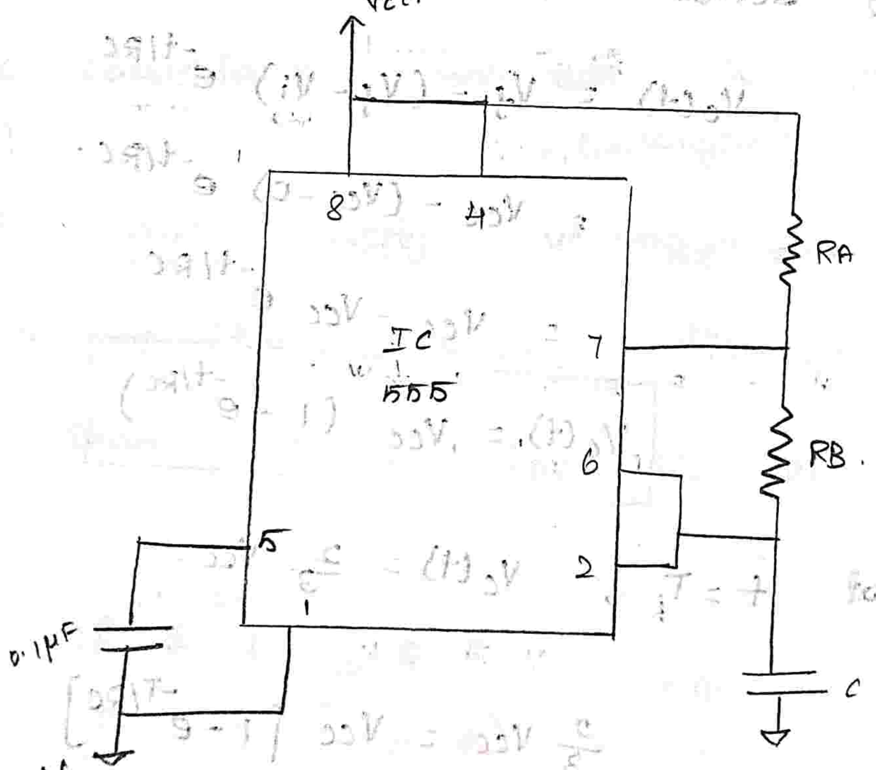
$$\boxed{T = 1.1\,RC.}$$

This is the time period of the o/p generated by the mono stable multivibrator.

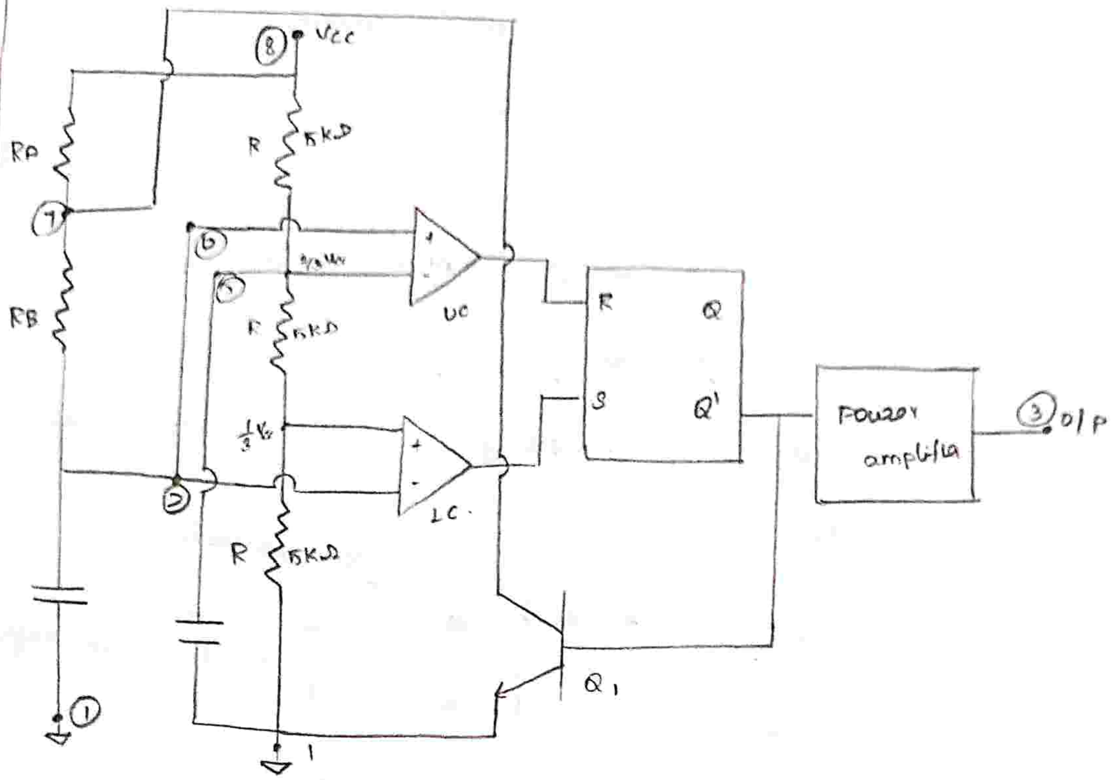
excitation table:

	Reset R	set S	present state Q	next state Q_{n+1}
	0	0	0	0
don't care (x)	0	1	0	1
	1	0(x)	1	0
	0	0	1	1
	1	1	x	x. → Invalid.

Astable operation using IC 555 timer.



*. capacitor is charged through two resistor RA & RB



If we don't want the pin 5 (control voltage) so we connect with ground through capacitor.

First:

UC → $\frac{2}{3}$ Vcc in -ve terminal. No voltage in +ve " so o/p of UC → 0.

LC → $\frac{1}{3}$ Vcc in +ve, No volt in -ve so o/p of LC → 1

$$\left.\begin{matrix} UC \to 0 & R = 0 \\ LC \to 1 & S = 1 \end{matrix}\right\} \to \left.\begin{matrix} Q = 1 \\ \bar{Q} = 0 \end{matrix}\right\} \to \text{o/p '1'}$$

(there no connection b/w C & ground)

If $\bar{Q} = 0$ → transistor (Q) is OFF, so capacitor is connected to Vcc through RA & RB.

... capacitor to get charge through RA & RB so the voltage at capacitor ↑ so upto V_{cc} but, after charge beyond $\frac{2}{3}$ (or) $\frac{1}{3} V_{cc}$

* whenever it is crossing $\frac{1}{3} V_{cc}$,

LC ⇒ -ve terminal more than +ve.

so o/p of LC ⇒ 0.

But UC unchange bcoz $\frac{1}{3}V_{cc} \rightarrow \frac{2}{3} V_{cc}$ it takes some time.

previous state.

Now, R = 0, S = 0 } Q = 1, Q' = 0 } → o/p = 1 (unchanged)

LC ⇒ 0

* when crossing $\frac{2}{3} V_{cc}$,

UC ⇒ +ve terminal more than -ve.

so o/p of UC ⇒ 1

R = 1, S = 0 } → Q = 0, $\overline{Q}$ = 1 } → o/p = 0

$\overline{Q}$ = 1 → transistor is ON so capacitor start discharging through RB.

* Now, whenever ↓se below $\frac{2}{3} V_{cc}$.

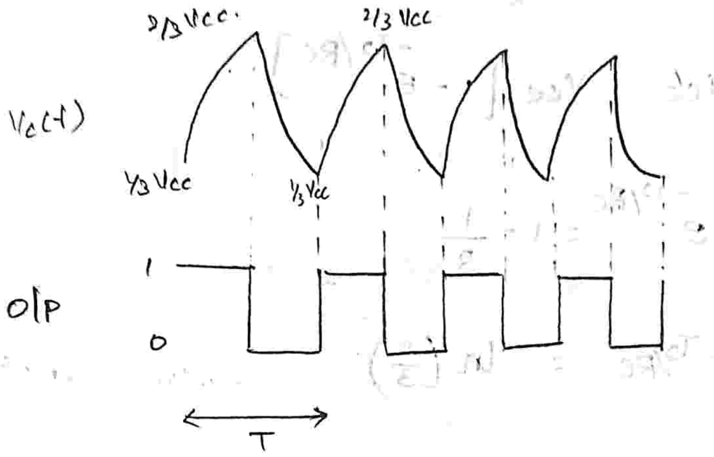
UC ⇒ become +V < -Voltage so o/p : 0

whenever ↓ses below $\frac{1}{3} V_{cc}$.

LC ⇒ become -ve < +ve terminal so o/p:1.

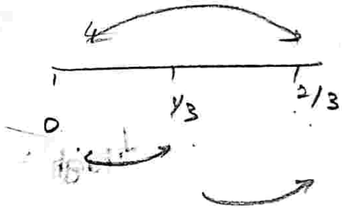
$\left.\begin{array}{l} R = 0 \\ S = 1 \end{array}\right\} \rightarrow \left.\begin{array}{l} Q = 1 \\ Q' = 0 \end{array}\right\} \rightarrow O/P = 1.$

$\Rightarrow \bar{Q} = 0$, (Q) transistor OFF, charging process occurs.



General equation.

$$V_c(t) = V_f - (V_f - V_i) \cdot e^{-t/RC}.$$



$[\because T = t_{high} + t_{low}]$

$t_1 \rightarrow$ capacitor : $0V \rightarrow \frac{2}{3} V_{cc}$.

$t_2 \rightarrow$ capacitor : $0V \rightarrow \frac{1}{3} V_{cc}$.

$t_{high} = t_1 - t_2 = \frac{1}{3} \rightarrow \frac{2}{3} V_{cc}$. [charging time]

$$V_c(t) = V_{cc} - (V_{cc} - 0)\, e^{-t/RC_{eq}} \qquad 0 - (0 - V_{cc}) e^{-t/5}.$$

$$V_c(t) = V_{cc}\left[1 - e^{-t/RC}\right]$$

$t_1 \quad \frac{2}{3} V_{cc} \rightarrow 0.$

at $t = T_1 \quad 0 \rightarrow \frac{2}{3} V_{cc}$.

$t_2 \quad \frac{1}{3} V_{cc} \rightarrow 0$

$$\frac{2}{3} \cancel{V_{cc}} = \cancel{V_{cc}}\left[1 - e^{-T_1/RC}\right]$$

$$e^{-T_1/RC} = 1 - \frac{2}{3}.$$

$$e^{-T_1/RC} = 1/3.$$

$-T_1/RC = -1.098$

$+T_1 = 1.1\,RC$

at $t \to T_2$. $\quad 0 \to \frac{1}{3}V_{cc}$

$\frac{1}{3}\cancel{V_{cc}} = \cancel{V_{cc}}\left[1 - e^{-T_2/RC}\right]$

$e^{-T_2/RC} = 1 - \frac{1}{3}$

take ln,

$-T_2/RC = \ln\left(\frac{2}{3}\right)$

$\frac{-T_2}{RC} = -0.405$

$\boxed{T_2 \doteq 0.41\,RC}$

$t_{high} = T_1 - T_2$ $\qquad t_{high} = t_{ON}$

$= (1.1 - 0.41)\,RC$

$\boxed{t_{high} = 0.69\,RC} \Longrightarrow T_{ON} = 0.69\,(R_A + R_B)\,C$

$\because t_{low} \to$ discharging time $\quad t_2 - t_1$. $\left(\frac{2}{3}V_{cc} \to \frac{1}{3}V_{cc}\right)$

$V_c(t) = V_f - (V_f - V_i)\,e^{-t/RC}$ $\qquad [V_{cc} \to 0V]$

$= 0 - (0 - V_{cc})\,e^{-t/RC}$

$V_c(t) = V_{cc}\,e^{-t/RC}$

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$\therefore$ $t_{low} \rightarrow$ discharging time (T_d).

$\frac{2}{3}V_{cc} \rightarrow \frac{1}{3}V_{cc}$

$$V_c(t) = V_f - (V_f - V_i)\, e^{-t/RC}$$

$$= 0 - \left(0 - \frac{2}{3}V_{cc}\right) e^{-t/RC}.$$

$$V_c(t) = \frac{2}{3}V_{cc} \cdot e^{-t/RC}.$$

at $t = t_d$.

$$\frac{1}{3}V_{cc} = \frac{2}{3}V_{cc}\; e^{-t_d/RC}.$$

$$\frac{1}{2} = e^{-t_d/RC}.$$

$$\ln(0.5) = -t_d/RC.$$

$$+0.693 = +\frac{t_d}{RC}.$$

$$\therefore\; T_d = 0.69\, RC.$$

$$\boxed{t_{low} = t_d = 0.69\, R_B\, C.}$$

$$T = T_{high} + T_{low}.$$

$$= T_c + T_d.$$

$$\boxed{T = 0.69\,[R_A + 2R_B]\,C.}$$

$$f = \frac{1}{T} \simeq \frac{1.44}{(R_A + 2R_B)C}.$$

To find Duty cycle: (D)

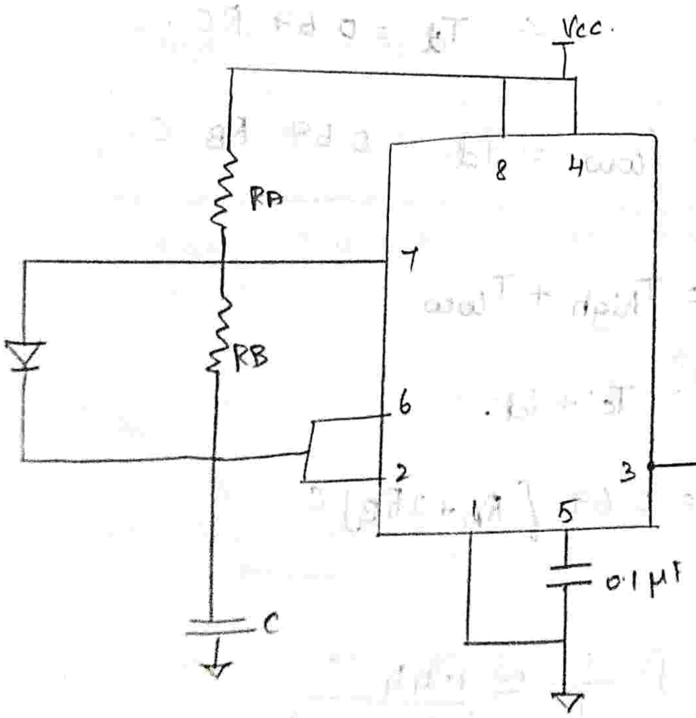
It is the ratio b/w ON time and total time period.

$$D = \frac{T_{ON}}{T} = \frac{0.69\,(R_A + R_B)\cancel{C}}{0.69\,(R_A + 2R_B)\cancel{C}}$$

$$\boxed{D = \frac{R_A + R_B}{R_A + 2R_B}}$$

This is suitable for duty cycle is greater than 50%.

To generate < 50%, = 50% we modify the circuit.



The capacitor charge through R_A & [illegible] R_B

$\therefore T_{ON} = 0.69\,(R_A)\,C$ (charging) design an astable multivibrator using IC555 timer for o/p of frequency 1KHz and duty cycle of 75%.

<u>Note:</u>

design a astable using IC-555.

Duty cycle $> 50\%$.

Duty cycle $< 50\%$.

$T_{ON} = 0.69 (R_A + R_B) C.$

$T_{OFF} = 0.69 R_B C.$

$T_{ON} = 0.69 R_A C.$

$T_{OFF} = 0.69 R_B C.$

if $D = 50\%$, then $R_A, R_B = R$.

<u>Sol:-</u>

Given, $f = 1KHz = 10^3 Hz \Rightarrow \boxed{T = 1ms.}$

$D = 75\% = \frac{75}{100} = 0.75.$

$D = \frac{T_{ON}}{T} \Rightarrow T_{ON} = T \times D = \frac{D}{f}.$

$= \frac{0.75}{10^3}$

$\boxed{T_{ON} = 0.75 \times 10^{-3} s}$

$T = T_{ON} + T_{OFF}.$

$T_{OFF} = 1 ms - 0.75 ms.$

$$\boxed{T_{OFF} = 0.25 \text{ ms}}$$

$$D = \frac{(R_A + R_B)C}{(R_A + 2R_B)C} = 0.75$$

$$R_A + R_B = 0.75 R_A + 1.5 R_B$$

$$0.25 R_A = 0.5 R_B$$

$$\boxed{R_A = 2R_B}$$

to find R_A & R_B:

$$T_{ON} = 0.69 (R_A + R_B) C = 0.75 \times 10^{-3} s.$$

$$(\because R_A = 2R_B)$$

$$(2R_B + R_B) C = 1.0869 \times 10^{-3}$$

$$3 R_B C = 1.0869.$$

Let $C = 0.1 \times 10^{-6}$ F.

$$R_B = \frac{1.0869 \times 10^{-3}}{0.3 \times 10^{-6}}$$

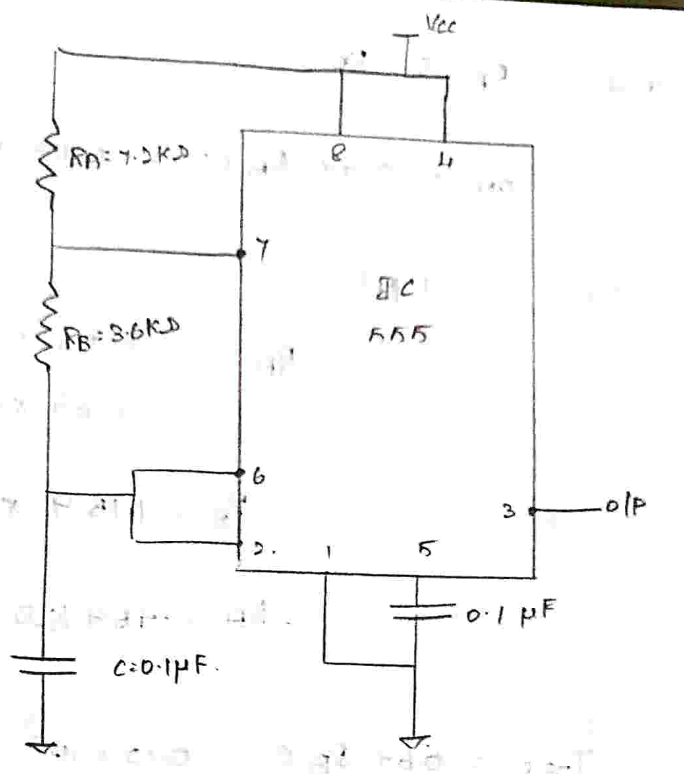
$$R_B = 3.623 \times 10^{3} \Omega.$$

$\because R_A = 2 \times R_B$.

$$R_A = 7.246 \times 10^{3} \Omega.$$

$$\boxed{\therefore R_A = 7.24 \text{ k}\Omega, \quad R_B = 3.623 \text{ k}\Omega.}$$

Design
to pr
duty
sol:



Design an astable multivibrator using IC555 to produce 5 KHz o/p waveform with duty cycle of 40%.

<u>sol:-</u>

Given,

$f = 5\,KHz \Rightarrow T = 0.2\,ms.$

$D = 40\% = \frac{40}{100} = 0.4$

$D = \frac{T_{ON}}{T} \Rightarrow T_{ON} = T \cdot D = 0.4 \times 0.2 \times 10^{-3}$

$\boxed{T_{ON} = 0.08\,ms.}$

$T = T_{ON} + T_{OFF}.$

$T_{OFF} = (0.2 - 0.08) \times 10^{-3}$

$\boxed{T_{OFF} = 0.12\,ms.}$

to find R_A & R_B.

$$T_{ON} = 0.69\, R_A C = 0.08 \times 10^{-3}.$$

Let $C = 0.1\ \mu F$.

$$R_A = \frac{0.08 \times 10^{-3}}{0.69 \times 10^{-7}}$$

$$R_A = 1.159 \times 10^{3}\ \Omega.$$

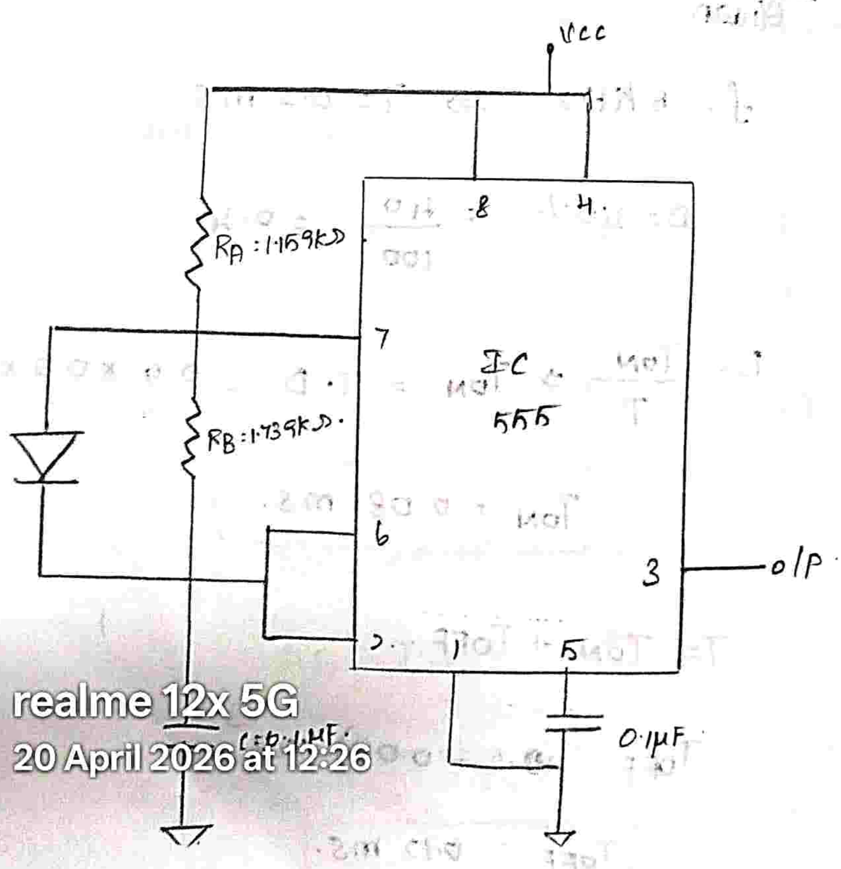
$$\boxed{R_A = 1.159\ K\Omega.}$$

$$T_{OFF} = 0.69\, R_B C = 0.12 \times 10^{-3}.$$

Let $C = 0.1\ \mu F$.

$$R_B = \frac{0.12 \times 10^{-3}}{0.069 \times 10^{-3}}$$

$$\boxed{R_B = 1.739\ K\Omega.}$$



Design

IC 555

2kHz.

Sol:-

To f

Design a astable multivibrator using IC 555 to generate a o/p of 2kHz. with duty cycle of 50%.

Sol:

Given,

$$f = 2\,kHz. \Rightarrow T = 0.5\,ms.$$

$$D = 50\% = \frac{50}{100} = 0.5$$

$$D = \frac{T_{ON}}{T} \Rightarrow T_{ON} = D \cdot T = 0.5 \times 0.5 \times 10^{-3}.$$

$$\boxed{T_{ON} = 0.25\,ms.}$$

$$T = T_{ON} + T_{OFF}.$$

$$T_{OFF} = (0.5 - 0.25) \times 10^{-3}$$

$$\boxed{T_{OFF} = 0.25\,ms.}$$

To find R_A & R_B.

$$T_{ON} = 0.69\,R_A C = 0.25 \times 10^{-3}.$$

Let $C = 0.1\,\mu F$.

$$R_A = \frac{0.25 \times 10^{-3}}{0.069 \times 10^{-6}}$$

$$R_A = 3.623\,k\Omega.$$

IIIly.
$$\boxed{R_A = R_B = 3.623\,k\Omega.}$$

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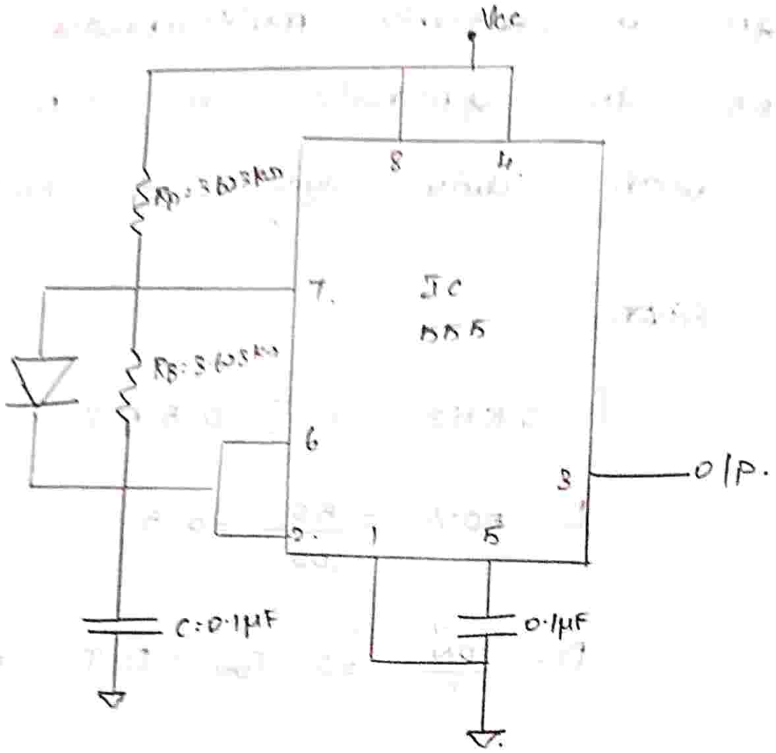


Applications of IC 555 timer in astable mode.

FSK generator circuit: [Frequency shift keying].

free running ramp generator (X) and it's derivation

refer PDF (11-03-26)
↓
in ECE family.

Voltage Controlled Oscillator (VCO).

(X) 16m * o/p frequency is strongly depends on i/p voltage.

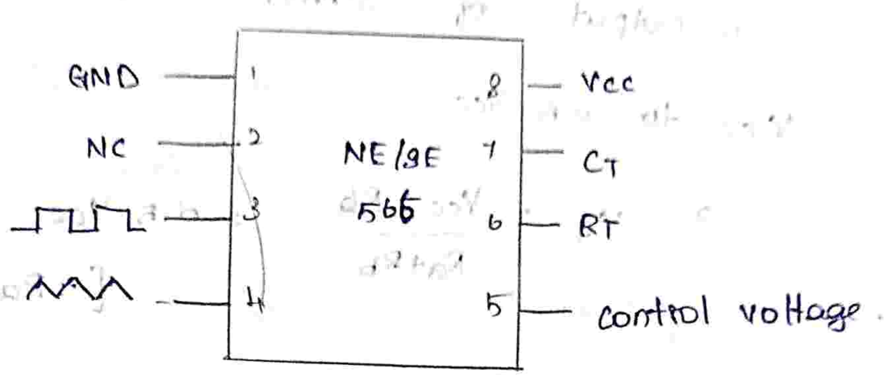
* The frequency of o/p of VCO is controlled by voltage applied

VCO [NE / SE 566].

manufacturer Name

IC Number

It is 8-pin package.



3 → used to get square wave o/p.

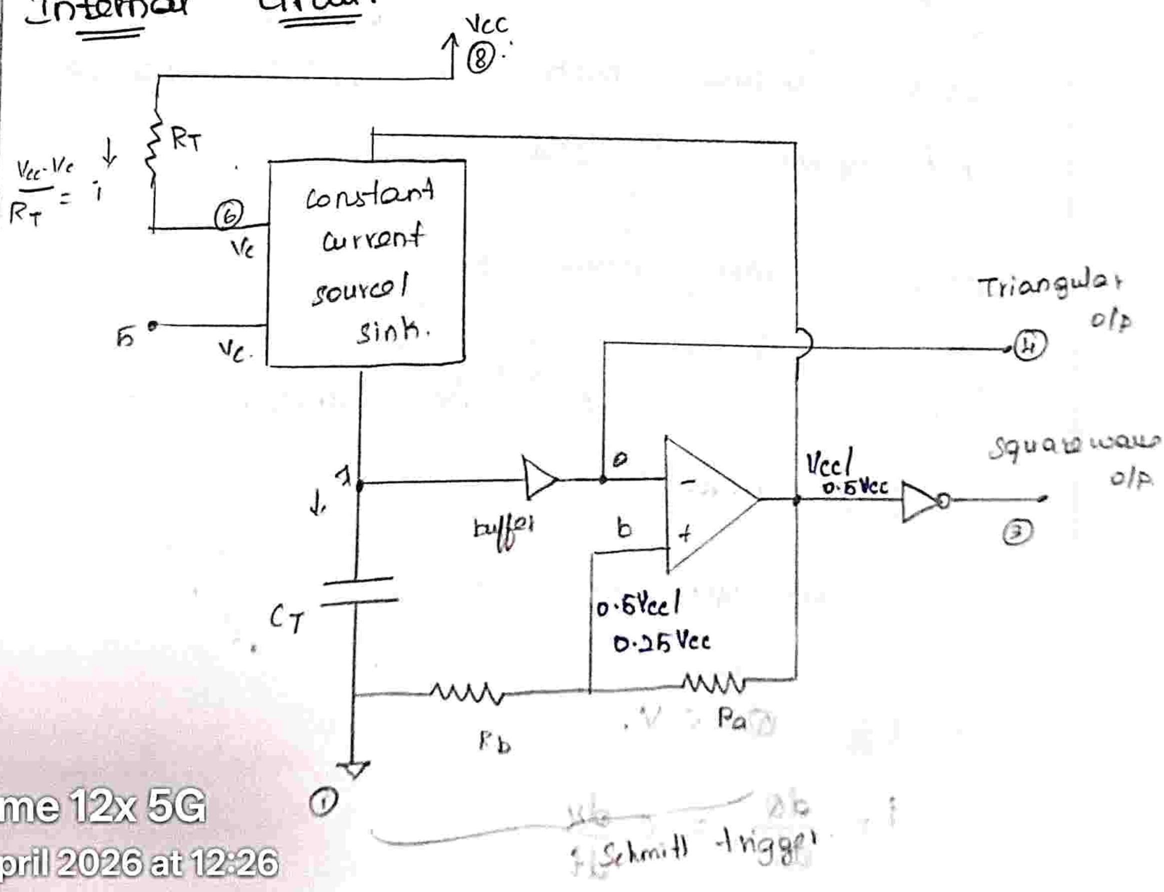
4 → " " " triangle " "

R_T → Timing resistor   NC → No connection

C_T → Timing capacitor

buffer → high i/p impedance logic. (same i/p get at o/p)

Internal circuit:



conditions :

1. Output of schmit - - trigger vary V_{cc} to $0.5\ V_{cc}$.

2. $V_+ = \dfrac{V_{cc}\ R_b}{R_a + R_b} = 0.5\ V_{cc}$ $\quad [\because R_a = R_b]$

o/p of ST = Vcc.

$V_+ = \dfrac{0.5\ V_{cc}\ R_b}{R_a + R_b} = 0.25\ V_{cc}.$

o/p of ST = 0.5 Vcc

*. when o/p of ST is Vcc, capacitor charge , when beyond 0.5 Vcc charges ↑se , the o/p goes 0.5 Vcc.

*. The capacitor discharge , when it goes below 0.25 Vcc , o/p changes Vcc and again it charges.

without constant current source,

o/p will be.

by using current source.

we get linear o/p

$Q = CV.$

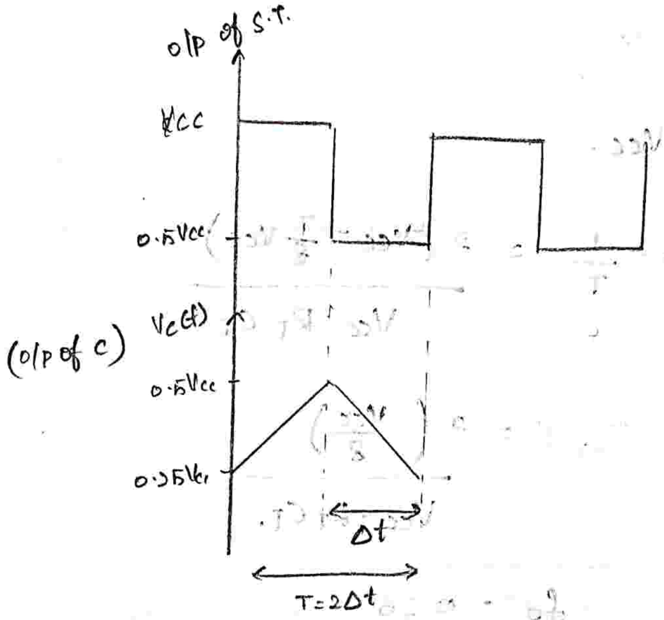
$i = \dfrac{dQ}{dt} = C\dfrac{dv}{dt}$

$$\boxed{i = C\frac{\Delta V}{\Delta t}}$$

This relates the current & voltage difference of capacitor.

$$i = \frac{V_{cc} - V_c}{R_T}$$

$\Delta V \rightarrow$ voltage difference at capacitor.



$$\left[\Delta V = 0.5V_{cc} - 0.25V_{cc}\right.$$
$$\left.\Delta V = 0.25V_{cc}\right]$$

$$\frac{V_{cc} - V_c}{R_T} = \frac{C_T\left[0.25V_{cc}\right]}{\Delta t}$$

$$\Delta t = \frac{0.25V_{cc} \cdot C_T \cdot R_T}{(V_{cc} - V_c)}$$

$$T = \Delta t + \Delta t = 2\Delta t = \frac{0.5\,V_{cc}\,C_T R_T}{(V_{cc} - V_c)}$$

$\Rightarrow$ it takes Δt for charging & Δt for discharging

$$f_o = \frac{1}{T} = \frac{2(V_{cc} - V_c)}{V_{cc} \cdot R_T \cdot C_T} \longrightarrow (1)$$

General expression of VCO.

Here, all are constant (V_{cc}, R_T, C_T) except V_c - control voltage.

The O/P frequency is controlled by voltage (V_c)

if $V_c = \frac{7}{8} V_{cc}$.

$$f_o = \frac{1}{T} = \frac{2\left(V_{cc} - \frac{7}{8}V_{cc}\right)}{V_{cc} \cdot R_T C_T}$$

$$= \frac{\cancel{2}\left(\frac{\cancel{V_{cc}}}{\cancel{8}\,4}\right)}{\cancel{V_{cc}} \cdot R_T C_T}$$

$$\boxed{f_o = \frac{0.25}{R_T C_T}} \longrightarrow (2)$$

Voltage-frequency conversion ratio (K_v):

To change the frequency we should change the V_c.

$$f_1 = \frac{2(V_{cc} - V_c + \Delta V_c)}{R_T C_T V_{cc}}$$

W.K.T $f_o = \frac{2(V_{cc}-V_c)}{R_T \cdot C_T \cdot V_{cc}}$.

$$f_1 - f_o = \Delta f = \frac{2\cancel{V_{cc}} - 2\cancel{V_c} + 2\Delta V_c}{V_{cc} \cdot R_T C_T} - \frac{2\cancel{V_{cc}} + 2\cancel{V_c}}{V_{cc} R_T C_T}$$

$$\Delta f = \frac{2\Delta V_c}{V_{cc} R_T C_T}.$$

$$\frac{\Delta f}{\Delta V_c} = \frac{2}{R_T C_T V_{cc}}. \longrightarrow (3).$$

Sub (2) in (3).

$$\frac{\Delta f}{\Delta V_c} = \frac{2}{\frac{0.25}{f_o} V_{cc}}.$$

$$\boxed{\therefore \frac{\Delta f}{\Delta V_c} = \frac{8 f_o}{V_{cc}} \Rightarrow K_v.}$$

Phase - Locked Loop: (PLL) : refer pdf [16/3/26]

It can able to extract the i/p signal.

(or). It is used to fetch transmitter frequency when i/p is applied.

⇒ when difference b/w f_o & f_s is 90° then the phase detector will be 0 error voltage

* Main blocks of PLL. (NE/SE 565)

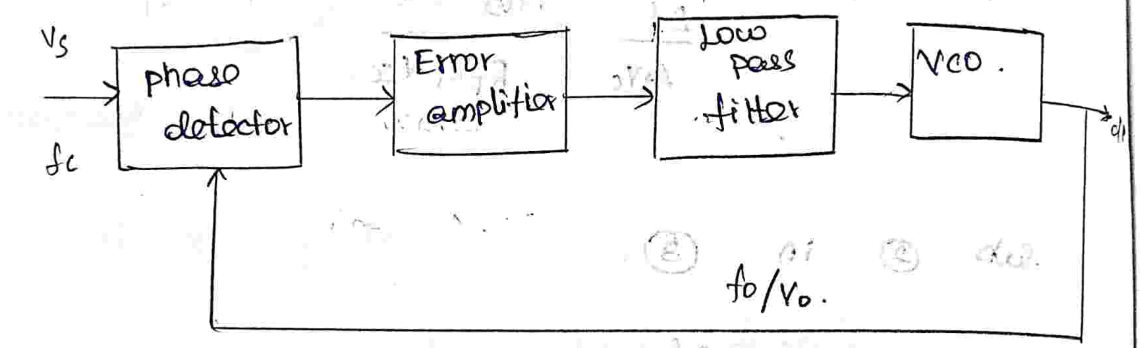
1) Phase detector.

2). error amplifier.

3). Low pass filter.

4) VCO.

Block diagram:



* PLL can be operated in 3 Modes.

1. Free Running Mode. (i/p is not given)

2. Capture Mode. - capture Range

3. Lock Mode. Lock-in-Range.

Pull - in Time

* In Free Running Mode the PLL act as VCO.

* Remaining 2 Modes are used to find when the o/p freq of VCO is equal to

$f_o = f_s$.

error amplifier - It depends upon o/p of phase detector (stregthen i/p signal)

phase detector: - Generally a multiplier circuit

↓

o/p is — $(f_s + f_o)$ sum of i/p freq
— difference of two freq $(f_s - f_o)$

Capture mode → PLL starts to track applied i/p.

Lock mode → PLL can exactly maintain the same relationship $f_o = f_s$ (i/p freq) (f_o: VCO freq)

capture range - over a period of time when it tracing the i/p signal.

Lock in range > Capture Range , $\Delta f_L > \Delta f_c$

For better performance

pull-in time: Time taken by the PLL to reach (attain) the lock mode.